MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3520 NEUTRALIZATION AND THICKENING GRADE SLAB & PIPE RACK FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	Ma'aden Project Management Team
'MMGP Project'	Mansourah Massarah Gold Processing Project
Attribute	Information element that describes a document or tag.
Supplier	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	Quality Assurance
QC	Quality Control
Works	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of Neutralization and Thickening area (3520) - pipe rack & stair case foundation, drain, pit and miscellaneous foundations. The member sizes and reinforcements are to be calculated. The report describes the methodology and the design parameters used in the design calculation.
The analysis & design is carried out in accordance with project specifications,codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD Pro. The STAAD modelling has been done with member & plate elements. Plate mat with subgrade modulus has been considered as per L&T approved Soil Report. Base Pressure check has been done from STAAD Pro output. Seismic load combinations are not shown as seismic loads are mush less compared to wind loads.
For strength calculation Bending moment & Shear Force values are taken from STAAD output & designed in the excel sheet.

4. DESIGN REQUIREMENTS

4.1. Design References

Reference is made in this standard as indicated in Civil,Structural and Architectural Design Criteria Doc No. : MD-216-1000-EG-ST-CRT-1001-A02 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

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4.2. Materials Property

Density of Material

- Density of Dry Soil : 1.9 T/m^3
- Density of Concrete : 2.5 T/m^3
- Density of Water : 1.0 T/m^3

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

Grout Grade

- Grout shall be cementitious under steel column bases of Strength 40 MPa

4.3. Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	Minimum Cove (mm)
Exposed to Soil	Foundation Bottom & Side	75
	Foundation Top	50
	Column & Beam	50
Exposed to Weather	Column	50
	Beam	50
	Slab	40

4.5. Allowable Soil Bearing Capacity

Recommended net allowable bearing capacity as per L&T soil report MD-216-3000-EG-CV-RPT-1008-A02, bore hole BH-13.

BH-no	Proposed Structures	Depth of foundation below FGL(m)	Recommended net allowable bearing capacity(KN/m2)	Vertical modulus of subgrage reaction(KN/m2/m)
BH-13	Neutralization and thickening	1.5	980	117600

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5. SAFETY FACTOR AGAINST STABILITY

5.1. Overturning : For foundation design a minimum factor of safety against overturning is consider 2.0
5.2. Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

6. LOAD CALCULATIONS

6.1. Primary Load Cases for pipe rack & stair foundation

1	FOUNDATION SELF WEIGHT(D)
2	DEAD LOAD(D1)
3	STRUCTURAL SELF WEIGHT(D2)
4	LIVE LOAD_5KN/SQM (L1)
5	LIVE LOAD_10KN/SQM (L2)
8	LIVE LOAD_20KN/SQM (L3)
11	WIND LOAD +Z(WL+Z)
12	WIND LOAD -Z(WL-Z)
13	WIND LOAD +X(WL+X)
14	WIND LOAD -X(WL-X)
15	TEMPERATURE LOAD_+20(T+20)
16	TEMPERATURE LOAD_-20(T-20)
17	MAXIMUM SEISMIC LOAD (E_MAX)
18	MINIMUM SEISMIC LOAD (E_MIN)

For pipe rack, primary loads are taken from OTE input drawing no. MD-216-3520-EG-ME-GAD-1047-A01
For stair case, primary loads are taken from structural reaction of STAAD_pro file
'MD-216-3520-EG-ST-CAL-2263_ST-1&ST-2.std' & 'MD-216-3520-EG-ST-CAL-2263_ST-3.std'

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6.2. Load Combinations for Stress Calculation(pipe rack & stair foundation)

LOAD CASE	LOAD COMBINATION
101	D+(D1+D2)
102	(D+D1+D2)+T(+20)+L1
103	(D+D1+D2)+T(+20)+L2
104	(D+D1+D2)+T(+20)+L3
105	(D+D1+D2)+T(-20)+L1
106	(D+D1+D2)+T(-20)+L2
107	(D+D1+D2)+T(-20)+L3
108	D+D1+D2+W(+X)
109	D+D1+D2+W(-X)
110	D+D1+D2+W(+Z)
111	D+D1+D2+W(-Z)
112	D+D1+D2+0.75L1+0.75W(+X)
113	D+D1+D2+0.75L1+0.75W(-X)
114	D+D1+D2+0.75L1+0.75W(+Z)
115	D+D1+D2+0.75L1+0.75W(-Z)
116	D+D1+D2+0.75L2+0.75W(+X)
117	D+D1+D2+0.75L2+0.75W(-X)
118	D+D1+D2+0.75L2+0.75W(+Z)
119	D+D1+D2+0.75L2+0.75W(-Z)
120	D+D1+D2+0.75L3+0.75W(+X)
121	D+D1+D2+0.75L3+0.75W(-X)
122	D+D1+D2+0.75L3+0.75W(+Z)
123	D+D1+D2+0.75L3+0.75W(-Z)
124	0.6(D1+D2)+W(+X)+D
125	0.6(D1+D2)+W(-X)+D
126	0.6(D1+D2)+W(+Z)+D
127	0.6(D1+D2)+W(-Z)+D
128	D+D1+D2+0.7E_MAX
129	D+D1+D2+0.7E_MIN
130	D+D1+D2+0.75L1+0.75*0.7E_MAX
131	D+D1+D2+0.75L1+0.75*0.7E_MIN
132	D+D1+D2+0.75L2+0.75*0.7E_MAX
133	D+D1+D2+0.75L2+0.75*0.7E_MIN
134	D+D1+D2+0.75L3+0.75*0.7E_MAX
135	D+D1+D2+0.75L3+0.75*0.7E_MIN
136	0.6(D1+D2)+0.7E_MAX+D
137	0.6(D1+D2)+0.7E_MIN+D

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6.3 Load Combinations for Strength Calculation(pipe rack & stair foundation)

LOAD CASE	LOAD COMBINATION
201	1.4D+1.4(D1+D2)
202	1.4(D+D1+D2)+1.4T(+20)+1.7L1
203	1.4(D+D1+D2)+1.4T(+20)+1.7L2
204	1.4(D+D1+D2)+1.4T(+20)+1.7L3
205	1.4(D+D1+D2)+1.4T(-20)+1.7L1
206	1.4(D+D1+D2)+1.4T(-20)+1.7L2
207	1.4(D+D1+D2)+1.4T(-20)+1.7L3
208	1.2(D+D1+D2)+0.8W(+X)
209	1.2(D+D1+D2)+0.8W(-X)
210	1.2(D+D1+D2)+0.8W(+Z)
211	1.2(D+D1+D2)+0.8W(-Z)
212	1.2(D+D1+D2)+1.0L1+1.6W(+X)
213	1.2(D+D1+D2)+1.0L1+1.6W(-X)
214	1.2(D+D1+D2)+1.0L1+1.6W(+Z)
215	1.2(D+D1+D2)+1.0L1+1.6W(-Z)
216	1.2(D+D1+D2)+1.0L2+1.6W(+X)
217	1.2(D+D1+D2)+1.0L2+1.6W(-X)
218	1.2(D+D1+D2)+1.0L2+1.6W(+Z)
219	1.2(D+D1+D2)+1.0L2+1.6W(-Z)
220	1.2(D+D1+D2)+1.0L3+1.6W(+X)
221	1.2(D+D1+D2)+1.0L3+1.6W(-X)
222	1.2(D+D1+D2)+1.0L3+1.6W(+Z)
223	1.2(D+D1+D2)+1.0L3+1.6W(-Z)
224	0.9(D1+D2)+W(+X)+D
225	0.9(D1+D2)+W(-X)+D
226	0.9(D1+D2)+W(+Z)+D
227	0.9(D1+D2)+W(-Z)+D
228	1.2(D+D1+D2)+1.0L1+1.0E_MAX
229	1.2(D+D1+D2)+1.0L1+1.0E_MIN
230	1.2(D+D1+D2)+1.0L2+1.0E_MAX
231	1.2(D+D1+D2)+1.0L2+1.0E_MIN
232	1.2(D+D1+D2)+1.0L3+1.0E_MAX
233	1.2(D+D1+D2)+1.0L3+1.0E_MIN
234	0.9(D1+D2)+1.0E_MAX(+X)+D
235	0.9(D1+D2)+1.0E_MIN(+X)+D

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6.4. Primary Load Cases for pit design

1 DEAD LOAD(D)
2 SOIL PRESSURE(DRY CONDITION) (H)
3 SURCHARGE PRESSURE (L)

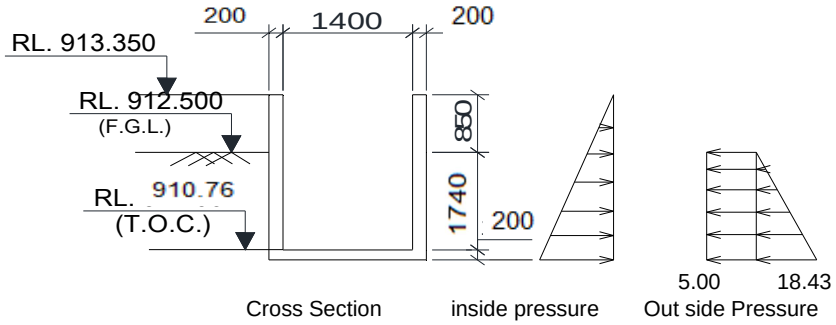
6.5. Load Combinations for Stress Calculation(pit design)

LOAD CASE	LOAD COMBINATION
101	D
102	D+H+L

6.6. Load Combinations for Strength Calculation(pit design)

LOAD CASE	LOAD COMBINATION
201	1.4D
202	1.4D+1.7(H+L)

6.7. Primary Load Calculation for pit design



Input Data :

Clear height of pit from FGL =	1740	mm	C/C distance along width of Pit =	1600	mm
Clear width of Pit =	1400	mm	C/C distance along length of Pit =	1600	mm
Clear length of Pit =	1400	mm	Bottom of pit from FGL,h =	1940	mm
Vertical wall thickness =	200	mm			
Bottom slab thickness =	200	mm			
Pressure coefficient, C_p =	0.5			1840	
Surcharge load at side of pit, P =	10	kN/m^2		190	
Dry unit weight of soil, $\gamma_{dry\ soil}$ =	19	kN/m^3			

1 DEAD LOAD(D)
Selfweight of structure

2 SOIL PRESSURE(DRY CONDITION) (H)
Lateral earth pressure at side wall due to dry soil = C_p x $\gamma_{dry\ soil}$ x h 18.43 kN/m^2

3 SURCHARGE PRESSURE (L)
Lateral earth pressure at side wall due to surcharged load = C_p x P 5.00 kN/m^2

4 WATER PRESSURE (F)
Water pressure at side wall = 27.90 kN/m^2

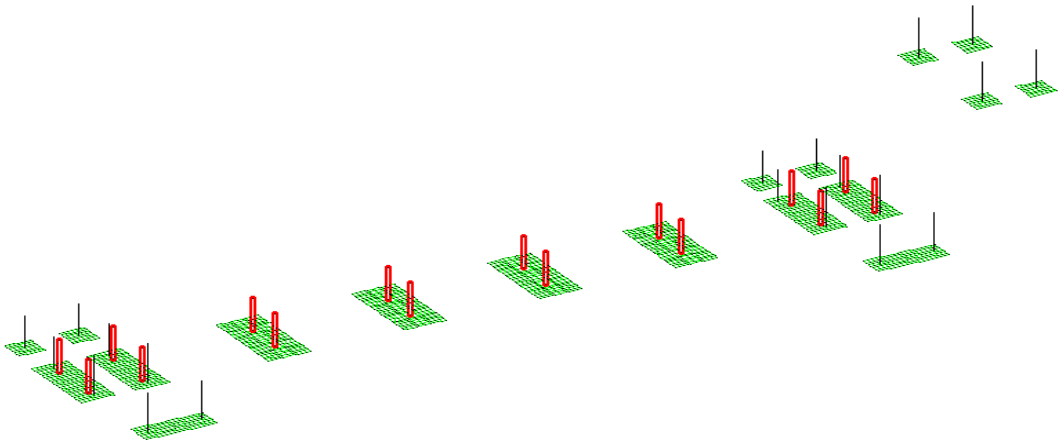
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7. Design Calculations

7.1. Design of Columns



Column group as per STAAD_pro

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Design of column C1

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.90

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 700 mm

Depth parallel to Z-Z axis, b = 700 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.79

Y_b = (b-d')/b = 0.79

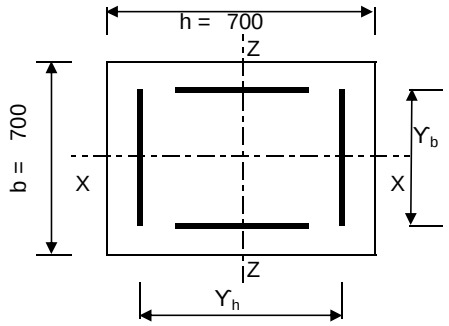
Unsupported length of column parallel to X-X axis, l_{ux} = 1.85 m kl_{ux}/r_b = 8.81 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.85 m kl_{uz}/r_h = 8.81

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 13501.8 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
Max Fx	1157	6.2	273.7	1286	6.9	304.1	0.075	0.070	840.4	0.01	0.070	840.4	0.36	1.61	0.2	safe
Min Fx	-543.8	0.0	0.0	-604	0.0	0.0	-0.04	0.070	840.4	0.00	0.070	840.4	0.00		0	safe
Max Fy	400.4	3.9	478.5	445	4.3	531.7	0.026	0.070	840.4	0.01	0.070	840.4	0.63		0.48	safe
Min Fy	830.4	15.3	494.6	923	17.0	549.6	0.054	0.070	840.4	0.02	0.070	840.4	0.65		0.51	safe
Max Fz	-508.2	451.5	49.2	-565	501.6	54.7	-0.03	0.070	840.4	0.60	0.070	840.4	0.07		0.45	safe
Min Fz	646.7	353.3	29.1	719	392.6	32.4	0.042	0.070	840.4	0.47	0.070	840.4	0.04		0.3	safe
Max Mx	155.0	1.2	0.6	172	1.4	0.6	0.01	0.070	840.4	0.00	0.070	840.4	0.00		0	safe
Min Mx	155.0	1.2	0.6	172	1.4	0.6	0.01	0.070	840.4	0.00	0.070	840.4	0.00		0	safe
Max My	646.7	353.3	29.1	719	392.6	32.4	0.042	0.070	840.4	0.47	0.070	840.4	0.04		0.3	safe
Min My	-508.2	451.5	49.2	-565	501.6	54.7	-0.03	0.070	840.4	0.60	0.070	840.4	0.07		0.45	safe
Max Mz	400.4	3.9	478.5	445	4.3	531.7	0.026	0.070	840.4	0.01	0.070	840.4	0.63		0.48	safe
Min Mz	830.4	15.3	494.6	923	17.0	549.6	0.054	0.070	840.4	0.02	0.070	840.4	0.65		0.51	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
4900	12	25				5893	1.2	10	400	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

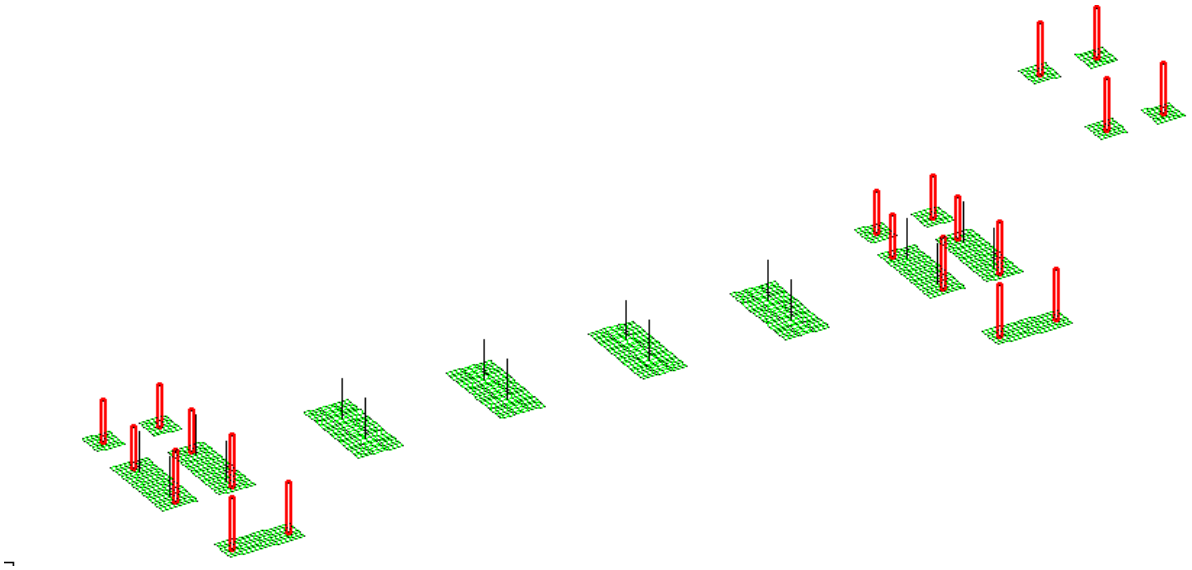
Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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Design of Column C2



Column group as per STAAD_pro

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Design of column C2

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.90

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.72

Y_b = (b-d')/b = 0.72

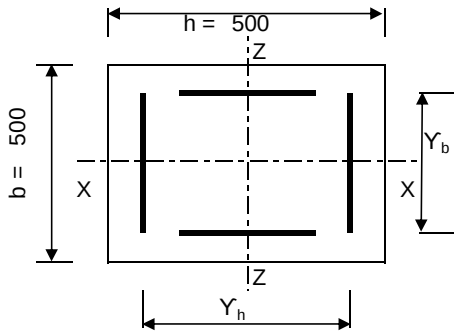
Unsupported length of column parallel to X-X axis, l_{ux} = 2.4 m kl_{ux}/r_b = 16 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.4 m kl_{uz}/r_h = 16

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 6735.18 KN safe

Factored Force Value				Design force Value			$K_n = P_n/f_c'bh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$		$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
Max Fx	213	18.5	81.2	237	20.6	90.2	0.027	0.060	262.5	0.08	0.060	262.5	0.34		0.2	safe
Max Fy	194.7	1.8	79.9	216	2.0	88.8	0.025	0.060	262.5	0.01	0.060	262.5	0.34		0.18	safe
Min Fy	213.0	18.5	81.2	237	20.6	90.2	0.027	0.060	262.5	0.08	0.060	262.5	0.34		0.2	safe
Max Fz	140.1	60.7	1.2	156	67.4	1.3	0.018	0.060	262.5	0.26	0.060	262.5	0.00		0.11	safe
Min Fz	154.6	60.6	27.9	172	67.4	30.9	0.02	0.060	262.5	0.26	0.060	262.5	0.12		0.15	safe
Max Mx	62.8	9.5	1.1	70	10.6	1.3	0.008	0.060	262.5	0.04	0.060	262.5	0.00	1.61	0.01	safe
Min Mx	62.8	9.5	1.1	70	10.6	1.3	0.008	0.060	262.5	0.04	0.060	262.5	0.00		0.01	safe
Max My	154.6	60.6	27.9	172	67.4	30.9	0.02	0.060	262.5	0.26	0.060	262.5	0.12		0.15	safe
Min My	140.1	60.7	1.2	156	67.4	1.3	0.018	0.060	262.5	0.26	0.060	262.5	0.00		0.11	safe
Max Mz	194.7	1.8	79.9	216	2.0	88.8	0.025	0.060	262.5	0.01	0.060	262.5	0.34		0.18	safe
Min Mz	213.0	18.5	81.2	237	20.6	90.2	0.027	0.060	262.5	0.08	0.060	262.5	0.34		0.2	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars, db				A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2500	8	20 +	0	0	2515	1.01	10	320	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.7

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7.2. Design of pipe rack & stair foundation Foundations

Foundation Base Pressure :

refer staad file : MD-216-3520-EG-CV-CAL-1285_pipe rack fdn in Neutralization area

	Node	L/C	Horizontal Fx KN/m2	Vertical Fy KN/m2	Horizontal Fz KN/m2
Max Px	2000	101	0	76.048	0
Min Px	2000	101	0	76.048	0
Max Py	2000	104	0	280.717	0
Min Py	3215	102	0	0	0
Max Pz	2000	101	0	76.048	0
Min Pz	2000	101	0	76.048	0

Gross Maximum Base pressure, p_{max} = 280.7 KN/m^2

Net Allowable Safe Bearing Capacity = 980 KN/m^2 (considered)

Depth of footing from FGL = 2 m

Unit weight of soil = 19 KN/m^3

Gross Safe Bearing Capacity = 1018 KN/m^2

> 280.7 KN/m^2 safe

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Design of Foundation F1:

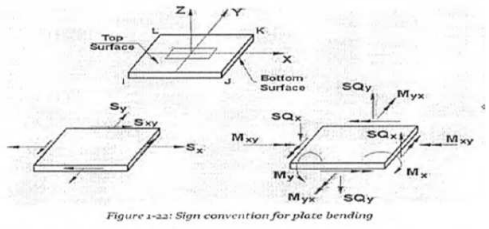
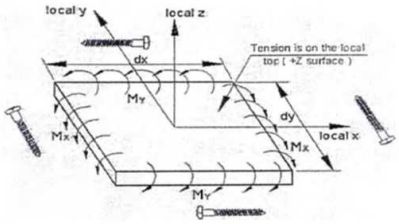


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G1 : F1_MOMENT At the face of the column
refer staad file : MD-216-3520-EG-CV-CAL-1285_pipe rack fdn in Neutralization area

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	2746	0.185	-0.052	0	0	0	-19.75	-10.96	2.652	-22.4	-13.61	212 1.2(D+D1+D2)+1.0L1+1.6W(+X)
Min Qx	2713	-0.203	-0.049	0	0	0	-20.88	-11.78	0	-20.88	-11.78	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Max Qy	2510	-0.03	0.106	0	0	0	-7.214	-10.88	1.185	-8.399	-12.07	215 1.2(D+D1+D2)+1.0L1+1.6W(-Z)
Min Qy	2718	-0.018	-0.134	0	0	0	-6.917	-12.75	-3.252	-10.17	-16	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Max Sx	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Min Sx	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Max Sy	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Min Sy	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Max Sxy	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Min Sxy	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Max Mx	2704	-0.083	-0.024	0	0	0	12.62	-5.036	4.471	17.09	-9.507	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Min Mx	2713	-0.203	-0.049	0	0	0	-20.88	-11.78	-3.751	-24.63	-15.53	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Max My	2697	0	-0.051	0	0	0	-6.124	7.534	3.733	-9.857	11.27	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Min My	2707	-0.187	0.061	0	0	0	-19.93	-13.61	0.718	-20.65	-14.33	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Max Mxy	2698	-0.042	-0.06	0	0	0	3.468	3.006	7.725	11.19	10.73	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Min Mxy	2737	0.031	-0.052	0	0	0	2.814	1.932	-6.901	9.715	8.833	212 1.2(D+D1+D2)+1.0L1+1.6W(+X)

PLATE CENTER STRESS ; G2 : F1_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	2747	0.061	-0.024	0	0	0	-3.435	-9.48	0.256	-3.691	-9.736	212 1.2(D+D1+D2)+1.0L1+1.6W(+X)
Min Qx	2714	-0.066	-0.033	0	0	0	-3.558	-11.08	0	-3.558	-11.08	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Max Qy	2735	0.04	0.04	0	0	0	-2.587	-5.156	-0.233	-2.82	-5.389	212 1.2(D+D1+D2)+1.0L1+1.6W(+X)
Min Qy	2720	-0.046	-0.053	0	0	0	-2.78	-6.191	-0.749	-3.529	-6.94	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Max Sx	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Min Sx	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Max Sy	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Min Sy	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Max Sxy	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Min Sxy	2762	0	0	0	0	0	-0.221	-0.245	0.28	-0.501	-0.525	201 1.4D+1.4(D1+D2)
Max Mx	2705	-0.026	-0.004	0	0	0	1.981	-0.768	0.768	2.749	-1.536	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Min Mx	2724	-0.036	-0.038	0	0	0	-9.864	-1.406	-2.313	-12.18	-3.719	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Max My	2536	0.018	0.006	0	0	0	1.073	2.168	0.196	1.269	2.364	224 0.9(D1+D2)+W(+X)+D
Min My	2714	-0.066	-0.033	0	0	0	-3.558	-11.08	-0.962	-4.52	-12.04	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Max Mxy	2760	0.013	-0.005	0	0	0	-4.807	-0.296	2.605	-7.412	-2.901	212 1.2(D+D1+D2)+1.0L1+1.6W(+X)
Min Mxy	2721	-0.012	-0.019	0	0	0	-6.309	-1.096	-2.963	-9.272	-4.059	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

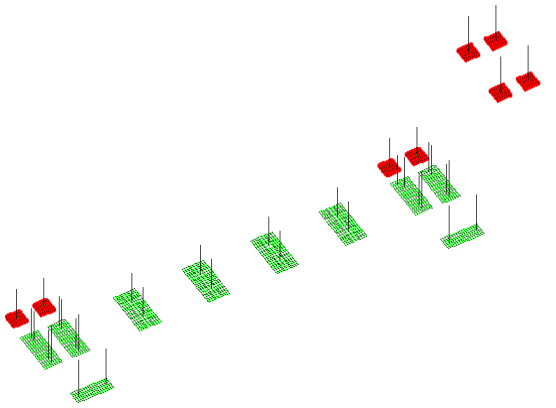
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c'bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	17.1	top face	50	444	17.09	0.014	102.0		12	251.43	200	565.8	7.978	9.972	0.131	tension control
Min Mxy	9.7	top face	50	442	9.715	*As	58.2	900	16	223.49	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.061	1	30.5	3.691	0.004	0.444	518.3	447	761.8	446.99	1635	> 30.5	ok	335.2	φVc>Vu	not required
	Min Qx	0.066	1	33	3.558			530.3			446.99	1635	> 33	ok	335.2	φVc>Vu	not required

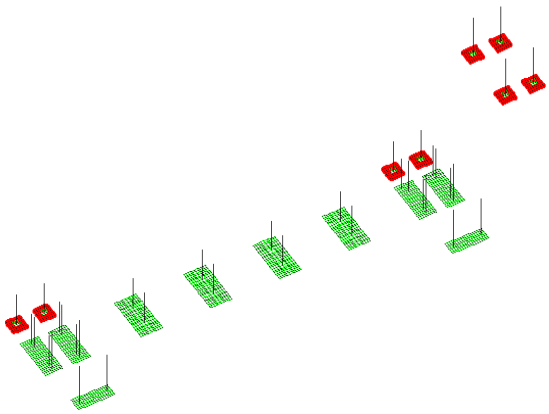
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c'bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	11.3	top face	50	444	11.27	0.014	67.2	900	12	251.43	200	565.8	7.978	9.972	0.131	tension control
Min My	-14.3	bottom face	75	417	14.33		91.0		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.04	1	20	5.389	0.001	0.444	464.3	447	761.8	446.99	465	> 20	ok	335.2	φVc>Vu	not required
	Min Qy	0.053	1	26.5	6.94			465.6			446.99	465	> 26.5	ok	335.2	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F2:

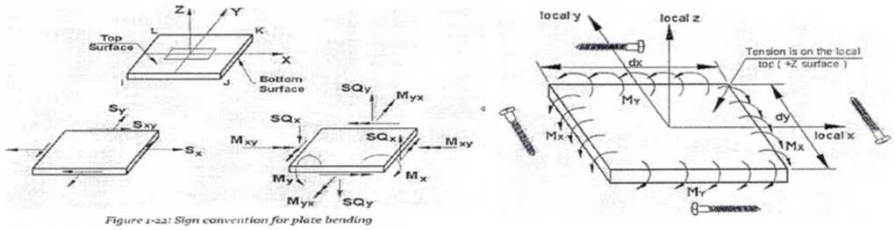


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G3 : F2_MOMENT At the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	1149	0.487	-0.322	0	0	0	-86.35	5.948	-59.55	-145.9	65.5	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Min Qx	1227	-0.878	0.137	0	0	0	-117.5	-106.5	0	-117.5	-106.5	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Max Qy	1226	-0.474	0.601	0	0	0	-91.53	-56.18	-22.24	-113.8	-78.42	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Min Qy	1229	-0.492	-0.799	0	0	0	-84.24	-42.45	-24.08	-108.3	-66.53	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Max Sx	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Min Sx	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Max Sy	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Min Sy	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Max Sxy	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Min Sxy	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Max Mx	1107	-0.11	-0.105	0	0	0	50.59	-9.664	-18.62	69.21	-28.28	202 1.4(D+D1+D2)+1.4T(+20)+1.7L1
Min Mx	1227	-0.878	0.137	0	0	0	-117.5	-106.5	-26.52	-144	-133	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Max My	2432	-0.023	-0.202	0	0	0	-3.483	80.51	-22.66	-26.15	103.2	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Min My	1138	0.148	0.276	0	0	0	-20.34	-193.3	-9.536	-29.87	-202.8	227 0.9(D1+D2)+W(-Z)+D
Max Mxy	1229	0.012	0.047	0	0	0	7.435	23.15	26.71	34.14	49.86	207 1.4(D+D1+D2)+1.4T(-20)+1.7L3
Min Mxy	1109	-0.009	0.175	0	0	0	23.08	21.92	-82.44	105.5	104.4	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3

PLATE CENTER STRESS ; G4 : F2_SHEAR At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	1096	0.094	0.188	0	0	0	-17.37	-59.51	4.444	-21.81	-63.96	223 1.2(D+D1+D2)+1.0L3+1.6W(-Z)
Min Qx	1307	-0.212	-0.058	0	0	0	-13.21	-76.43	0	-13.21	-76.43	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Max Qy	1136	0.05	0.252	0	0	0	-29.29	-61.93	2.829	-32.12	-64.75	223 1.2(D+D1+D2)+1.0L3+1.6W(-Z)
Min Qy	1310	-0.097	-0.426	0	0	0	-10.67	-0.853	-12.19	-22.86	-13.05	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Max Sx	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Min Sx	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Max Sy	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Min Sy	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Max Sxy	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Min Sxy	2464	-0.004	-0.002	0	0	0	-0.842	19.24	-2.019	-2.861	21.26	201 1.4D+1.4(D1+D2)
Max Mx	1028	-0.044	0.165	0	0	0	8.24	10.08	-25.31	33.55	35.4	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Min Mx	1191	0.012	-0.276	0	0	0	-43.49	46.64	-46.79	-90.28	93.43	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Max My	2432	-0.023	-0.202	0	0	0	-3.483	80.51	-22.66	-26.15	103.2	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Min My	1176	-0.038	0.118	0	0	0	-19.88	-147.5	-0.696	-20.57	-148.2	227 0.9(D1+D2)+W(-Z)+D
Max Mxy	1191	-0.024	-0.026	0	0	0	-4.796	25.33	20.7	-25.5	46.03	206 1.4(D+D1+D2)+1.4T(-20)+1.7L2
Min Mxy	1111	0.06	-0.076	0	0	0	-10.74	38.25	-59.38	-70.12	97.63	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

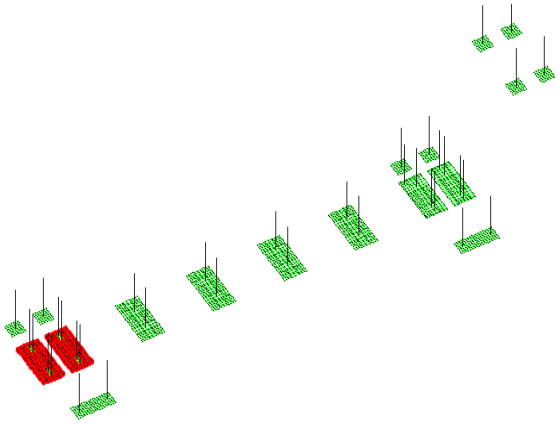
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	69.2	top face	50	444	69.21	0.014	415.1	900	12	251.43	200	565.8	7.978	9.972	0.131	tension control
Min Mx	-144.0	bottom face	75	415	144		932.7		20	336.95	200	1571.5	22.16	27.7	0.042	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.094	1	47	21.81	0.004	0.444	445.8	447	761.8	445.84	1635	> 47	ok	334.4	φVc>Vu	not required
	Min Qx	0.212	1	106	13.21			515.5			446.99	1635	> 106	ok	335.2	φVc>Vu	not required

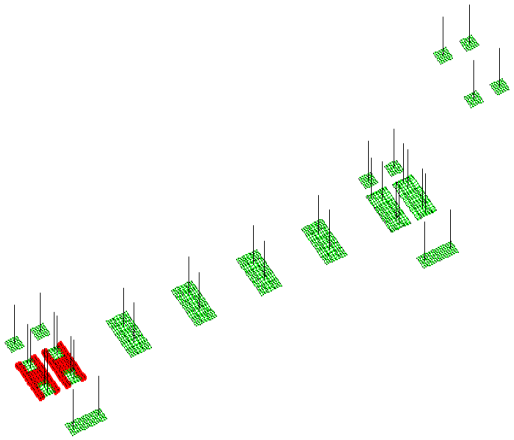
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	103.2	top face	50	442	103.2	0.014	623.7	900	16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min My	-202.8	bottom face	75	415	202.8		1322.5		20	237.65	200	1571.5	22.16	27.7	0.042	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.252	1	126	64.75	0.002	0.442	443.4	447	761.8	443.36	463	> 126	ok	332.5	φVc>Vu	not required
	Min Qy	0.426	1	213	13.05			613.9			446.99	465	> 213	ok	335.2	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F2A & F2B :

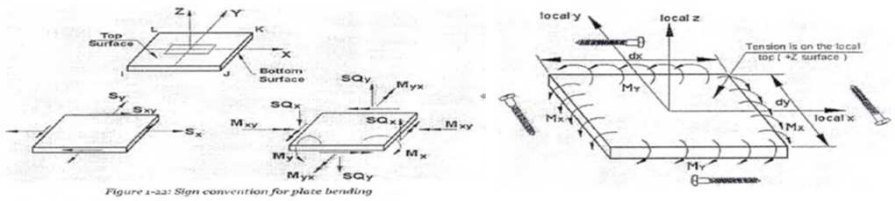


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G9 : F2A & F2B_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	873	0.688	-0.246	0	0	0	-111.3	-71.11	26.66	-138	-97.77	202 1.4(D+D1+D2)+1.4T(+20)+1.7L1
Min Qx	817	-0.528	-0.191	0	0	0	-54.67	-32.21	0	-54.67	-32.21	222 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max Qy	881	-0.268	0.55	0	0	0	59.39	60.81	-49.26	108.6	110.1	206 1.4(D+D1+D2)+1.4T(-20)+1.7L2
Min Qy	881	0.389	-0.678	0	0	0	-80.47	-40.46	38.26	-118.7	-78.73	202 1.4(D+D1+D2)+1.4T(+20)+1.7L1
Max Sx	2885	0.015	0.025	0	0	0	-0.17	-1.114	-0.526	-0.696	-1.64	201 1.4D+1.4(D1+D2)
Min Sx	2885	0.015	0.025	0	0	0	-0.17	-1.114	-0.526	-0.696	-1.64	201 1.4D+1.4(D1+D2)
Max Sy	2885	0.015	0.025	0	0	0	-0.17	-1.114	-0.526	-0.696	-1.64	201 1.4D+1.4(D1+D2)
Min Sy	2885	0.015	0.025	0	0	0	-0.17	-1.114	-0.526	-0.696	-1.64	201 1.4D+1.4(D1+D2)
Max Sxy	2885	0.015	0.025	0	0	0	-0.17	-1.114	-0.526	-0.696	-1.64	201 1.4D+1.4(D1+D2)
Min Sxy	2885	0.015	0.025	0	0	0	-0.17	-1.114	-0.526	-0.696	-1.64	201 1.4D+1.4(D1+D2)
Max Mx	873	-0.514	0.169	0	0	0	86.82	77.22	-34.11	120.9	111.3	207 1.4(D+D1+D2)+1.4T(-20)+1.7L3
Min Mx	865	0.665	0.156	0	0	0	-113.1	-76.26	20.01	-133.1	-96.26	202 1.4(D+D1+D2)+1.4T(+20)+1.7L1
Max My	792	0.077	-0.099	0	0	0	-20.64	129.3	-20.77	-41.4	150.1	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Min My	788	0.126	0.237	0	0	0	-14.61	-176.5	-10.33	-24.93	-186.9	227 0.9(D1+D2)+W(-Z)+D
Max Mxy	885	0.135	0.366	0	0	0	43.89	27.21	66.07	110	93.28	202 1.4(D+D1+D2)+1.4T(+20)+1.7L1
Min Mxy	885	-0.196	-0.438	0	0	0	-59.25	-11.88	-61.43	-120.7	-73.31	206 1.4(D+D1+D2)+1.4T(-20)+1.7L2

PLATE CENTER STRESS ; G10 : F2A & F2B_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	867	0.187	-0.056	0	0	0	-16.56	-58.76	15.99	-32.55	-74.75	202 1.4(D+D1+D2)+1.4T(+20)+1.7L1
Min Qx	871	-0.124	-0.082	0	0	0	-14.23	-22.22	0	-14.23	-22.22	207 1.4(D+D1+D2)+1.4T(-20)+1.7L3
Max Qy	891	-0.05	0.318	0	0	0	7.897	39.69	-24.15	32.04	63.83	206 1.4(D+D1+D2)+1.4T(-20)+1.7L2
Min Qy	891	0.089	-0.327	0	0	0	-12.52	-8.374	16.82	-29.35	-25.2	202 1.4(D+D1+D2)+1.4T(+20)+1.7L1
Max Sx	2884	-0.002	-0.033	0	0	0	-0.154	17.64	-2.097	-2.251	19.74	201 1.4D+1.4(D1+D2)
Min Sx	2884	-0.002	-0.033	0	0	0	-0.154	17.64	-2.097	-2.251	19.74	201 1.4D+1.4(D1+D2)
Max Sy	2884	-0.002	-0.033	0	0	0	-0.154	17.64	-2.097	-2.251	19.74	201 1.4D+1.4(D1+D2)
Min Sy	2884	-0.002	-0.033	0	0	0	-0.154	17.64	-2.097	-2.251	19.74	201 1.4D+1.4(D1+D2)
Max Sxy	2884	-0.002	-0.033	0	0	0	-0.154	17.64	-2.097	-2.251	19.74	201 1.4D+1.4(D1+D2)
Min Sxy	2884	-0.002	-0.033	0	0	0	-0.154	17.64	-2.097	-2.251	19.74	201 1.4D+1.4(D1+D2)
Max Mx	2849	0.003	-0.001	0	0	0	20.29	0.27	-2.026	22.32	2.296	202 1.4(D+D1+D2)+1.4T(+20)+1.7L1
Min Mx	732	0.031	-0.254	0	0	0	-43.92	24.27	-12.89	-56.81	37.16	207 1.4(D+D1+D2)+1.4T(-20)+1.7L3
Max My	776	0.006	-0.017	0	0	0	-10.03	99.4	-9.887	-19.92	109.3	222 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Min My	772	0.033	0.086	0	0	0	-17.92	-136.1	-1.15	-19.07	-137.3	227 0.9(D1+D2)+W(-Z)+D
Max Mxy	900	-0.071	-0.064	0	0	0	-12.06	19.57	54.5	-66.56	74.07	202 1.4(D+D1+D2)+1.4T(+20)+1.7L1
Min Mxy	900	0.062	-0.008	0	0	0	-9.247	14.34	-58.74	-67.99	73.09	206 1.4(D+D1+D2)+1.4T(-20)+1.7L2

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

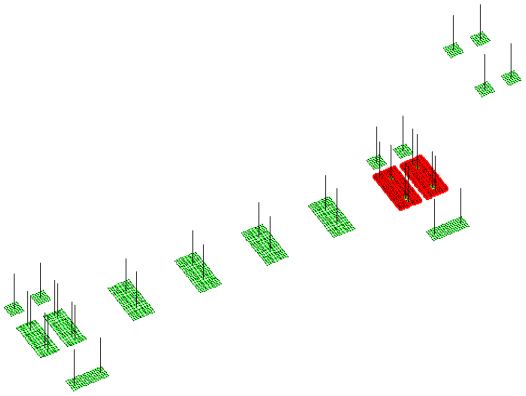
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	120.9	top face	50	442	120.9	0.014	732.4	900	16	274.64	200	1005.8	14.18	17.73	0.072	tension control
Min Mx	-133.1	bottom face	75	417	133.1		856.8		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.187	1	93.5	32.55	0.002	0.442	440.1	435.5	758.3	435.48	1621	> 93.5	ok	326.6	φVc>Vu	not required
	Min Qx	0.124	1	62	14.23			451.3			435.48	1621	> 62	ok	326.6	φVc>Vu	not required

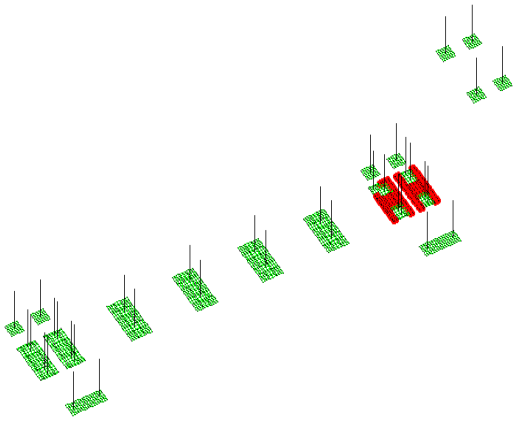
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	150.1	top face	50	442	150.1	0.014	911.4	900	16	220.7	200	1005.8	14.18	17.73	0.072	tension control
Min My	-186.9	bottom face	75	415	186.9		1216.3		20	258.39	200	1571.5	22.16	27.7	0.042	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.318	1	159	63.83	0.002	0.442	437.2	435.5	758.3	435.48	456	> 159	ok	326.6	φVc>Vu	not required
	Min Qy	0.327	1	163.5	25.2			467.4			435.48	456	> 163.5	ok	326.6	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F3:

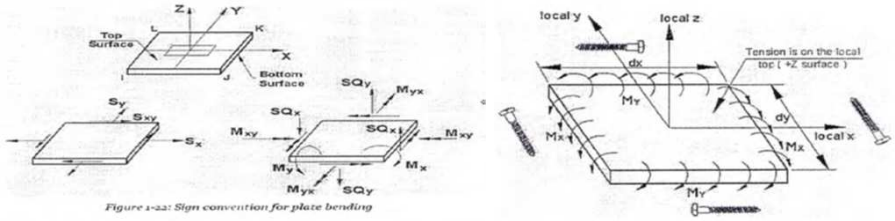


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G5 : F3_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	372	0.753	0.299	0	0	0	68.84	50.38	5.148	73.99	55.53	227 0.9(D1+D2)+W(-Z)+D
Min Qx	367	-1.044	-0.242	0	0	0	-153.1	-135.9	0	-153.1	-135.9	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Max Qy	369	-0.593	0.896	0	0	0	-113.1	-89.48	7.781	-120.9	-97.26	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Min Qy	372	-0.606	-0.915	0	0	0	-115.3	-92.45	-7.228	-122.5	-99.67	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Max Sx	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Min Sx	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Max Sy	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Min Sy	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Max Sxy	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Min Sxy	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Max Mx	287	-0.274	0.586	0	0	0	90.82	77.49	-1.368	92.19	78.86	227 0.9(D1+D2)+W(-Z)+D
Min Mx	358	-1.031	0.281	0	0	0	-153.5	-134.3	-10.61	-156.1	-144.9	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Max My	287	-0.051	0.534	0	0	0	39.19	86.61	-3.158	42.35	89.77	223 1.2(D+D1+D2)+1.0L3+1.6W(-Z)
Min My	366	-0.535	0.044	0	0	0	-74.04	-356.3	14.33	-88.36	-370.6	227 0.9(D1+D2)+W(-Z)+D
Max Mxy	301	0.008	-0.294	0	0	0	23.65	-23.05	64.31	87.96	-87.36	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Min Mxy	304	0.015	0.292	0	0	0	23.54	-24.76	-65.13	88.67	-89.89	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3

PLATE CENTER STRESS ; G6 : F3_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	198	0.181	0.127	0	0	0	-5.78	-116.9	4.318	-10.1	-121.3	223 1.2(D+D1+D2)+1.0L3+1.6W(-Z)
Min Qx	401	-0.346	-0.098	0	0	0	-25.21	-93.07	0	-25.21	-93.07	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Max Qy	1979	-0.034	0.599	0	0	0	-1.344	-171.8	3.414	-4.758	-175.2	227 0.9(D1+D2)+W(-Z)+D
Min Qy	392	0.058	-0.376	0	0	0	3.647	-101.9	11.83	15.47	-113.7	227 0.9(D1+D2)+W(-Z)+D
Max Sx	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Min Sx	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Max Sy	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Min Sy	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Max Sxy	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Min Sxy	2329	-0.006	-0.003	0	0	0	-0.662	-0.483	0.219	-0.881	-0.702	201 1.4D+1.4(D1+D2)
Max Mx	2210	0.064	0.193	0	0	0	63.89	12.28	-6.846	70.73	19.13	227 0.9(D1+D2)+W(-Z)+D
Min Mx	2210	-0.035	-0.332	0	0	0	-70.08	-28.56	-3.653	-73.73	-32.22	222 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max My	468	-0.041	-0.011	0	0	0	2.081	40.65	0.545	2.626	41.19	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Min My	399	-0.149	-0.04	0	0	0	-14.03	-327.8	16.25	-30.28	-344.1	227 0.9(D1+D2)+W(-Z)+D
Max Mxy	2006	0.086	0.13	0	0	0	-39.55	-14.56	33.61	-73.16	-48.17	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
Min Mxy	2206	0.089	-0.136	0	0	0	-40.6	-15.23	-33.88	-74.49	-49.11	204 1.4(D+D1+D2)+1.4T(+20)+1.7L3

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

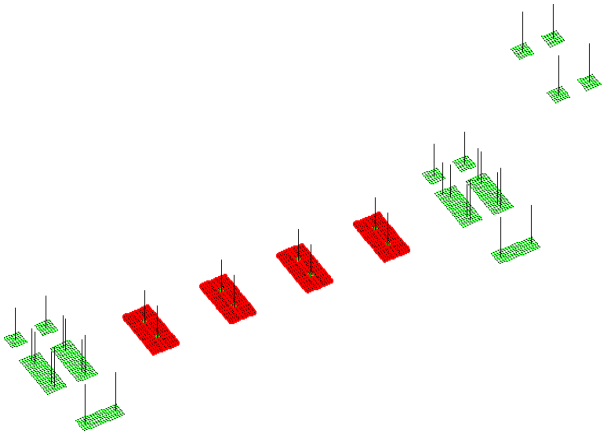
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	92.2	top face	50	444	92.19	0.014	554.2	900	12	204.17	200	565.8	7.978	9.972	0.131	tension control
Min Mx	-156.1	bottom face	75	415	156.1		1012.3		20	310.47	200	1571.5	22.16	27.7	0.042	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.181	1	90.5	10.1	0.001	0.444	458.6	429.9	761.8	429.9	1623	> 90.5	ok	322.4	φVc>Vu	not required
	Min Qx	0.346	1	173	25.21			449.6			429.9	1623	> 173	ok	322.4	φVc>Vu	not required

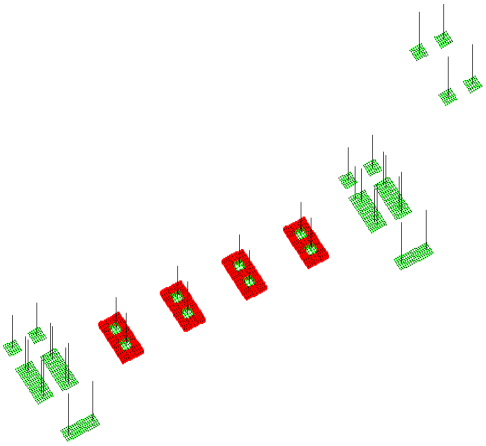
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	89.8	top face	50	444	89.77	0.014	539.5	900	12	209.71	200	565.8	7.978	9.972	0.131	tension control
Min My	-370.6	bottom face	75	415	370.6		2465.9		20	127.45	100	3142.9	44.31	55.39	0.019	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.599	1	299.5	175.2	0.007	0.444	427.6	429.9	761.8	427.58	451	> 299.5	ok	320.7	φVc>Vu	not required
	Min Qy	0.376	1	188	113.7			427.3			427.34	451	> 188	ok	320.5	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F4:

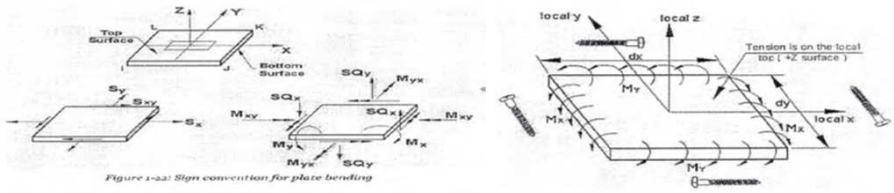


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G9 : F4_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	1695	0.168	-0.065	0	0	0	24.56	-9.217	6.352	30.92	-15.57	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Min Qx	1762	-0.146	-0.077	0	0	0	19.19	-5.308	0	19.19	-5.308	212 1.2(D+D1+D2)+1.0L1+1.6W(+X)
Max Qy	1675	-0.013	0.204	0	0	0	-0.438	-14.85	4.548	-4.986	-19.4	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Min Qy	1603	0.028	-0.211	0	0	0	-0.57	-22.67	3.281	-3.851	-25.95	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Max Sx	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Min Sx	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Max Sy	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Min Sy	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Max Sxy	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Min Sxy	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Max Mx	1569	0.034	-0.019	0	0	0	62.88	-12.82	10.83	73.71	-23.65	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Min Mx	1687	0.028	-0.022	0	0	0	-35.13	1.499	4.146	-31.27	5.645	224 0.9(D1+D2)+W(+X)+D
Max My	1562	-0.055	0.052	0	0	0	3.84	11.47	-2.306	6.146	13.78	225 0.9(D1+D2)+W(-X)+D
Min My	1617	-0.129	0.128	0	0	0	-7.62	-31.72	4.233	-11.85	-35.95	213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
Max Mxy	1568	0.011	-0.032	0	0	0	32.13	2.858	17.3	49.43	20.16	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Min Mxy	1608	-0.027	-0.027	0	0	0	30.23	1.622	-11.79	42.02	13.42	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)

PLATE CENTER STRESS ; G10 : F4_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	2776	0.112	-0.006	0	0	0	48.25	-0.761	4.112	52.37	-4.873	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Min Qx	1772	-0.109	-0.009	0	0	0	38.67	-1.023	0	38.67	-1.023	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Max Qy	2840	0	0.019	0	0	0	3.469	0.171	0.663	4.132	0.834	227 0.9(D1+D2)+W(-Z)+D
Min Qy	2840	-0.002	-0.024	0	0	0	51.36	-0.5	2.773	54.13	-3.273	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Max Sx	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Min Sx	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Max Sy	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Min Sy	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Max Sxy	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Min Sxy	2848	-0.01	-0.001	0	0	0	21.04	-0.077	0.724	21.76	-0.801	201 1.4D+1.4(D1+D2)
Max Mx	2837	0.022	0	0	0	0	60.52	-0.098	1.841	62.36	-1.939	202 1.4(D+D1+D2)+1.4T(+20)+1.7L1
Min Mx	1748	-0.019	0.005	0	0	0	-30.17	-0.772	0.229	-30.4	-1.001	225 0.9(D1+D2)+W(-X)+D
Max My	1756	0.014	0.005	0	0	0	2.659	1.197	6.606	9.265	7.803	227 0.9(D1+D2)+W(-Z)+D
Min My	1756	-0.076	-0.011	0	0	0	39.65	-5.196	-4.415	44.06	-9.611	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Max Mxy	2767	0.041	-0.013	0	0	0	44.77	-2.45	9.323	54.1	-11.77	214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Min Mxy	1748	-0.035	-0.003	0	0	0	17.12	-2.015	-6.58	23.7	-8.595	226 0.9(D1+D2)+W(+Z)+D

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

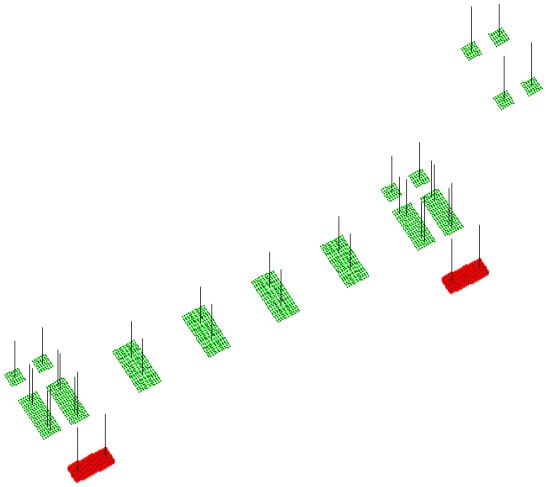
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	73.7	top face	50	444	73.71	0.014	442.3	900	12	251.43	200	565.8	7.978	9.972	0.131	tension control
Min Mx	-31.3	bottom face	75	417	31.27		199.1		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.112	1	56	52.37	0.001	0.444	424.8	429.9	761.8	424.85	1619	> 56	ok	318.6	φVc>Vu	not required
	Min Qx	0.109	1	54.5	38.67			426.3			426.3	1620	> 54.5	ok	319.7	φVc>Vu	not required

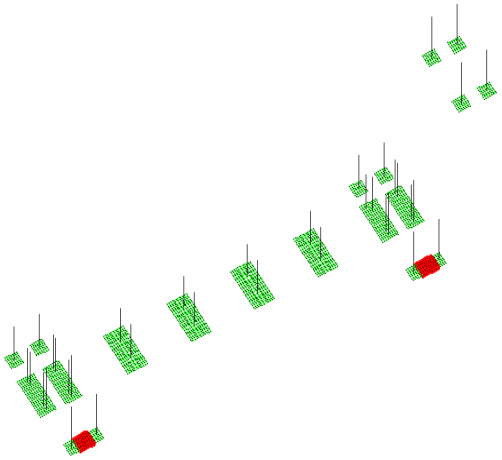
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	13.8	top face	50	444	13.78	0.014	82.2	900	12	251.43	200	565.8	7.978	9.972	0.131	tension control
Min My	-36.0	bottom face	75	417	35.95		229.0		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.019	1	9.5	0.834	0.002	0.444	468.9	429.9	761.8	429.9	452	> 9.5	ok	322.4	φVc>Vu	not required
	Min Qy	0.024	1	12	3.273			435.9			429.9	452	> 12	ok	322.4	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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7.3. Design of Pit

Foundation Base Pressure :

refer staad file : MD-216-3520-EG-CV-CAL-1285_pit in Neutralization area

	Node	L/C	Horizontal Fx KN/m2	Vertical Fy KN/m2	Horizontal Fz KN/m2
Max Px	1	101	0	39.904	0
Min Px	1	101	0	39.904	0
Max Py	1	103	0	67.905	0
Min Py	27	101	0	36.673	0
Max Pz	1	101	0	39.904	0
Min Pz	1	101	0	39.904	0

Gross Maximum Base pressure, p_{max} = 67.9 KN/m^2

Net Allowable Safe Bearing Capacity = 980 KN/m^2 (considered)

Depth of footing from FGL = 2 m

Unit weight of soil = 19 KN/m^3

Gross Safe Bearing Capacity = 1018 KN/m^2

> 67.9 KN/m^2 safe

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Design of Pit Wall :

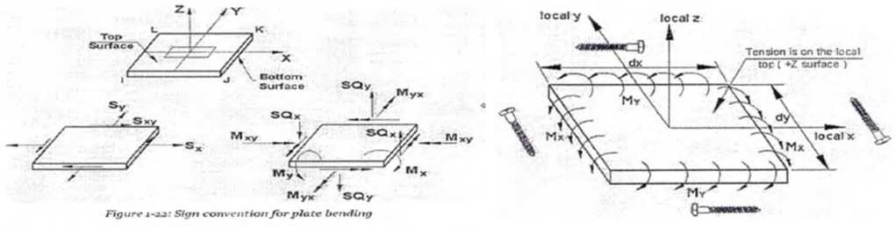


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards inside of pit

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G2 : WALL At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	158	0.048	-0.005	-0.08	-0.06	0.006	-0.407	0.32	-0.159	-0.566	0.479	202 1.4DL + 1.7(SDC+SP)
Min Qx	89	-0.048	-0.005	-0.08	-0.06	-0.006	-0.407	0.32	0	-0.407	0.32	202 1.4DL + 1.7(SDC+SP)
Max Qy	148	0.007	0.086	-0.02	-0.13	-0.016	-0.248	-2.406	0.143	-0.391	-2.549	202 1.4DL + 1.7(SDC+SP)
Min Qy	178	-0.009	-0.013	-0.04	-0.03	0.002	0.851	0.144	-0.194	1.045	0.338	202 1.4DL + 1.7(SDC+SP)
Max Sx	154	0.007	0.015	0.032	-0.11	-0.01	-0.547	-0.747	-0.046	-0.593	-0.793	203 1.4(DL+WP) + 1.7(SDC+SP)
Min Sx	158	0.048	-0.005	-0.08	-0.06	0.006	-0.407	0.32	-0.159	-0.566	0.479	202 1.4DL + 1.7(SDC+SP)
Max Sy	248	-0.002	0.002	0.01	-0	0	-0.087	-0.021	0.111	-0.198	-0.132	202 1.4DL + 1.7(SDC+SP)
Min Sy	148	0.007	0.028	0.006	-0.14	-0.007	-0.581	-2.374	-0.263	-0.844	-2.637	201 1.4DL
Max Sxy	74	-0.006	0.04	-0.02	-0.07	0.027	-0.292	-1.326	-0.211	-0.503	-1.537	202 1.4DL + 1.7(SDC+SP)
Min Sxy	149	0.006	0.04	-0.02	-0.07	-0.027	-0.292	-1.326	0.211	-0.503	-1.537	202 1.4DL + 1.7(SDC+SP)
Max Mx	130	-0.014	-0.01	-0.07	-0.04	0.001	1.434	0.843	-0.079	1.513	0.922	202 1.4DL + 1.7(SDC+SP)
Min Mx	148	0.007	0.027	0.006	-0.14	-0.007	-0.588	-2.34	-0.28	7.132	-2.62	203 1.4(DL+WP) + 1.7(SDC+SP)
Max My	160	-0.015	-0.001	-0.07	-0.06	-0.004	1.37	0.962	0.107	1.477	1.069	202 1.4DL + 1.7(SDC+SP)
Min My	148	0.007	0.086	-0.02	-0.13	-0.016	-0.248	-2.406	0.143	-0.391	-2.549	202 1.4DL + 1.7(SDC+SP)
Max Mxy	38	-0.019	0.023	0.01	-0.07	0.013	-0.173	-1.355	0.51	-0.683	-1.865	203 1.4(DL+WP) + 1.7(SDC+SP)
Min Mxy	149	0.019	0.023	0.01	-0.07	-0.013	-0.173	-1.355	-0.51	-0.683	-1.865	203 1.4(DL+WP) + 1.7(SDC+SP)

PLATE CENTER STRESS ; G2 : WALL At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	158	0.048	-0.005	-0.08	-0.06	0.006	-0.407	0.32	-0.159	-0.566	0.479	202 1.4DL + 1.7(SDC+SP)
Min Qx	89	-0.048	-0.005	-0.08	-0.06	-0.006	-0.407	0.32	0	-0.407	0.32	202 1.4DL + 1.7(SDC+SP)
Max Qy	148	0.007	0.086	-0.02	-0.13	-0.016	-0.248	-2.406	0.143	-0.391	-2.549	202 1.4DL + 1.7(SDC+SP)
Min Qy	178	-0.009	-0.013	-0.04	-0.03	0.002	0.851	0.144	-0.194	1.045	0.338	202 1.4DL + 1.7(SDC+SP)
Max Sx	154	0.007	0.015	0.032	-0.11	-0.01	-0.547	-0.747	-0.046	-0.593	-0.793	203 1.4(DL+WP) + 1.7(SDC+SP)
Min Sx	158	0.048	-0.005	-0.08	-0.06	0.006	-0.407	0.32	-0.159	-0.566	0.479	202 1.4DL + 1.7(SDC+SP)
Max Sy	248	-0.002	0.002	0.01	-0	0	-0.087	-0.021	0.111	-0.198	-0.132	202 1.4DL + 1.7(SDC+SP)
Min Sy	148	0.007	0.028	0.006	-0.14	-0.007	-0.581	-2.374	-0.263	-0.844	-2.637	201 1.4DL
Max Sxy	74	-0.006	0.04	-0.02	-0.07	0.027	-0.292	-1.326	-0.211	-0.503	-1.537	202 1.4DL + 1.7(SDC+SP)
Min Sxy	149	0.006	0.04	-0.02	-0.07	-0.027	-0.292	-1.326	0.211	-0.503	-1.537	202 1.4DL + 1.7(SDC+SP)
Max Mx	130	-0.014	-0.01	-0.07	-0.04	0.001	1.434	0.843	-0.079	1.513	0.922	202 1.4DL + 1.7(SDC+SP)
Min Mx	148	0.007	0.027	0.006	-0.14	-0.007	-0.588	-2.34	-0.28	-0.868	-2.62	203 1.4(DL+WP) + 1.7(SDC+SP)
Max My	160	-0.015	-0.001	-0.07	-0.06	-0.004	1.37	0.962	0.107	1.477	1.069	202 1.4DL + 1.7(SDC+SP)
Min My	148	0.007	0.086	-0.02	-0.13	-0.016	-0.248	-2.406	0.143	-0.391	-2.549	202 1.4DL + 1.7(SDC+SP)
Max Mxy	38	-0.019	0.023	0.01	-0.07	0.013	-0.173	-1.355	0.51	-0.683	-1.865	203 1.4(DL+WP) + 1.7(SDC+SP)
Min Mxy	149	0.019	0.023	0.01	-0.07	-0.013	-0.173	-1.355	-0.51	-0.683	-1.865	203 1.4(DL+WP) + 1.7(SDC+SP)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	inner face	50	mm
	outer face	50	mm
Thicknes of slab =		200	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

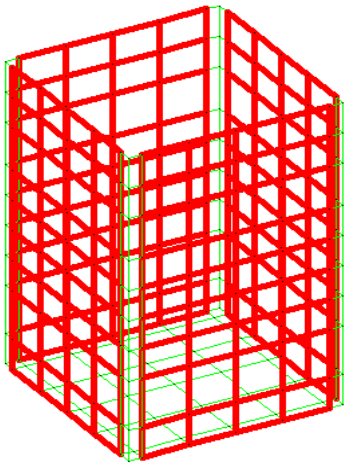
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	1.5	inner face	50	144	1.513	0.014	27.8	360	12	314.29	200	565.8	7.978	9.972	0.04	tension control
Min Mx	7.1	inner face	50	144	7.132		131.9		12	314.29	200	565.8	7.978	9.972	0.04	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.048	1	9.6	0.566	0.004	0.144	159.8	145.9	247.1	145.93	531	> 9.6	ok	109.4	φVc>Vu	not required
	Min Qx	0.048	1	9.6	0.407			169			145.93	531	> 9.6	ok	109.4	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	1.1	inner face	50	144	1.069	0.014	19.7	360	12	628.57	200	565.8	7.978	9.972	0.04	tension control
Min My	-2.5	outer face	50	144	2.549		46.9		12	314.29	200	565.8	7.978	9.972	0.04	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.086	1	17.2	2.549	0.007	0.144	145.7	145.9	247.1	145.65	151	> 17.2	ok	109.2	φVc>Vu	not required
	Min Qy	0.013	1	2.6	0.338			147			145.93	152	> 2.6	ok	109.4	φVc>Vu	not required



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Design of Pit Base :

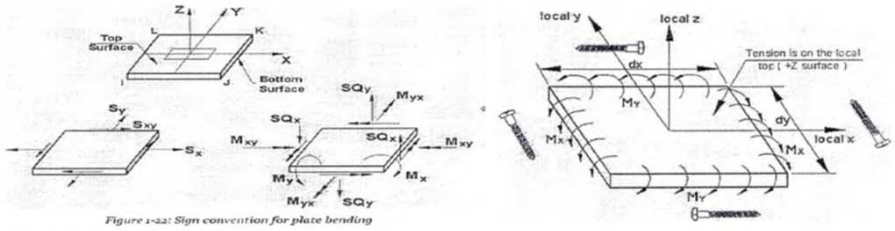


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G1 : FDN At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)Mpa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	20	0.073	-0.003	0	0	0	-0.103	0.953	0.446	-0.549	1.399	202 1.4DL + 1.7(SDC+SP)
Min Qx	23	-0.073	-0.003	0	0	0	-0.103	0.953	0	-0.103	0.953	202 1.4DL + 1.7(SDC+SP)
Max Qy	10	-0.003	0.073	0	0	0	0.953	-0.103	0.446	1.399	-0.549	202 1.4DL + 1.7(SDC+SP)
Min Qy	27	0.003	-0.073	0	0	0	0.953	-0.103	0.446	1.399	-0.549	202 1.4DL + 1.7(SDC+SP)
Max Sx	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Min Sx	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Max Sy	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Min Sy	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Max Sxy	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Min Sxy	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Max Mx	21	0.019	-0.019	0	0	0	2.774	2.774	0.269	3.043	3.043	203 1.4(DL+WP) + 1.7(SDC+SP)
Min Mx	11	-0.027	0.027	0	0	0	-0.223	-0.223	0.84	6.937	-1.063	202 1.4DL + 1.7(SDC+SP)
Max My	21	0.019	-0.019	0	0	0	2.774	2.774	0.269	3.043	3.043	203 1.4(DL+WP) + 1.7(SDC+SP)
Min My	11	-0.027	0.027	0	0	0	-0.223	-0.223	0.84	-1.063	-1.063	202 1.4DL + 1.7(SDC+SP)
Max Mxy	26	0.03	-0.03	0	0	0	0.333	0.333	1.229	1.562	1.562	203 1.4(DL+WP) + 1.7(SDC+SP)
Min Mxy	29	-0.03	-0.03	0	0	0	0.333	0.333	-1.229	1.562	1.562	203 1.4(DL+WP) + 1.7(SDC+SP)

PLATE CENTER STRESS ; G1 : FDN At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)Mpa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	20	0.073	-0.003	0	0	0	-0.103	0.953	0.446	-0.549	1.399	202 1.4DL + 1.7(SDC+SP)
Min Qx	23	-0.073	-0.003	0	0	0	-0.103	0.953	0	-0.103	0.953	202 1.4DL + 1.7(SDC+SP)
Max Qy	10	-0.003	0.073	0	0	0	0.953	-0.103	0.446	1.399	-0.549	202 1.4DL + 1.7(SDC+SP)
Min Qy	27	0.003	-0.073	0	0	0	0.953	-0.103	0.446	1.399	-0.549	202 1.4DL + 1.7(SDC+SP)
Max Sx	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Min Sx	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Max Sy	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Min Sy	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Max Sxy	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Min Sxy	29	-0.03	-0.03	0	0	0	0.321	0.321	-1.219	1.54	1.54	201 1.4DL
Max Mx	21	0.019	-0.019	0	0	0	2.774	2.774	0.269	3.043	3.043	203 1.4(DL+WP) + 1.7(SDC+SP)
Min Mx	11	-0.027	0.027	0	0	0	-0.223	-0.223	0.84	-1.063	-1.063	202 1.4DL + 1.7(SDC+SP)
Max My	21	0.019	-0.019	0	0	0	2.774	2.774	0.269	3.043	3.043	203 1.4(DL+WP) + 1.7(SDC+SP)
Min My	11	-0.027	0.027	0	0	0	-0.223	-0.223	0.84	-1.063	-1.063	202 1.4DL + 1.7(SDC+SP)
Max Mxy	26	0.03	-0.03	0	0	0	0.333	0.333	1.229	1.562	1.562	203 1.4(DL+WP) + 1.7(SDC+SP)
Min Mxy	29	-0.03	-0.03	0	0	0	0.333	0.333	-1.229	1.562	1.562	203 1.4(DL+WP) + 1.7(SDC+SP)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		200	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

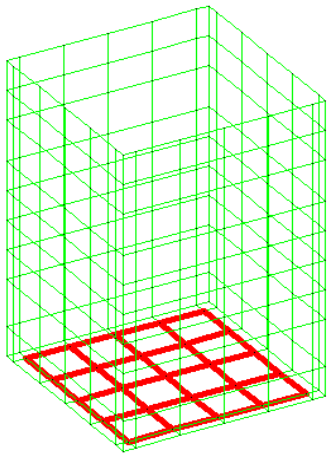
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	3.0	top face	50	144	3.043	0.014	56.1	360	12	314.29	200	565.8	7.978	9.972	0.04	tension control
Min Mx	6.9	top face	50	144	6.937		128.2		12	314.29	200	565.8	7.978	9.972	0.04	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.073	1	14.6	0.549	0.004	0.144	173.1	145.9	247.1	145.93	531	> 14.6	ok	109.4	φVc>Vu	not required
	Min Qx	0.073	1	14.6	0.103			332.6			145.93	531	> 14.6	ok	109.4	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	3.0	top face	50	144	3.043	0.014	56.1	360	12	628.57	200	565.8	7.978	9.972	0.04	tension control
Min My	-1.1	bottom face	50	144	1.063		19.5		12	314.29	200	565.8	7.978	9.972	0.04	tension control

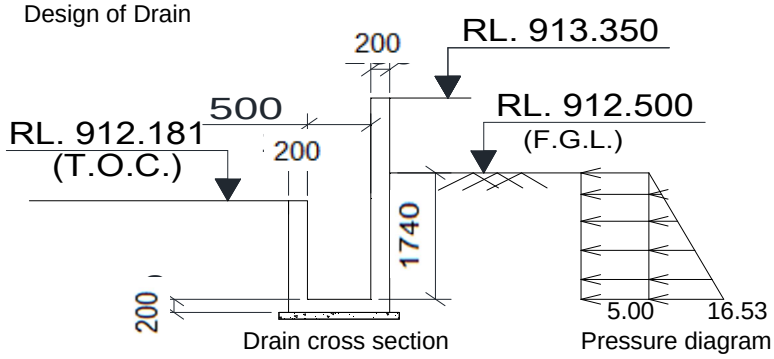
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.073	1	14.6	0.549	0.007	0.144	173.1	145.9	247.1	145.93	152	> 14.6	ok	109.4	φVc>Vu	not required
	Min Qy	0.073	1	14.6	0.549			173.1			145.93	152	> 14.6	ok	109.4	φVc>Vu	not required



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7.4. Design of Drain



Drain cross section Pressure diagram

Design Input :

Clear height ofdrain pit from FGL =	1740	mm
Clear width of Pit =	500	mm
Vertical wall thickness =	200	mm
Bottom slab thickness =	200	mm
Pressure coefficient, C_p =	0.5	
Surcharge load at side of pit, P =	10	kN/m^2
Dry unit weight of soil, $\gamma_{dry\ soil}$ =	19	kN/m^3

Load Calculation :

1 SOIL PRESSURE(DRY CONDITION) (H)

Lateral earth pressure at side wall due to dry soil = $C_p \times \gamma_{dry\ soil} \times h$ 16.53 kN/m^2

2 SURCHARGE PRESSURE (L)

Lateral earth pressure at side wall due to surcharged load = $C_p \times P$ 5.00 kN/m^2

Moment due lateral pressure -

5*1.74*1.74/2+0.5*16.53*1.74*1.74/3 = 15.91 kN-m

Depth of section=	200	mm	Factored moment =	27.05	kN-m
Effective depth =	144	mm	As =	509.63	mm^2
Dia of bar =	12	mm	As min = 0.18/100*200*1000	360	mm^2
Grade of concrete =	fc'35		Spacing required =	221.92	mm c/c
Grade of reinf. =	fy420				
Strength Reduction Factor,φ :	0.9		Spacing provided	200	mm c/c

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Check for shear :

Shear force at a distance d = 5*1.596 + 0.5*19*1.596*0.5*1.596 = 20.08 kN

Factored shear force = 34.13 kN

$V_{c1} = (0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d_s / M_{us}) b_w d_s$	138.1	kN					
$V_{c2} = (0.16\lambda\sqrt{f_c'} + 17\rho_w) b_w d_s$	145.9	kN	A_s=	565.49	mm^2	M_{us}=	27.05 kN-m
$V_{c3} = 0.29\lambda\sqrt{f_c'} b_w d_s$	247.1	kN	ρ_w=	0.00393		b_w =	1000 mm
V_c = Minimum of V_{c1}, V_{c2} and V_{c3}	138.1	kN	λ=	1		d_s =	144 mm

Strength Reduction Factor,φ : 0.75

$V_u \leq \varphi (V_c + 0.66\sqrt{f_c'} b_w d_s)$ 525.2 kN ok

φV_c = 103.5 kN

Remarks : shear reinforcement not reqd

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ATTACHMENT 1 OTE INPUT (CIVIL GUIDE DWG)

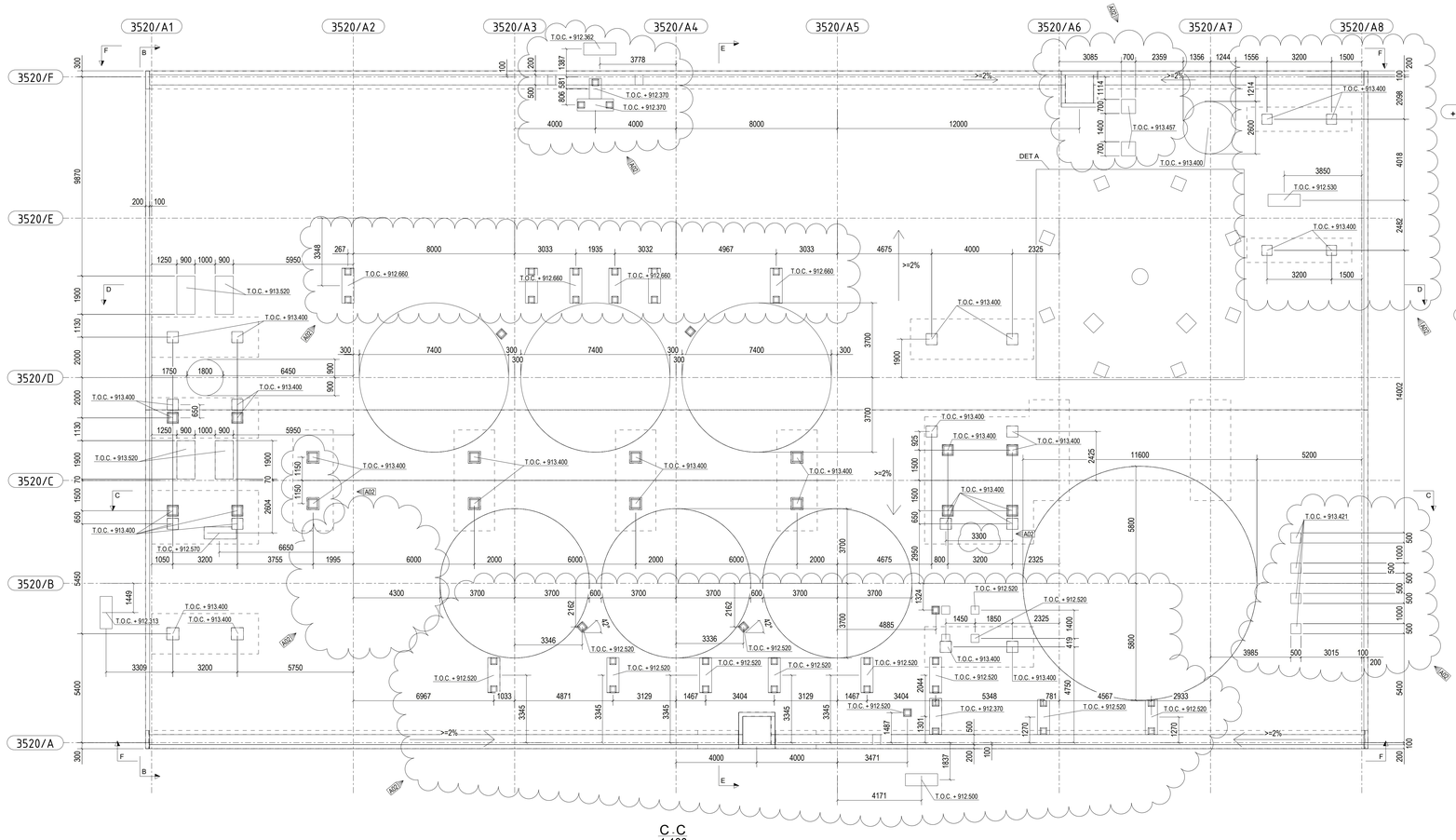
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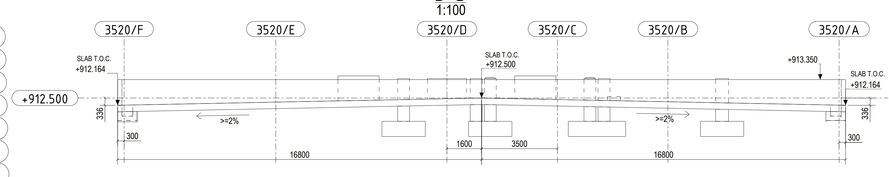
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DRAWING NUMBER: MD-216-3520-EG-CV-GAD-1001 | REV: A02

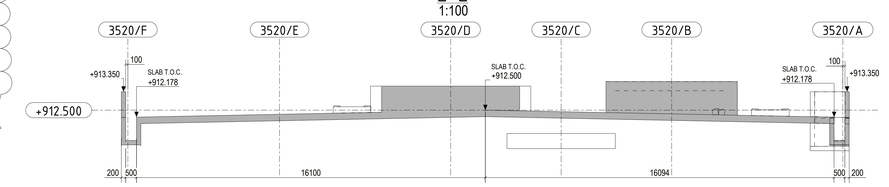
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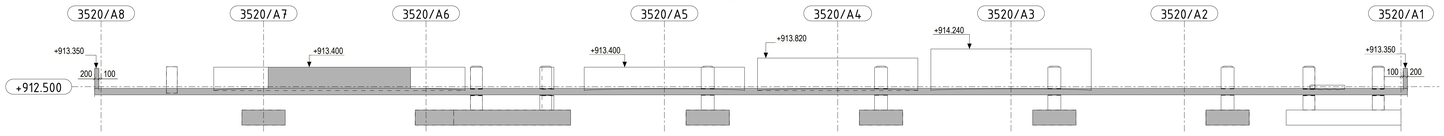
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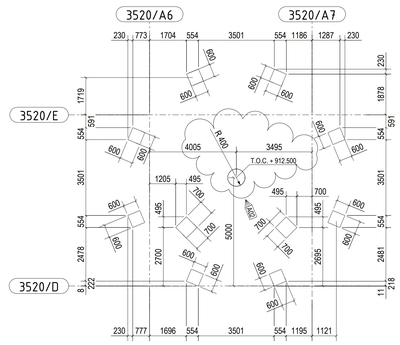
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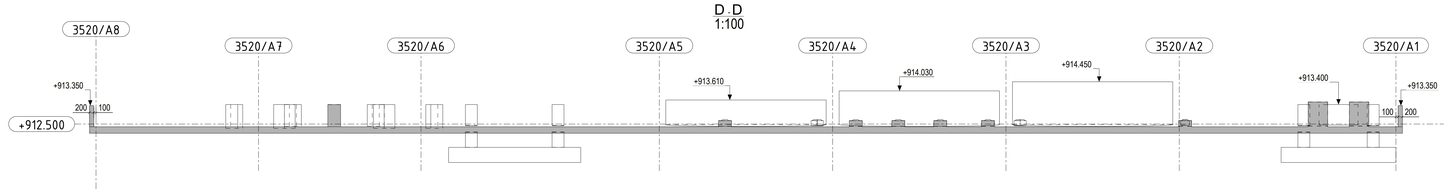
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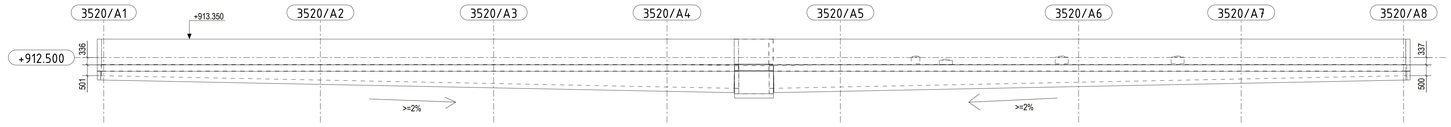
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F - F
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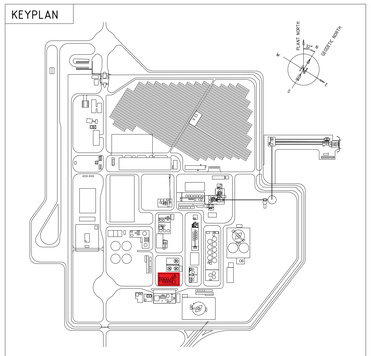
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3520 = RL(+)912.500 m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3520-EG-ST-GAD-1001	3D VIEW	MD-216-3520-EG-ST-GAD-1002	PLAN AND SECTIONS
MD-216-3520-EG-ME-GAD-1043	SECTION D-D	MD-216-1000-EG-ST-CRT-0001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN BASIS
MD-216-3520-EG-ME-GAD-1041	SECTION A-A AND B-B	MD-216-3520-EG-ME-GAD-1042	SECTION C-C
MD-216-3520-EG-ME-GAD-1039	PLAN AT EL. +912.500	MD-216-3520-EG-ME-GAD-1040	PLAN VIEW

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	30.03.2020	ISSUED FOR DETAIL DESIGN, FOUNDATIONS UPDATED	KirJak	MAkLai	LuiPiz	SouCha	
A01	11.12.2019	ISSUED FOR APPROVAL	AnnFor	MAkLai	LuiPiz	SouCha	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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OUTOTEC DRAWING NUMBER	900707_GAU01_RN13_CTA04_35201	DOCUMENT ID	OU100667800
		REVISION	1

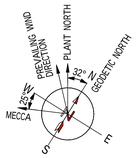
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CHECKED	MAkLai	11.12.2019
ENGINEER	LuiPiz	11.12.2019
APPROVED	SouCha	11.12.2019
EMPLOYER		
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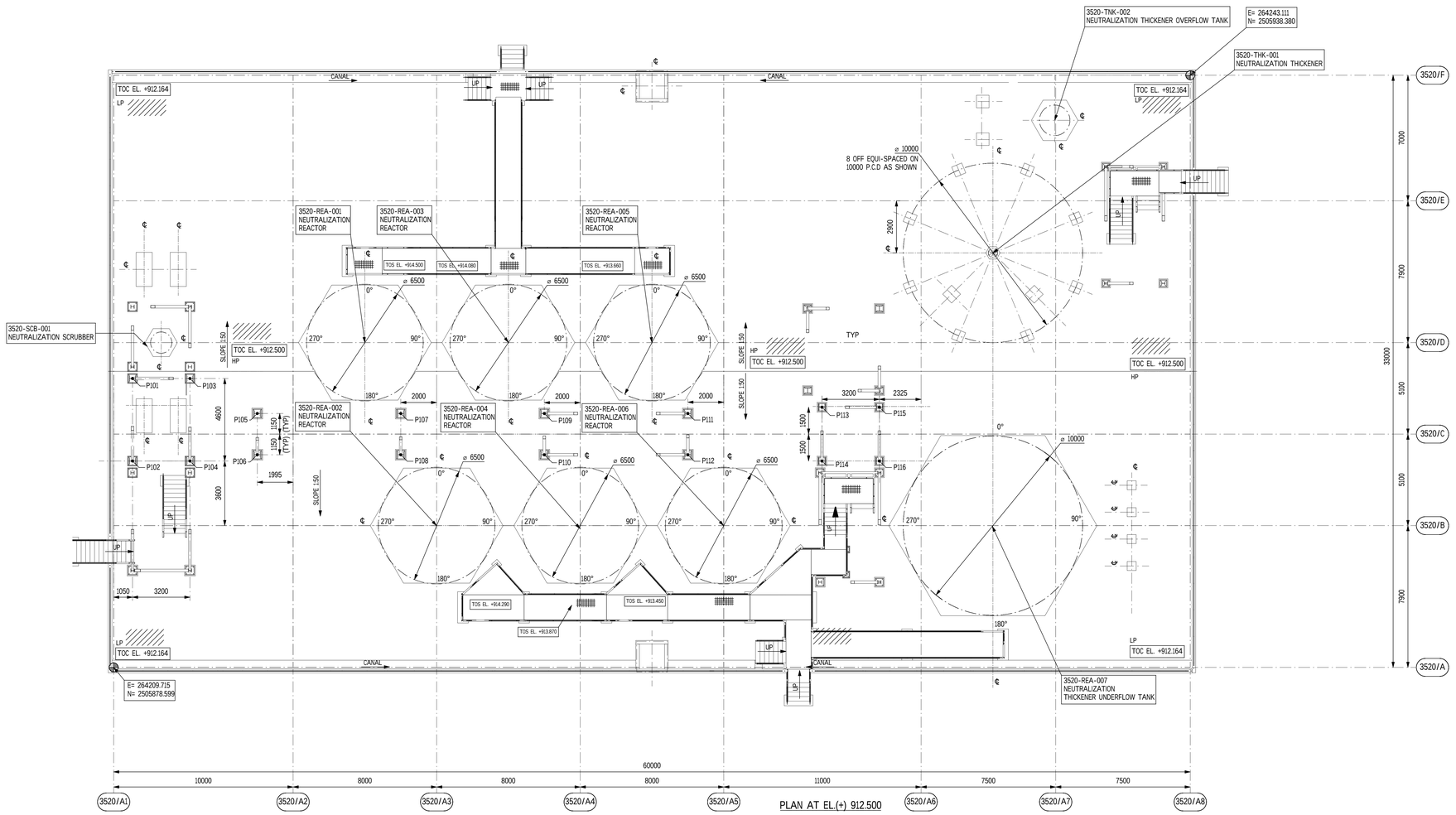
MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
3520 NEUTRALIZATION AND THICKENING
PLAN AT EL. +912.500

DRAWING NUMBER: MD-216-3520-EG-CV-GAD-1001 | REV: A02

Metric Structures

DRAWING NUMBER: MD-216-3520-EG-ME-GAD-1047 REV: A01





PLAN AT EL.(+) 912.500

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ALKALI/ ACID RESISTANT COATING
CHECKERED PLATE
GRATING
GRATING - FRP
ACID RESISTANT BRICKLINING
BRICK WALL
CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS: +Z, +Mz, +Y, +X, +My, +Mx

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM- GRATING 5 kN/m²
GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN.

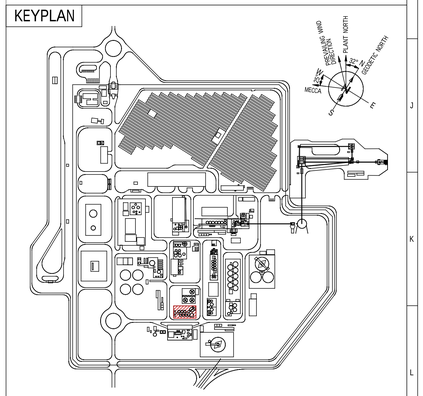
13 - REPORTED SEISMIC LOAD ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301-2007. HOWEVER, IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNER.

14 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)912.500 m.

15 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

16 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



0 1 10m

NODES - SUPPORT FORCES

Node	Support Forces [kN]				Node	Support Forces [kN]				Node	Support Forces [kN]				Node	Support Forces [kN]				Node	Support Forces [kN]			
No.	LC/CO	F_x	F_y	F_z	No.	LC/CO	F_x	F_y	F_z	No.	LC/CO	F_x	F_y	F_z	No.	LC/CO	F_x	F_y	F_z	No.	LC/CO	F_x	F_y	F_z
P101	LC2 - DEAD LOAD	-2,19	0,46	-28,16	P104	LC2 - DEAD LOAD	0,16	0,21	-30,33	P107	LC2 - DEAD LOAD	0,10	0,28	-43,44	P110	LC2 - DEAD LOAD	-7,55	-0,27	-51,17	P113	LC2 - DEAD LOAD	-0,28	0,07	-41,25	P116	LC2 - DEAD LOAD	0,31	-0,26	-12,40
	LC3 - SELF WEIGHT	-0,76	0,08	-29,61		LC3 - SELF WEIGHT	0,05	0,01	-25,95		LC3 - SELF WEIGHT	0,04	0,23	-50,94		LC3 - SELF WEIGHT	-4,27	-0,27	-53,49		LC3 - SELF WEIGHT	-0,25	0,06	-37,15		LC3 - SELF WEIGHT	0,12	-0,13	-24,86
	LC4 - LIVE LOAD 5 kN/m²	-5,56	1,13	-70,60		LC4 - LIVE LOAD 5 kN/m²	0,39	0,54	-76,11		LC4 - LIVE LOAD 5 kN/m²	0,24	0,71	-112,05		LC4 - LIVE LOAD 5 kN/m²	-18,97	-0,80	-131,11		LC4 - LIVE LOAD 5 kN/m²	-0,66	0,15	-96,08		LC4 - LIVE LOAD 5 kN/m²	0,80	-0,63	-26,83
	LC5 - LIVE LOAD 10 kN/m²	-11,12	2,26	-141,20		LC5 - LIVE LOAD 10 kN/m²	0,77	1,07	-152,22		LC5 - LIVE LOAD 10 kN/m²	0,48	1,42	-224,09		LC5 - LIVE LOAD 10 kN/m²	-37,95	-1,60	-262,22		LC5 - LIVE LOAD 10 kN/m²	-1,32	0,30	-192,17		LC5 - LIVE LOAD 10 kN/m²	1,59	-1,26	-53,67
	LC8 - LIVE LOAD 15-20 kN/m²	-17,06	3,17	-230,74		LC8 - LIVE LOAD 15-20 kN/m²	1,13	1,67	-247,24		LC8 - LIVE LOAD 15-20 kN/m²	0,72	1,87	-305,77		LC8 - LIVE LOAD 15-20 kN/m²	-56,90	-2,09	-361,05		LC8 - LIVE LOAD 15-20 kN/m²	-2,05	0,49	-299,10		LC8 - LIVE LOAD 15-20 kN/m²	2,32	-1,94	-92,51
	LC11 - WIND DIRECTION -Y	1,73	0,09	63,19		LC11 - WIND DIRECTION -Y	0,00	-33,45	-63,30		LC11 - WIND DIRECTION -Y	0,18	0,11	188,43		LC11 - WIND DIRECTION -Y	-11,26	-50,67	-241,84		LC11 - WIND DIRECTION -Y	0,59	-0,34	173,28		LC11 - WIND DIRECTION -Y	0,07	-33,64	-113,97
	LC12 - WIND DIRECTION +Y	-7,63	0,05	-79,89		LC12 - WIND DIRECTION +Y	0,09	85,24	171,56		LC12 - WIND DIRECTION +Y	-0,03	-0,34	-386,04		LC12 - WIND DIRECTION +Y	17,25	97,60	398,72		LC12 - WIND DIRECTION +Y	-0,76	0,14	-183,80		LC12 - WIND DIRECTION +Y	0,07	34,57	117,67
	LC13 - WIND DIRECTION +X	25,57	0,01	60,40		LC13 - WIND DIRECTION +X	2,10	-1,00	2,27		LC13 - WIND DIRECTION +X	0,54	1,51	-1,50		LC13 - WIND DIRECTION +X	17,23	0,04	27,02		LC13 - WIND DIRECTION +X	-0,18	-0,02	57,10		LC13 - WIND DIRECTION +X	-0,71	0,17	-1,57
	LC14 - WIND DIRECTION -X	-6,05	0,00	-22,26		LC14 - WIND DIRECTION -X	0,54	1,51	-1,50		LC14 - WIND DIRECTION -X	0,42	0,00	4,05		LC14 - WIND DIRECTION -X	-12,98	-0,03	-19,77		LC14 - WIND DIRECTION -X	0,11	0,01	-51,81		LC14 - WIND DIRECTION -X	1,20	-0,48	0,44
	LC15 - TEMPERATURE +20	-65,30	0,56	-230,33		LC15 - TEMPERATURE +20	-0,98	7,13	8,09		LC15 - TEMPERATURE +20	-1,63	0,38	17,26		LC15 - TEMPERATURE +20	-43,34	-0,68	-14,54		LC15 - TEMPERATURE +20	-4,22	-2,67	221,08		LC15 - TEMPERATURE +20	2,71	-3,18	-10,87
	LC16 - TEMPERATURE -20	63,78	-0,57	162,50		LC16 - TEMPERATURE -20	1,06	-9,09	-11,46		LC16 - TEMPERATURE -20	1,39	-0,38	-10,07		LC16 - TEMPERATURE -20	41,09	0,71	10,83		LC16 - TEMPERATURE -20	4,15	2,66	-189,26		LC16 - TEMPERATURE -20	-2,59	2,18	6,42
P102	LC2 - DEAD LOAD	0,14	-0,46	-12,08	P105	LC2 - DEAD LOAD	0,10	0,19	-38,12	P108	LC2 - DEAD LOAD	0,21	-0,61	-47,44	P111	LC2 - DEAD LOAD	-1,47	0,29	-39,32	P114	LC2 - DEAD LOAD	1,17	-0,13	-30,11					
	LC3 - SELF WEIGHT	0,05	-0,09	-25,06		LC3 - SELF WEIGHT	0,04	0,11	-37,08		LC3 - SELF WEIGHT	0,07	-0,30	-50,97		LC3 - SELF WEIGHT	3,43	0,24	-48,25		LC3 - SELF WEIGHT	0,51	-0,04	-33,49					
	LC4 - LIVE LOAD 5 kN/m²	0,35	-1,14	-85,21		LC4 - LIVE LOAD 5 kN/m²	0,25	0,47	-94,41		LC4 - LIVE LOAD 5 kN/m²	0,53	-1,54	-119,51		LC4 - LIVE LOAD 5 kN/m²	-3,37	0,71	-97,30		LC4 - LIVE LOAD 5 kN/m²	2,94	-0,31	-72,21					
	LC5 - LIVE LOAD 10 kN/m²	0,70	-2,28	-110,42		LC5 - LIVE LOAD 10 kN/m²	0,50	0,94	-188,82		LC5 - LIVE LOAD 10 kN/m²	1,06	-3,08	-239,01		LC5 - LIVE LOAD 10 kN/m²	-6,74	1,41	-194,60		LC5 - LIVE LOAD 10 kN/m²	5,89	-0,61	-144,42					
	LC8 - LIVE LOAD 15-20 kN/m²	1,04	-3,21	-184,08		LC8 - LIVE LOAD 15-20 kN/m²	0,75	1,39	-278,35		LC8 - LIVE LOAD 15-20 kN/m²	1,57	-4,26	-332,43		LC8 - LIVE LOAD 15-20 kN/m²	-11,55	1,85	-266,42		LC8 - LIVE LOAD 15-20 kN/m²	8,12	-0,92	-223,83					
	LC11 - WIND DIRECTION -Y	-0,02	-31,65	-57,96		LC11 - WIND DIRECTION -Y	-0,05	0,03	83,20		LC11 - WIND DIRECTION -Y	-0,16	-37,17	-188,37		LC11 - WIND DIRECTION -Y	-9,91	0,27	248,75		LC11 - WIND DIRECTION -Y	-0,06	-75,44	-189,90					
	LC12 - WIND DIRECTION +Y	0,14	31,50	57,85		LC12 - WIND DIRECTION +Y	-0,03	-0,16	-252,29		LC12 - WIND DIRECTION +Y	0,16	92,03	385,52		LC12 - WIND DIRECTION +Y	12,49	-0,31	-327,93		LC12 - WIND DIRECTION +Y	-0,25	79,31	202,00					
	LC13 - WIND DIRECTION +X	4,62	-0,06	-0,04		LC13 - WIND DIRECTION +X	-0,25	0,00	-1,61		LC13 - WIND DIRECTION +X	-0,51	0,57	3,41		LC13 - WIND DIRECTION +X	16,49	0,04	-23,34		LC13 - WIND DIRECTION +X	-0,61	0,44	2,32					
	LC14 - WIND DIRECTION -X	0,47	0,00	0,01		LC14 - WIND DIRECTION -X	0,39	-0,01	3,79		LC14 - WIND DIRECTION -X	0,75	-0,90	-5,26		LC14 - WIND DIRECTION -X	-11,05	-0,03	15,07		LC14 - WIND DIRECTION -X	0,77	-0,31	-2,60					
	LC15 - TEMPERATURE +20	-2,55	-0,45	0,25		LC15 - TEMPERATURE +20	-1,79	0,34	1,68		LC15 - TEMPERATURE +20	-0,66	-3,57	-15,88		LC15 - TEMPERATURE +20	31,89	0,35	6,95		LC15 - TEMPERATURE +20	2,43	3,96	7,08					
	LC16 - TEMPERATURE -20	2,62	0,50	-0,11		LC16 - TEMPERATURE -20	1,62	-0,35	-19,65		LC16 - TEMPERATURE -20	0,79	2,46	10,31		LC16 - TEMPERATURE -20	-23,33	-0,35	-23,37		LC16 - TEMPERATURE -20	-2,33	-4,24	-6,03					
P103	LC2 - DEAD LOAD	0,39	0,45	-20,92	P106	LC2 - DEAD LOAD	0,17	-0,50	-35,74	P109	LC2 - DEAD LOAD	-7,19	0,28	-47,65	P112	LC2 - DEAD LOAD	-2,07	-0,01	-36,68	P115	LC2 - DEAD LOAD	-0,40	0,03	-2,66					
	LC3 - SELF WEIGHT	0,34	0,08	-26,95		LC3 - SELF WEIGHT	0,06	-0,15	-35,51		LC3 - SELF WEIGHT	-4,61	0,24	-53,77		LC3 - SELF WEIGHT	3,65	-0,15	-48,59		LC3 - SELF WEIGHT	1,55	0,04	-27,36					
	LC4 - LIVE LOAD 5 kN/m²	0,97	1,11	-52,29		LC4 - LIVE LOAD 5 kN/m²	0,42	-1,23	-68,93		LC4 - LIVE LOAD 5 kN/m²	-18,10	0,77	-128,92		LC4 - LIVE LOAD 5 kN/m²	-6,15	0,02	-91,11		LC4 - LIVE LOAD 5 kN/m²	-1,10	0,05	3,50					
	LC5 - LIVE LOAD 10 kN/m²	1,93	2,23	-104,57		LC5 - LIVE LOAD 10 kN/m²	0,84	-2,46	-177,86		LC5 - LIVE LOAD 10 kN/m²	-36,19	1,54	-257,84		LC5 - LIVE LOAD 10 kN/m²	-10,30	0,04	-182,22		LC5 - LIVE LOAD 10 kN/m²	-2,20	0,09	7,00					
	LC8 - LIVE LOAD 15-20 kN/m²	3,05	3,09	-175,05		LC8 - LIVE LOAD 15-20 kN/m²	1,24	-3,54	-267,11		LC8 - LIVE LOAD 15-20 kN/m²	-52,14	2,00	-354,53		LC8 - LIVE LOAD 15-20 kN/m²	-17,19	0,24	-244,72		LC8 - LIVE LOAD 15-20 kN/m²	-3,11	0,19	-2,56					
	LC11 - WIND DIRECTION -Y	-0,43	-0,21	57,38		LC11 - WIND DIRECTION -Y	-0,01	-15,39	-82,37		LC11 - WIND DIRECTION -Y	14,42	0,12	247,16		LC11 - WIND DIRECTION -Y	13,41	-60,30	-254,98		LC11 - WIND DIRECTION -Y	-8,49	0,19	131,38					
	LC12 - WIND DIRECTION +Y	0,57	0,48	-147,66		LC12 - WIND DIRECTION +Y	0,04	56,46	251,16		LC12 - WIND DIRECTION +Y	-17,53	-0,31	-399,13		LC12 - WIND DIRECTION +Y	-13,08	86,93	327,95		LC12 - WIND DIRECTION +Y	8,48	-0,30	-136,11					
	LC13 - WIND DIRECTION +X	-0,77	-0,01	-56,79		LC13 - WIND DIRECTION +X	-0,34	0,44	-1,02		LC13 - WIND DIRECTION +X	16,42	-0,03	24,98		LC13 - WIND DIRECTION +X	18,10	-0,62	-30,28		LC13 - WIND DIRECTION +X	13,83	0,03	-60,35					
	LC14 - WIND DIRECTION -X	0,15	0,02	17,48		LC14 - WIND DIRECTION -X	0,58	-0,59	2,25		LC14 - WIND DIRECTION -X	-12,38	0,02	-17,90		LC14 - WIND DIRECTION -X	-12,53	0,80	23,35		LC14 - WIND DIRECTION -X	-11,01	-0,02	54,64					
	LC15 - TEMPERATURE +20	6,44	-1,19	223,87		LC15 - TEMPERATURE +20	-1,30	-2,80	0,05		LC15 - TEMPERATURE +20	-45,01	0,42	-9,07		LC15 - TEMPERATURE +20	25,25	0,58	19,10		LC15 - TEMPERATURE +20	118,06	0,82	-224,72					
	LC16 - TEMPERATURE -20	-5,72	1,30	-150,19		LC16 - TEMPERATURE -20	1,36	6,26	15,16		LC16 - TEMPERATURE -20	53,32	-0,42	21,52		LC16 - TEMPERATURE -20	-27,85	0,15	-12,06		LC16 - TEMPERATURE -20	-109,37	-0,79	195,44					

NODES - SEISMIC FORCES

Node	Support Forces [kN]			
No.		F_x	F_y	F_z
P101	Max	1,51	2,24	-60,04
	Min	-17,13	0,63	-193,98
P102	Max	1,51	2,22	-49,93
	Min	-0,53	-5,13	-155,25
P103	Max	2,62	2,29	-25,36
	Min	0,64	0,52	-177,06
P104	Max	1,63	13,24	-38,55
	Min	-0,56	-11,89	-220,15
P105	Max	0,72	1,08	-35,55
	Min	-0,01	0,35	-284,09
P106	Max	1,36	10,75	-31,31
	Min	-0,19	-14,08	-273,24
P107	Max	0,89	1,60	-42,52
	Min	0,00	0,56	-336,89
P108	Max	1,79	15,07	-46,89
	Min	-0,34	-19,38	-355,51
P109	Max	-5,21	1,70	-44,33
	Min	-49,22	0,59	-376,33
P110	Max	-3,76	17,53	-53,27
	Min	-52,06	-19,88	-378,42
P111	Max	7,47	1,69	-38,92
	Min	-8,54	0,51	-304,47
P112	Max	16,42	17,32	-17,72
	Min	-21,23	-17,59	-309,30
P113	Max	-0,56	0,49	-47,30
	Min	-1,74	0,07	-290,42
P114	Max	6,75	11,15	-41,68
	Min	1,33	-12,02	-222,87
P115	Max	5,29	0,23	-23,50
	Min	-4,08	0,03	-49,82
P116	Max	2,47	4,72	-25,50
	Min	-0,24	-6,62	-110,18

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	02-03-2020	ISSUED FOR APPROVAL	AS	TV	SC	RL	

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3520-EG-ME-GAD-1045	LOAD PLAN AT EL.+912.500
MD-216-3520-EG-ME-GAD-1008	GENERAL ARRANGEMENT DRAWING, PLAN AT EL. +912.500

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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MA'ADEN معادن Saudi Arabian Mining Company

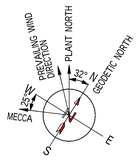
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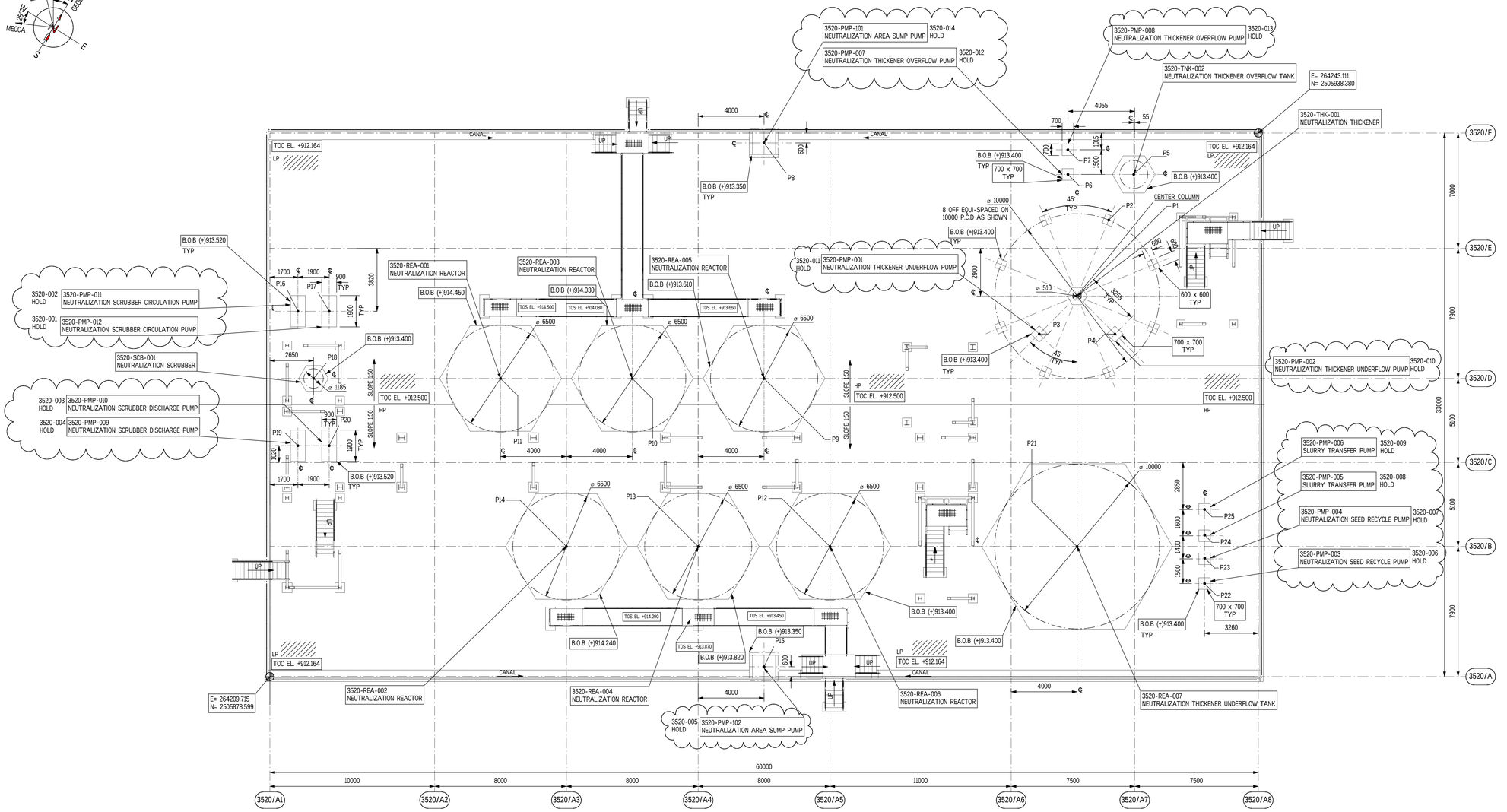
LARSEN & TOUBRO DRAWING NUMBER | REVISION: 0

OUTOTEC DRAWING NUMBER: 900707 _ GAU01 _ RN13 _ MLD14 _ 35203 | DOCUMENT ID: OU100674417 | REVISION: 0

DRAWN	AS	02-03-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	02-03-2020	LOAD PLAN
ENGINEER	SC	02-03-2020	NEUTRALIZATION AND THICKENING
APPROVED	RL	02-03-2020	LAUNDER STRUCTURE LOADING DIAGRAM
EMPLOYER			DRAWING NUMBER: MD-216-3520-EG-ME-GAD-1047
SCALE 1:100	A0		REV: A01

DRAWING NUMBER: MD-216-3520-EG-ME-GAD-1008 | REV: A01





GENERAL NOTES, LEGEND

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- A — CONCRETE FLOOR WITH ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- F — GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
LP = LOW POINT.
HP = HIGHT POINT.

DIRECTION OF FORCES AND MOMENTS: +Z, +Mz, +X, +Mx, +Y, +My

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT -WIND LOADS
- ADDITIONAL LOADS FOR ERECTION PURPOSES -EARTHQUAKE LOADS

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

GRATING 5 kN/m² | CONCRETE PLATFORMS 10 kN/m² | GROUNDFLOOR 25 kN/m² | CABLE FLOOR 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

DUST 0.5 kN/m² LIVE LOAD 0.75 kN/m²

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

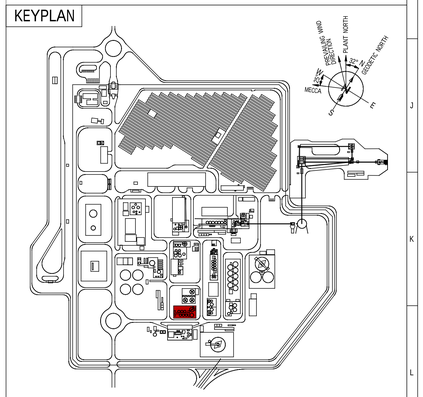
11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)912.500 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



PLAN AT EL.(+) 912.500

* :-
FOR,
P1 & P2 : REFER DWG. MD-216-3520-EG-ME-GAD-1002 FOR LOAD DETAILS
P9, P10 & P11 : REFER DWG. MD-216-3520-EG-ME-PLN-1001 FOR LOAD DETAILS
P12, P13 & P14 : REFER DWG. MD-216-3520-EG-ME-PLN-1002 FOR LOAD DETAILS
P21 : REFER DWG. MD-216-3520-EG-ME-PLN-1007 FOR LOAD DETAILS

LOADING POINT	P1*	P2*	P3	P4	P5	P6	P7	P8	P9*	P10*	P11*	P12*	P13*	P14*	P15	P16	P17	P18	P19	P20	P21*	P22	P23	P24	P25
LOAD CASE	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]
1 DEAD LOAD EQUIPMENT			-6	-6	-30	-6	-6	-6							-6	-3	-3	-15	-2	-2		-4	-4	-5	-5
2 DEAD LOAD (SECONDARY EQUIPMENT)																									
3 OPERATION LOAD 1. FILLING																									
4 OPERATION LOAD 2. PRESSURE																									
5 OPERATION LOAD 3. TEMPERATURE																									
6 OPERATION LOAD 4. DYNAMIC			-9	-9		-12	-12	-12							-12	-6	-6		-5	-5		-10	-10	-10	-10
7 OPERATION LOAD 5. OTHERS																									
8 TEST LOAD FILLING / EMERGENCY																									
9 LIVE LOAD					-55													-15							
10 WIND ±X - DIRECTION																									
11 WIND ±Y - DIRECTION																									
12 EARTHQUAKE ±X - DIRECTION																									
13 EARTHQUAKE ±Y - DIRECTION																									
EQUIPMENT - NO.	3520-THK-001	3520-THK-001	3520-PMP-001	3520-PMP-002	3520-TNK-002	3520-PMP-007	3520-PMP-008	3520-PMP-101	3520-REA-005	3520-REA-003	3520-REA-001	3520-REA-006	3520-REA-004	3520-REA-002	3520-PMP-102	3520-PMP-011	3520-PMP-012	3520-SCB-001	3520-PMP-009	3520-PMP-010	3520-REA-007	3520-PMP-003	3520-PMP-004	3520-PMP-005	3520-PMP-006
DESCRIPTION	NEUTRALIZATION THICKNER	NEUTRALIZATION THICKNER	NEUTRALIZATION THICKNER UNDERFLOW PUMP	NEUTRALIZATION THICKNER UNDERFLOW PUMP	NEUTRALIZATION THICKNER OVERFLOW TANK	NEUTRALIZATION THICKNER OVERFLOW PUMP	NEUTRALIZATION THICKNER OVERFLOW PUMP	NEUTRALIZATION REACTOR	NEUTRALIZATION REACTOR	NEUTRALIZATION REACTOR	NEUTRALIZATION REACTOR	NEUTRALIZATION REACTOR	NEUTRALIZATION REACTOR	NEUTRALIZATION REACTOR	NEUTRALIZATION AREA SUMP PUMP	NEUTRALIZATION SCRUBBER CIRCULATION PUMP	NEUTRALIZATION SCRUBBER CIRCULATION PUMP	NEUTRALIZATION SCRUBBER	NEUTRALIZATION SCRUBBER DISCHARGE PUMP	NEUTRALIZATION SCRUBBER DISCHARGE PUMP	NEUTRALIZATION THICKNER UNDERFLOW TANK	NEUTRALIZATION SEED RECYCLE PUMP	NEUTRALIZATION SEED RECYCLE PUMP	SLURRY TRANSFER PUMP	SLURRY TRANSFER PUMP

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3520-EG-ME-GAD-1039	GENERAL ARRANGEMENT DRAWING, PLAN AT EL +912.500
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3520-EG-ME-GAD-1009	PLAN AT EL +928.550

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	03-01-2020	ISSUED FOR APPROVAL	BS	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

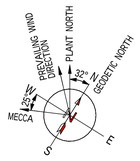
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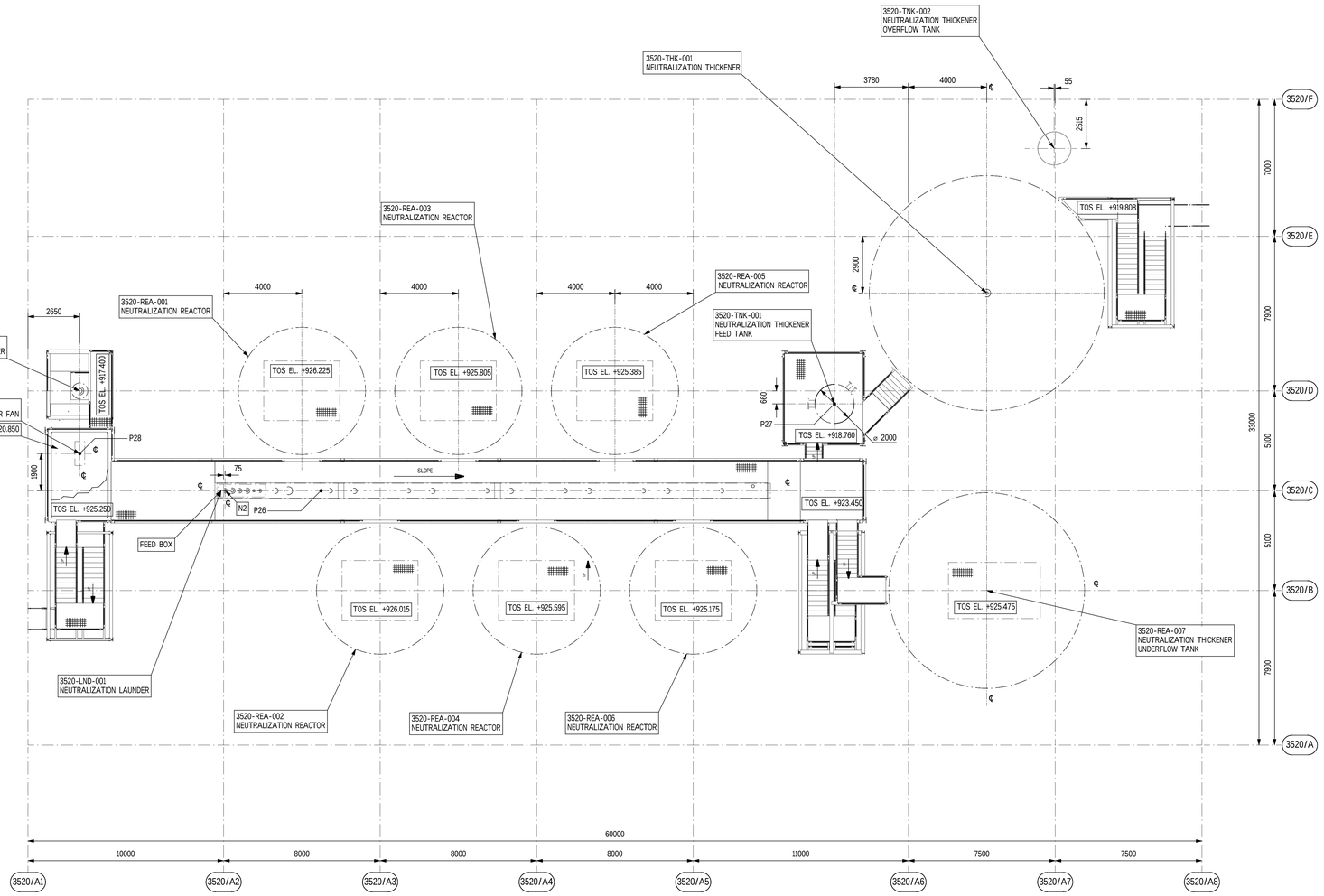
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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_450_M_D_02813_001			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707 _GAU01 _RN13 _MLD14 _35201		OU100667829	0
DRAWN	BS	03-01-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	03-01-2020	LOAD PLAN
ENGINEER	SC	03-01-2020	NEUTRALIZATION AND THICKENING
APPROVED	RL	03-01-2020	PLAN AT EL. +912.500
EMPLOYER			
SCALE 1:100	A0	DRAWING NUMBER: MD-216-3520-EG-ME-GAD-1008	REV: A01

DRAWING NUMBER: MD-216-3520-EG-ME-GAD-1009
REV: A01





PLAN AT EL.(+)925.280

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

GENERAL NOTES, LEGEND

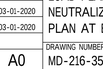
01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
LP = LOW POINT.
HP = HIGHT POINT.

DIRECTION OF FORCES AND MOMENTS



06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT -WIND LOADS
- ADDITIONAL LOADS FOR ERECTION PURPOSES -EARTHQUAKE LOADS

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

GRATING 5 kN/m² | CONCRETE PLATFORMS 10 kN/m² | GROUNDFLOOR 25 kN/m² | CABLE FLOOR 10 kN/M²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

DUST 0.5 kN/m² LIVE LOAD 0.75 kN/m²

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)912.500 m.

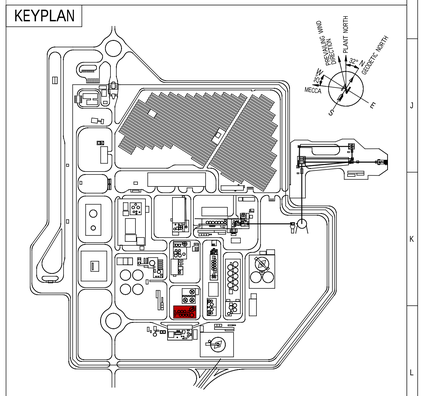
13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

* :-
FOR,
P26 : REFER DWG. MD-216-3520-EG-ME-GAD-1010 FOR LOAD DETAILS

	LOADING POINT	P26*				P27				P28			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]	My[kN]	Vz[kN]	Hx[kN]	Hy[kN]	My[kN]	Vz[kN]	Hx[kN]	Hy[kN]	My[kN]
1	DEAD LOAD EQUIPMENT					- 21				- 30			
2	DEAD LOAD (SECONDARY EQUIPMENT)												
3	OPERATION LOAD 1. FILLING												
4	OPERATION LOAD 2. PRESSURE												
5	OPERATION LOAD 3. TEMPERATURE												
6	OPERATION LOAD 4. DYNAMIC									- 40			
7	OPERATION LOAD 5. OTHERS												
8	TEST LOAD FILLING / EMERGENCY												
9	LIVE LOAD					- 125							
10	WIND ±X - DIRECTION												
11	WIND ±Y - DIRECTION												
12	EARTHQUAKE ±X - DIRECTION												
13	EARTHQUAKE ±Y - DIRECTION												
	EQUIPMENT - NO.	3520-LND-001				3520-TNK-001				3520-FAN-001			
	DESCRIPTION	NEUTRALIZATION LAUNDER				NEUTRALIZATION THICKENER FEED TANK				NEUTRALIZATION SCRUBBER FAN			

KEYPLAN



0 1 10m

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	03-01-2020	ISSUED FOR APPROVAL	BS	TV	SC	RL	

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3520-EG-ME-GAD-1039	GENERAL ARRANGEMENT DRAWING, PLAN AT EL. +912.500
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3520-EG-ME-GAD-1008	PLAN AT EL. +912.150

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_M_D_02813_002 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 _ GAU01 _ RN13 _ MLD14 _ 35202 — DOCUMENT ID: OU100667830 — REVISION: 0

DRAWN	BS	03-01-2020
CHECKED	TV	03-01-2020
ENGINEER	SC	03-01-2020
APPROVED	RL	03-01-2020
EMPLOYER		
SCALE	1:100	A0

MANSOURAH-MASSARAH GOLD PROJECT
LOAD PLAN
NEUTRALIZATION AND THICKENING
PLAN AT EL. +925.280

DRAWING NUMBER: MD-216-3520-EG-ME-GAD-1009 — REV: A01

ATTACHMENT 2 STAAD OUTPUT : MD-216-3520-EG-CV-CAL-1285_PIPE RACK FDN IN NEUTRALIZATION AREA

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```
**************************************************
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*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAY 18, 2020                *
*           Time=    16:34:10                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
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```
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     2. START JOB INFORMATION
     3. ENGINEER DATE 04-FEB-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
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     9. 6 3 0.9 2.3; 7 11 0.9 0; 8 11 0.9 2.3; 9 19 0.9 0; 10 19 0.9 2.3; 11 27 0.9 0
    10. 12 27 0.9 2.3; 13 35 0.9 -0.35; 14 35 0.9 2.65; 15 38.2 0.9 -0.35
    11. 16 38.2 0.9 2.65; 101 -8.2 0.9 3.25; 102 -5 0.9 3.25; 103 -5 0.9 8.65
    12. 104 -8.2 0.9 8.65; 201 35 0.9 3.25; 202 38.2 0.9 3.25; 203 38.2 0.9 8.65
    13. 204 35 0.9 8.65; 301 -8.2 0.9 -3.85; 302 -5 0.9 -3.85; 303 -5 0.9 -0.95
    14. 304 -8.2 0.9 -0.95; 401 35 0.9 -3.35; 402 38.2 0.9 -3.35; 403 38.2 0.9 -1.275
    15. 404 34.2 0.9 -1.275; 501 50 0.9 -13.35; 502 53.2 0.9 -13.35
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1326. 2818 5098 5084 5083 5097; 2819 5100 5086 5085 5099; 2820 5101 5087 5086 5100

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1327. 2821 5102 5088 5087 5101; 2822 5103 5089 5088 5102; 2823 5104 5090 5089 5103
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1330. 2830 5112 5098 5097 5111; 2831 5114 5100 5099 5113; 2832 5115 5101 5100 5114
1331. 2833 5116 5102 5101 5115; 2834 5117 5103 5102 5116; 2835 5118 5104 5103 5117
1332. 2836 5119 5105 5104 5118; 2837 5070 5112 5111 5069; 2838 5069 5111 5110 5068
1333. 2839 5068 5110 5109 5067; 2840 5067 5109 5108 5066; 2841 5066 5108 5107 5065
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1336. 2848 5072 5114 5113 5071; 2849 3110 3131 5125 5124; 2850 3131 3152 5126 5125
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1338. 2854 3068 3089 5123 5122; 2855 3047 3068 5122 5121; 2856 3026 3047 5121 5120
1339. 2857 3299 3320 5134 5133; 2858 3320 3341 5135 5134; 2859 3341 3362 5136 5135
1340. 2860 3362 3383 5137 5136; 2861 3278 3299 5133 5132; 2862 3257 3278 5132 5131
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1344. 2872 5146 3034 3033 5145; 2873 5148 3039 3038 5147; 2874 5149 3040 3039 5148
1345. 2875 5150 3041 3040 5149; 2876 5151 3042 3041 5150; 2877 5152 3043 3042 5151
1346. 2878 5153 3044 3043 5152; 2879 5154 3045 3044 5153; 2880 5155 3046 3045 5154
1347. 2881 5156 4548 3034 5146; 2882 5157 4584 4548 5156; 2883 5158 4620 4584 5157
1348. 2884 5147 3038 4620 5158; 2885 5138 3026 5120 5159
1349. START GROUP DEFINITION
1350. ELEMENT
1351. _F1_MOMENT 2467 TO 2492 2495 TO 2498 2501 TO 2504 2507 TO 2510 2513 TO 2538 -
1352. 2543 TO 2554 2556 2557 2559 2560 2562 2563 2565 TO 2590 2592 2593 2595 2596 -
1353. 2598 2599 2601 TO 2614 2617 TO 2628 2630 2631 2633 2634 2636 2637 -
1354. 2639 TO 2664 2666 2667 2669 2670 2672 2673 2675 TO 2688 2691 TO 2702 2704 -
1355. 2705 2707 2708 2710 2711 2713 TO 2738 2740 2741 2743 2744 2746 2747 2749 -
1356. 2750 TO 2762
1357. _F1_SHEAR 2467 TO 2479 2484 2485 2490 2491 2496 2497 2502 2503 2508 2509 2514 -
1358. 2515 2520 2521 2526 TO 2538 2543 TO 2548 2551 2554 2557 2560 2563 2566 2569 -
1359. 2572 TO 2584 2587 2590 2593 2596 2599 2602 2605 2608 TO 2614 2617 TO 2622 -
1360. 2625 2628 2631 2634 2637 2640 2643 2646 TO 2658 2661 2664 2667 2670 2673 -
1361. 2676 2679 2682 TO 2688 2691 TO 2696 2699 2702 2705 2708 2711 2714 2717 2720 -
1362. 2721 TO 2732 2735 2738 2741 2744 2747 2750 2753 2756 TO 2762
1363. _F2_MOMENT 1002 TO 1009 1014 TO 1021 1024 TO 1031 1036 TO 1051 1056 TO 1071 -
1364. 1076 TO 1091 1096 TO 1111 1116 TO 1124 1129 TO 1131 1136 TO 1138 1143 1144 -
1365. 1149 TO 1151 1156 TO 1158 1163 1164 1169 TO 1171 1176 TO 1178 1183 1184 1189 -
1366. 1190 TO 1191 1196 TO 1198 1203 TO 1211 1216 TO 1231 1236 TO 1251 1256 TO 1271 -
1367. 1276 TO 1291 1296 TO 1311 1316 TO 1323 2353 TO 2368 2385 TO 2400 -
1368. 2417 TO 2432 2449 TO 2464
1369. _F2_SHEAR 1002 TO 1009 1014 TO 1021 1024 TO 1031 1036 TO 1043 1051 1056 1071 -
1370. 1076 1091 1096 1111 1116 1131 1136 1151 1156 1171 1176 1191 1196 1211 1216 -
1371. 1231 1236 1251 1256 1271 1276 1284 TO 1291 1296 TO 1311 1316 TO 1323 2353 -
1372. 2354 TO 2368 2385 TO 2400 2417 TO 2432 2449 TO 2464
1373. _F3_MOMENT 2 TO 8 12 15 18 TO 29 32 35 TO 46 49 52 TO 63 66 70 TO 76 80 83 -
1374. 86 TO 97 100 103 TO 114 117 120 TO 131 134 138 TO 144 148 151 154 TO 165 -
1375. 168 171 TO 182 185 188 TO 199 202 206 TO 212 216 219 222 TO 233 236 -
1376. 239 TO 250 253 256 TO 267 270 274 TO 280 284 287 290 TO 301 304 307 TO 318 -
1377. 321 324 TO 335 338 342 TO 348 352 355 358 TO 369 372 375 TO 386 389 -
1378. 392 TO 403 406 410 TO 416 420 423 426 TO 437 440 443 TO 454 457 460 TO 471 -
1379. 474 478 TO 484 488 491 494 TO 505 508 511 TO 522 525 528 TO 539 542 1356 -
1380. 1357 TO 1368 1371 TO 1383 1386 TO 1424 1427 TO 1465 1940 TO 1979 1990 TO 2029 -
1381. 2040 TO 2079 2090 TO 2129 2140 TO 2179 2190 TO 2229 2240 TO 2279 -
1382. 2290 TO 2329
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1383. _F3_SHEAR 4 TO 6 21 TO 23 38 TO 40 52 TO 63 66 72 TO 74 89 TO 91 106 TO 108 -
1384. 120 TO 131 134 140 TO 142 157 TO 159 174 TO 176 188 TO 199 202 208 TO 210 -
1385. 225 TO 227 242 TO 244 256 TO 267 270 276 TO 278 293 TO 295 310 TO 312 324 -
1386. 325 TO 335 338 344 TO 346 361 TO 363 378 TO 380 392 TO 403 406 412 TO 414 -
1387. 429 TO 431 446 TO 448 460 TO 471 474 480 TO 482 497 TO 499 514 TO 516 528 -
1388. 529 TO 539 542 1356 TO 1368 1371 TO 1383 1386 TO 1424 1427 TO 1465 1943 1947 -
1389. 1951 1955 1959 1963 1967 1971 TO 1979 1990 TO 2029 2040 TO 2079 2090 TO 2129 -
1390. 2143 2147 2151 2155 2159 2163 2167 2171 TO 2179 2190 TO 2229 2240 TO 2279 -
1391. 2290 TO 2329
1392. _F4_MOMENT 1551 TO 1556 1559 TO 1564 1567 TO 1572 1575 1576 1579 1580 1583 -
1393. 1584 1587 1588 1591 1592 1595 1596 1599 1600 1603 1604 1607 TO 1612 1615 -
1394. 1616 TO 1620 1623 TO 1628 1662 TO 1664 1666 1667 1670 TO 1672 1674 1675 1679 -
1395. 1680 1683 1687 1688 1691 1694 TO 1696 1698 1699 1702 TO 1704 1706 1707 1726 -
1396. 1727 1730 TO 1732 1734 1735 1738 TO 1740 1743 1747 1748 1751 1755 1756 1758 -
1397. 1759 1762 TO 1764 1766 1767 1770 TO 1772 2765 TO 2848
1398. _F4_SHEAR 1732 1740 1748 1756 1764 1772 2765 TO 2848
1399. _F2A_F2B_MOMENT 564 581 598 615 632 649 684 TO 689 692 TO 699 701 TO 703 705 -
1400. 706 TO 707 709 TO 711 713 TO 739 772 TO 795 797 TO 799 801 TO 803 805 TO 807 -
1401. 809 TO 811 813 TO 815 817 TO 819 821 TO 823 825 TO 835 841 TO 843 -
1402. 845 TO 847 849 TO 851 853 TO 863 865 TO 867 869 TO 871 873 TO 875 -
1403. 877 TO 903 936 TO 959 961 TO 963 965 TO 967 969 TO 971 973 TO 975 -
1404. 977 TO 979 981 TO 983 985 TO 987 989 TO 999 2337 TO 2352 2369 TO 2384 2401 -
1405. 2402 TO 2416 2433 TO 2448 2849 TO 2885
1406. _F2A_F2B_SHEAR 564 581 598 615 687 695 703 711 715 719 723 727 731 TO 739 -
1407. 772 TO 779 783 787 791 795 799 803 807 811 815 819 823 827 831 835 843 847 -
1408. 851 855 859 863 867 871 875 879 883 887 891 895 TO 903 936 TO 943 947 951 -
1409. 955 959 963 967 971 975 979 983 987 991 995 999 2337 TO 2352 2369 TO 2384 -
1410. 2401 TO 2416 2433 TO 2448 2849 TO 2852 2860 2864 2869 TO 2884
1411. MEMBER
1412. _C1 545 TO 552 836 TO 839 1000 1001 1022 1023
1413. _C2 1518 TO 1525 2465 2466 2539 TO 2542 2615 2616 2689 2690 2763 2764
1414. END GROUP DEFINITION
1415. ELEMENT PROPERTY
1416. 564 581 598 615 632 649 666 683 TO 739 772 TO 835 840 TO 903 936 TO 999 1002 -
1417. 1003 TO 1009 1014 TO 1021 1024 TO 1031 1036 TO 1051 1056 TO 1071 1076 TO 1091 -
1418. 1096 TO 1111 1116 TO 1131 1136 TO 1151 1156 TO 1171 1176 TO 1191 -
1419. 1196 TO 1211 1216 TO 1231 1236 TO 1251 1256 TO 1271 1276 TO 1291 -
1420. 1296 TO 1311 1316 TO 1323 1551 TO 1556 1559 TO 1564 1567 TO 1572 -
1421. 1575 TO 1580 1583 TO 1588 1591 TO 1596 1599 TO 1604 1607 TO 1612 -
1422. 1615 TO 1620 1623 TO 1628 1662 TO 1664 1666 1667 1670 TO 1672 1674 1675 1678 -
1423. 1679 TO 1680 1682 1683 1686 TO 1688 1690 1691 1694 TO 1696 1698 1699 1702 -
1424. 1703 TO 1704 1706 1707 1726 1727 1730 TO 1732 1734 1735 1738 TO 1740 1742 -
1425. 1743 1746 TO 1748 1750 1751 1754 TO 1756 1758 1759 1762 TO 1764 1766 1767 -
1426. 1770 TO 1772 2337 TO 2464 2467 TO 2538 2543 TO 2614 2617 TO 2688 -
1427. 2691 TO 2762 2765 TO 2885 THICKNESS 0.5
1428. 1 TO 12 15 18 TO 29 32 35 TO 46 49 52 TO 63 66 69 TO 80 83 86 TO 97 100 103 -
1429. 104 TO 114 117 120 TO 131 134 137 TO 148 151 154 TO 165 168 171 TO 182 185 -
1430. 188 TO 199 202 205 TO 216 219 222 TO 233 236 239 TO 250 253 256 TO 267 270 -
1431. 273 TO 284 287 290 TO 301 304 307 TO 318 321 324 TO 335 338 341 TO 352 355 -
1432. 358 TO 369 372 375 TO 386 389 392 TO 403 406 409 TO 420 423 426 TO 437 440 -
1433. 443 TO 454 457 460 TO 471 474 477 TO 488 491 494 TO 505 508 511 TO 522 525 -
1434. 528 TO 539 542 1356 TO 1368 1371 TO 1383 1386 TO 1424 1427 TO 1465 -
1435. 1940 TO 1979 1990 TO 2029 2040 TO 2079 2090 TO 2129 2140 TO 2179 -
1436. 2190 TO 2229 2240 TO 2279 2290 TO 2329 THICKNESS 0.5
1437. DEFINE MATERIAL START
1438. ISOTROPIC CONCRETE
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1439. E 2.17184E+007
1440. POISSON 0.17
1441. DENSITY 25
1442. ALPHA 5.5E-006
1443. DAMP 0.05
1444. TYPE CONCRETE
1445. STRENGTH FCU 35000
1446. END DEFINE MATERIAL
1447. MEMBER PROPERTY AMERICAN
1448. 545 TO 552 836 TO 839 1000 1001 1022 1023 PRIS YD 0.7 ZD 0.7
1449. MEMBER PROPERTY AMERICAN
1450. 1518 TO 1525 2465 2466 2539 TO 2542 2615 2616 2689 2690 2763 -
1451. 2764 PRIS YD 0.5 ZD 0.5
1452. CONSTANTS
1453. MATERIAL CONCRETE ALL
1454. SUPPORTS
1455. 1 TO 12 15 18 TO 29 32 35 TO 46 49 52 TO 63 66 69 TO 80 83 86 TO 97 100 103 -
1456. 104 TO 114 117 120 TO 131 134 137 TO 148 151 154 TO 165 168 171 TO 182 185 -
1457. 188 TO 199 202 205 TO 216 219 222 TO 233 236 239 TO 250 253 256 TO 267 270 -
1458. 273 TO 284 287 290 TO 301 304 307 TO 318 321 324 TO 335 338 341 TO 352 355 -
1459. 358 TO 369 372 375 TO 386 389 392 TO 403 406 409 TO 420 423 426 TO 437 440 -
1460. 443 TO 454 457 460 TO 471 474 477 TO 488 491 494 TO 505 508 511 TO 522 525 -
1461. 528 TO 539 542 564 581 598 615 632 649 666 683 TO 739 772 TO 835 840 TO 903 -
1462. 936 TO 999 1002 TO 1009 1014 TO 1021 1024 TO 1031 1036 TO 1051 1056 TO 1071 -
1463. 1076 TO 1091 1096 TO 1111 1116 TO 1131 1136 TO 1151 1156 TO 1171 -
1464. 1176 TO 1191 1196 TO 1211 1216 TO 1231 1236 TO 1251 1256 TO 1271 -
1465. 1276 TO 1291 1296 TO 1311 1316 TO 1323 1356 TO 1368 1371 TO 1383 -
1466. 1386 TO 1424 1427 TO 1465 1551 TO 1556 1559 TO 1564 1567 TO 1572 -
1467. 1575 TO 1580 1583 TO 1588 1591 TO 1596 1599 TO 1604 1607 TO 1612 -
1468. 1615 TO 1620 1623 TO 1628 1662 TO 1664 1666 1667 1670 TO 1672 1674 1675 1678 -
1469. 1679 TO 1680 1682 1683 1686 TO 1688 1690 1691 1694 TO 1696 1698 1699 1702 -
1470. 1703 TO 1704 1706 1707 1726 1727 1730 TO 1732 1734 -
1471. 1735  PLATE MAT DIRECT Y SUBGRADE 117600 PRINT COMPRESSION
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JOINT	INFLUENCE AREA
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2001	0.035000
2002	0.043750
2003	0.052500
2004	0.061250
2005	0.043750
2006	0.035000
2007	0.043750
2008	0.021875
2012	0.021875
2013	0.043750
2014	0.035000
2015	0.043750
2016	0.061250
2017	0.052500
2018	0.043750
2019	0.035000
2020	0.013125
2021	0.018750

2022	0.050000
2023	0.062500
2024	0.075000
2025	0.087500
2026	0.062500
2027	0.050000
2028	0.062500
2029	0.031250
2033	0.031250
2034	0.062500
2035	0.050000
2036	0.062500
2037	0.087500
2038	0.075000
2039	0.062500
2040	0.050000
2041	0.018750
2042	0.015000
2043	0.040000
2044	0.050000
2045	0.060000
2046	0.070000
2047	0.050000
2048	0.040000
2049	0.050000
2050	0.025000
2054	0.025000
2055	0.050000
2056	0.040000
2057	0.050000
2058	0.070000
2059	0.060000
2060	0.050000
2061	0.040000
2062	0.015000
2063	0.022500
2064	0.060000
2065	0.075000
2066	0.090000
2067	0.105000
2068	0.075000
2069	0.060000
2070	0.075000
2071	0.037500
2075	0.037500
2076	0.075000
2077	0.060000
2078	0.075000
2079	0.105000
2080	0.090000
2081	0.075000
2082	0.060000
2083	0.022500
2084	0.026250
2085	0.070000
2086	0.087500

2087	0.105000
2088	0.122500
2089	0.087500
2090	0.070000
2091	0.087500
2092	0.043750
2096	0.043750
2097	0.087500
2098	0.070000
2099	0.087500
2100	0.122500
2101	0.105000
2102	0.087500
2103	0.070000
2104	0.026250
2105	0.022500
2106	0.060000
2107	0.075000
2108	0.090000
2109	0.105000
2110	0.075000
2111	0.060000
2112	0.075000
2113	0.037500
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2118	0.075000
2119	0.060000
2120	0.075000
2121	0.105000
2122	0.090000
2123	0.075000
2124	0.060000
2125	0.022500
2126	0.015000
2127	0.040000
2128	0.050000
2129	0.060000
2130	0.070000
2131	0.050000
2132	0.040000
2133	0.050000
2134	0.025000
2138	0.025000
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2141	0.050000
2142	0.070000
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2145	0.040000
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2147	0.018750
2148	0.050000
2149	0.062500
2150	0.075000
2151	0.087500

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2153	0.050000
2154	0.062500
2155	0.031250
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2162	0.062500
2163	0.087500
2164	0.075000
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2167	0.018750
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2169	0.035000
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2171	0.052500
2172	0.061250
2173	0.043750
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2180	0.021875
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2189	0.013125
2190	0.035000
2191	0.043750
2192	0.052500
2193	0.061250
2194	0.043750
2195	0.035000
2196	0.043750
2197	0.021875
2201	0.021875
2202	0.043750
2203	0.035000
2204	0.043750
2205	0.061250
2206	0.052500
2207	0.043750
2208	0.035000
2209	0.013125
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2213	0.075000
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2216	0.050000

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2222	0.031250
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2226	0.087500
2227	0.075000
2228	0.062500
2229	0.050000
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2234	0.060000
2235	0.070000
2236	0.050000
2237	0.040000
2238	0.050000
2239	0.025000
2243	0.025000
2244	0.050000
2245	0.040000
2246	0.050000
2247	0.070000
2248	0.060000
2249	0.050000
2250	0.040000
2251	0.015000
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2253	0.060000
2254	0.075000
2255	0.090000
2256	0.105000
2257	0.075000
2258	0.060000
2259	0.075000
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2264	0.037500
2265	0.075000
2266	0.060000
2267	0.075000
2268	0.105000
2269	0.090000
2270	0.075000
2271	0.060000
2272	0.022500
2273	0.026250
2274	0.070000
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2297	0.090000
2298	0.105000
2299	0.075000
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2301	0.075000
2302	0.037500
2306	0.037500
2307	0.075000
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2310	0.105000
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2313	0.060000
2314	0.022500
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2317	0.050000
2318	0.060000
2319	0.070000
2320	0.050000
2321	0.040000
2322	0.050000
2323	0.025000
2327	0.025000
2328	0.050000
2329	0.040000
2330	0.050000
2331	0.070000
2332	0.060000
2333	0.050000
2334	0.040000
2335	0.015000
2336	0.018750
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2339	0.075000
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2342	0.050000
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2348	0.031250
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2357	0.013125
2358	0.035000
2359	0.043750
2360	0.052500
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2362	0.043750
2363	0.035000
2364	0.043750
2365	0.021875
2369	0.021875
2370	0.043750
2371	0.035000
2372	0.043750
2373	0.061250
2374	0.052500
2375	0.043750
2376	0.035000
2377	0.013125
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2379	0.087445
2380	0.063889
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2383	0.063889
2384	0.063889
2385	0.063889
2386	0.052139
2387	0.063889
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2389	0.062500
2390	0.018750
2393	0.018750
2396	0.087445
2397	0.087445
2398	0.063889
2399	0.052139
2400	0.063889
2401	0.063889
2402	0.063889
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2404	0.052139
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2407	0.062500
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2411	0.018750
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2415	0.069956
2416	0.051111

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2422	0.041711
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2424	0.050000
2425	0.050000
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3893	0.031250
3894	0.015625
TOTAL	76.076210

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1472. 1738 TO 1740 1742 1743 1746 TO 1748 1750 1751 1754 TO 1756 1758 1759 1762 -
1473. 1763 TO 1764 1766 1767 1770 TO 1772 1940 TO 1979 1990 TO 2029 2040 TO 2079 -
1474. 2090 TO 2129 2140 TO 2179 2190 TO 2229 2240 TO 2279 2290 TO 2329 -
1475. 2337 TO 2464 2467 TO 2538 2543 TO 2614 2617 TO 2688 2691 TO 2762 -
1476. 2765 TO 2885 PLATE MAT DIRECT Y SUBGRADE 117600 PRINT COMPRESSION
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JOINT	INFLUENCE AREA
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2012	0.021875
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2033	0.031250

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2054	0.025000
2071	0.037500
2075	0.037500
2092	0.043750
2096	0.043750
2113	0.037500
2117	0.037500
2134	0.025000
2138	0.025000
2155	0.031250
2159	0.031250
2176	0.021875
2180	0.021875
2197	0.021875
2201	0.021875
2218	0.031250
2222	0.031250
2239	0.025000
2243	0.025000
2260	0.037500
2264	0.037500
2281	0.043750
2285	0.043750
2302	0.037500
2306	0.037500
2323	0.025000
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2344	0.031250
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2365	0.021875
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2447	0.031250
2462	0.043750
2465	0.043750
2480	0.031250
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2501	0.025000
2516	0.031250
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2609	0.031250
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2627	0.043750
2642	0.031250
2645	0.031250
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2681	0.031250
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3097	0.034375
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3110	0.021875
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3202	0.021875
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4595	0.050000
4596	0.075000
4597	0.087500
4598	0.075000
4599	0.050000
4600	0.075000
4601	0.056250
4602	0.043750
4603	0.062500
4604	0.050000

4605	0.075000
4606	0.087500
4607	0.075000
4608	0.050000
4609	0.062500
4610	0.043750
4611	0.043750
4612	0.062500
4613	0.050000
4614	0.075000
4615	0.087500
4616	0.075000
4617	0.050000
4618	0.062500
4619	0.043750
4620	0.050000
4621	0.062500
4622	0.056250
4623	0.068750
4624	0.087500
4625	0.062500
4626	0.062500
4627	0.087500
4628	0.043750
4629	0.056250
4630	0.075000
4631	0.050000
4632	0.075000
4633	0.087500
4634	0.075000
4635	0.050000
4636	0.075000
4637	0.056250
4640	0.062500
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4998	0.031250
4999	0.031250
5000	0.015625
5001	0.062500
5002	0.062500

5003	0.062500
5004	0.031250
5005	0.062500
5006	0.062500
5007	0.031250
5008	0.062500
5009	0.062500
5010	0.062500
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5034	0.031250
5035	0.015625
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5037	0.062500
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5058	0.062500

5059	0.062500
5060	0.062500
5061	0.062500
5062	0.062500
5063	0.031250
5064	0.028125
5065	0.056250
5066	0.056250
5067	0.056250
5068	0.056250
5069	0.056250
5070	0.028125
5071	0.028125
5072	0.056250
5073	0.056250
5074	0.056250
5075	0.056250
5076	0.056250
5077	0.028125
5078	0.031250
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5093	0.062500
5094	0.062500
5095	0.062500
5096	0.062500
5097	0.062500
5098	0.031250
5099	0.031250
5100	0.062500
5101	0.062500
5102	0.062500
5103	0.062500
5104	0.062500
5105	0.031250
5106	0.028125
5107	0.056250
5108	0.056250
5109	0.056250
5110	0.056250
5111	0.056250
5112	0.028125
5113	0.028125
5114	0.056250

5115	0.056250
5116	0.056250
5117	0.056250
5118	0.056250
5119	0.028125
5120	0.012500
5121	0.015625
5122	0.014063
5123	0.017188
5124	0.021875
5125	0.015625
5126	0.015625
5127	0.021875
5128	0.010938
5129	0.014063
5130	0.018750
5131	0.012500
5132	0.018750
5133	0.021875
5134	0.018750
5135	0.012500
5136	0.018750
5137	0.014063
5138	0.014063
5139	0.018750
5140	0.021563
5141	0.025313
5142	0.026250
5143	0.018750
5144	0.015000
5145	0.018750
5146	0.018750
5147	0.018750
5148	0.018750
5149	0.015000
5150	0.018750
5151	0.026250
5152	0.022500
5153	0.018750
5154	0.015000
5155	0.005625
5156	0.018750
5157	0.018750
5158	0.018750
5159	0.004688
TOTAL	54.963745

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 1477. DEFINE REFERENCE LOADS
  **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
             MASTER/SLAVE OR IF UNCONNECTED JOINTS.

**WARNING, 1 or more joints are in multiple SUPPORT commands, First Joint  2008
The subsequent entries will add to support direction specs. See Tech Ref Manual.
 1478. LOAD R1 LOADTYPE DEAD  TITLE FOUNDATION SELF WEIGHT(D)
 1479. SELFWEIGHT Y -1 LIST ALL
 1480. ELEMENT LOAD
 1481. 1 TO 12 15 18 TO 29 32 35 TO 46 49 52 TO 63 66 69 TO 80 83 86 TO 97 100 103 -
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1482. 104 TO 114 117 120 TO 131 134 137 TO 148 151 154 TO 165 168 171 TO 182 185 -
1483. 188 TO 199 202 205 TO 216 219 222 TO 233 236 239 TO 250 253 256 TO 267 270 -
1484. 273 TO 284 287 290 TO 301 304 307 TO 318 321 324 TO 335 338 341 TO 352 355 -
1485. 358 TO 369 372 375 TO 386 389 392 TO 403 406 409 TO 420 423 426 TO 437 440 -
1486. 443 TO 454 457 460 TO 471 474 477 TO 488 491 494 TO 505 508 511 TO 522 525 -
1487. 528 TO 539 542 564 581 598 615 632 649 666 683 TO 739 772 TO 835 840 TO 903 -
1488. 936 TO 999 1002 TO 1009 1014 TO 1021 1024 TO 1031 1036 TO 1051 1056 TO 1071 -
1489. 1076 TO 1091 1096 TO 1111 1116 TO 1131 1136 TO 1151 1156 TO 1171 -
1490. 1176 TO 1191 1196 TO 1211 1216 TO 1231 1236 TO 1251 1256 TO 1271 -
1491. 1276 TO 1291 1296 TO 1311 1316 TO 1323 1356 TO 1368 1371 TO 1383 -
1492. 1386 TO 1424 1427 TO 1465 1551 TO 1556 1559 TO 1564 1567 TO 1572 -
1493. 1575 TO 1580 1583 TO 1588 1591 TO 1596 1599 TO 1604 1607 TO 1612 -
1494. 1615 TO 1620 1623 TO 1628 1662 TO 1664 1666 1667 1670 TO 1672 1674 1675 1678 -
1495. 1679 TO 1680 1682 1683 1686 TO 1688 1690 1691 1694 TO 1696 1698 1699 1702 -
1496. 1703 TO 1704 1706 1707 1726 1727 1730 TO 1732 1734 1735 PR GY -30
1497. 2849 TO 2880 2885 PR GY -30
1498. 1738 TO 1740 1742 1743 1746 TO 1748 1750 1751 1754 TO 1756 1758 1759 1762 -
1499. 1763 TO 1764 1766 1767 1770 TO 1772 1940 TO 1979 1990 TO 2029 2040 TO 2079 -
1500. 2090 TO 2129 2140 TO 2179 2190 TO 2229 2240 TO 2279 2290 TO 2329 -
1501. 2337 TO 2464 2467 TO 2538 2543 TO 2614 2617 TO 2688 2691 TO 2762 -
1502. 2765 TO 2848 2881 TO 2884 PR GY -30
1503. LOAD R2 LOADTYPE DEAD  TITLE DEAD LOAD(D1)
1504. **LOAD FROM PIPE RACK
1505. JOINT LOAD
1506. 1 FX -2.19 FY -28.16 FZ -0.45
1507. 2 FX 0.14 FY -22.08 FZ 0.46
1508. 3 FX 0.39 FY -20.92 FZ -0.45
1509. 4 FX 0.16 FY -30.33 FZ -0.21
1510. 5 FX 0.1 FY -38.12 FZ -0.19
1511. 6 FX 0.17 FY -35.74 FZ 0.5
1512. 7 FX 0.1 FY -43.44 FZ -0.28
1513. 8 FX 0.21 FY -47.44 FZ 0.61
1514. 9 FX -7.19 FY -47.65 FZ -0.28
1515. 10 FX -7.55 FY -51.17 FZ 0.27
1516. 11 FX -1.47 FY -39.32 FZ -0.29
1517. 12 FX -2.07 FY -36.68 FZ 0.01
1518. 13 FX -0.28 FY -41.25 FZ -0.07
1519. 14 FX 1.17 FY -30.11 FZ 0.13
1520. 15 FX -0.4 FY -2.66 FZ -0.03
1521. 16 FX 0.32 FY -12.4 FZ 0.26
1522. **LOAD FROM STAIR CASE
1523. JOINT LOAD
1524. 101 FX -0.278 FY -26.712 FZ -2.35
1525. 102 FX 0.278 FY -24.782 FZ -0.498
1526. 103 FX 2.326 FY -24.729 FZ 0.498
1527. 104 FX -2.326 FY -29.436 FZ 2.35
1528. JOINT LOAD
1529. 201 FX -0.278 FY -26.712 FZ -2.35
1530. 202 FX 0.278 FY -24.782 FZ -0.498
1531. 203 FX 2.326 FY -24.729 FZ 0.498
1532. 204 FX -2.326 FY -29.436 FZ 2.35
1533. JOINT LOAD
1534. 301 FX -0.244 FY -11.7 FZ -2.892
1535. 302 FX 0.244 FY -9.89 FZ -0.751
1536. 303 FX 3.312 FY -10.188 FZ 0.751
1537. 304 FX -3.312 FY -13.836 FZ 2.892
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1538. JOINT LOAD
1539. 401 FX -0.244 FY -11.7 FZ -2.892
1540. 402 FX 0.244 FY -9.89 FZ -0.751
1541. 403 FX 3.312 FY -10.188 FZ 0.751
1542. 404 FX -3.312 FY -13.836 FZ 2.892
1543. JOINT LOAD
1544. 501 FX -0.244 FY -11.7 FZ -2.892
1545. 502 FX 0.244 FY -9.89 FZ -0.751
1546. 503 FX 3.312 FY -10.188 FZ 0.751
1547. 504 FX -3.312 FY -13.836 FZ 2.892
1548. LOAD R3 LOADTYPE DEAD  TITLE STRUCTURAL SELF WEIGHT(D2)
1549. **LOAD FROM PIPE RACK
1550. JOINT LOAD
1551. 1 FX -0.76 FY -29.61 FZ -0.08
1552. 2 FX 0.05 FY -25.06 FZ 0.09
1553. 3 FX 0.34 FY -26.95 FZ -0.08
1554. 4 FX 0.05 FY -25.95 FZ -0.01
1555. 5 FX 0.04 FY -37.08 FZ -0.11
1556. 6 FX 0.06 FY -35.51 FZ 0.15
1557. 7 FX 0.04 FY -50.94 FZ -0.23
1558. 8 FX 0.07 FY -50.97 FZ 0.3
1559. 9 FX -4.63 FY -53.77 FZ -0.26
1560. 10 FX -4.27 FY -53.49 FZ 0.27
1561. 11 FX 3.43 FY -48.25 FZ -0.24
1562. 12 FX 3.65 FY -48.59 FZ 0.15
1563. 13 FX -0.25 FY -37.15 FZ -0.06
1564. 14 FX 0.51 FY -33.49 FZ 0.04
1565. 15 FX 1.55 FY -27.36 FZ -0.04
1566. 16 FX 0.12 FY -24.86 FZ 0.13
1567. LOAD R4 LOADTYPE LIVE  TITLE LIVE LOAD_5KN/SQM (L1)
1568. **LOAD FROM PIPE RACK
1569. JOINT LOAD
1570. 1 FX -5.56 FY -70.6 FZ -1.13
1571. 2 FX 0.35 FY -55.21 FZ 1.14
1572. 3 FX 0.97 FY -52.29 FZ -1.11
1573. 4 FX 0.39 FY -76.11 FZ -0.54
1574. 5 FX 0.25 FY -94.41 FZ -0.47
1575. 6 FX 0.42 FY -88.93 FZ 1.23
1576. 7 FX 0.24 FY -112.05 FZ -0.71
1577. 8 FX 0.53 FY -119.51 FZ 1.54
1578. 9 FX -18.1 FY -128.92 FZ -0.77
1579. 10 FX -18.97 FY -131.11 FZ 0.8
1580. 11 FX -3.37 FY -97.3 FZ -0.71
1581. 12 FX -5.15 FY -91.11 FZ -0.02
1582. 13 FX -0.66 FY -96.08 FZ -0.15
1583. 14 FX 2.94 FY -72.21 FZ 0.31
1584. 15 FX -1.1 FY 3.5 FZ -0.05
1585. 16 FX 0.8 FY -26.83 FZ 0.63
1586. **LOAD FROM STAIR CASE
1587. 101 FX -0.041 FY -43.829 FZ -2.665
1588. 102 FX 0.041 FY -42.365 FZ -0.003
1589. 103 FX 6.67 FY -42.365 FZ 0.003
1590. 104 FX -6.67 FY -55.022 FZ 2.665
1591. 201 FX -0.041 FY -43.829 FZ -2.665
1592. 202 FX 0.041 FY -42.365 FZ -0.003
1593. 203 FX 6.67 FY -42.365 FZ 0.003
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1594. 204 FX -6.67 FY -55.022 FZ 2.665
1595. 301 FX -0.01 FY -15.2 FZ -3.082
1596. 302 FX 0.01 FY -14.122 FZ -0.001
1597. 303 FX 9.718 FY -14.122 FZ 0.001
1598. 304 FX -9.718 FY -27.164 FZ 3.082
1599. 401 FX -0.01 FY -15.2 FZ -3.082
1600. 402 FX 0.01 FY -14.122 FZ -0.001
1601. 403 FX 9.718 FY -14.122 FZ 0.001
1602. 404 FX -9.718 FY -27.164 FZ 3.082
1603. JOINT LOAD
1604. 501 FX -0.01 FY -15.2 FZ -3.082
1605. 502 FX 0.01 FY -14.122 FZ -0.001
1606. 503 FX 9.718 FY -14.122 FZ 0.001
1607. 504 FX -9.718 FY -27.164 FZ 3.082
1608. LOAD R5 LOADTYPE LIVE  TITLE LIVE LOAD_10KN/SQM (L2)
1609. **LOAD FROM PIPE RACK
1610. JOINT LOAD
1611. 1 FX -11.12 FY -141.2 FZ -2.26
1612. 2 FX 0.7 FY -110.42 FZ 2.28
1613. 3 FX 1.93 FY -104.57 FZ -2.23
1614. 4 FX 0.77 FY -152.22 FZ -1.07
1615. 5 FX 0.5 FY -188.82 FZ -0.94
1616. 6 FX 0.84 FY -177.86 FZ 2.46
1617. 7 FX 0.48 FY -224.09 FZ -1.42
1618. 8 FX 1.06 FY -239.01 FZ 3.08
1619. 9 FX -36.19 FY -257.84 FZ -1.54
1620. 10 FX -37.95 FY -262.22 FZ 1.6
1621. 11 FX -6.74 FY -194.6 FZ -1.41
1622. 12 FX -10.3 FY -182.22 FZ -0.04
1623. 13 FX -1.32 FY -192.17 FZ -0.3
1624. 14 FX 5.89 FY -144.42 FZ 0.61
1625. 15 FX -2.2 FY 7 FZ -0.09
1626. 16 FX 1.59 FY -53.67 FZ 1.26
1627. LOAD R8 LOADTYPE LIVE  TITLE LIVE LOAD_20KN/SQM (L3)
1628. **LOAD FROM PIPE RACK
1629. JOINT LOAD
1630. 1 FX -17.06 FY -230.74 FZ -3.17
1631. 2 FX 1.04 FY -184.08 FZ 3.21
1632. 3 FX 3.05 FY -175.05 FZ -3.09
1633. 4 FX 1.13 FY -247.26 FZ -1.67
1634. 5 FX 0.75 FY -278.35 FZ -1.39
1635. 6 FX 1.24 FY -267.11 FZ 3.54
1636. 7 FX 0.72 FY -305.77 FZ -1.87
1637. 8 FX 1.57 FY -332.43 FZ 4.26
1638. 9 FX -52.14 FY -354.53 FZ -2
1639. 10 FX -54.9 FY -361.05 FZ 2.09
1640. 11 FX -11.55 FY -266.42 FZ -1.85
1641. 12 FX -17.19 FY -244.72 FZ -0.24
1642. 13 FX -2.05 FY -299.1 FZ -0.49
1643. 14 FX 8.12 FY -223.83 FZ 0.92
1644. 15 FX -3.11 FY -2.56 FZ -0.19
1645. 16 FX 2.32 FY -92.51 FZ 1.94
1646. LOAD R11 LOADTYPE WIND  TITLE WIND LOAD +Z(WL+Z)
1647. **LOAD FROM PIPE RACK
1648. JOINT LOAD
1649. 1 FX 1.73 FY 63.19 FZ -0.09
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1650. 2 FX -0.02 FY -57.96 FZ 31.65
1651. 3 FX -0.43 FY 57.38 FZ 0.21
1652. 4 FX 0 FY -63.3 FZ 33.45
1653. 5 FX -0.05 FY 83.2 FZ -0.03
1654. 6 FX -0.01 FY -82.37 FZ 15.39
1655. 7 FX 0.18 FY 188.43 FZ -0.11
1656. 8 FX -0.16 FY -188.37 FZ 37.17
1657. 9 FX 14.42 FY 247.16 FZ -0.12
1658. 10 FX -11.26 FY -241.84 FZ 50.67
1659. 11 FX -9.91 FY 248.75 FZ -0.27
1660. 12 FX 13.41 FY -254.98 FZ 60.3
1661. 13 FX 0.59 FY 173.28 FZ 0.34
1662. 14 FX -0.06 FY -189.98 FZ 75.44
1663. 15 FX -8.49 FY 131.38 FZ -0.19
1664. 16 FX 0.07 FY -113.97 FZ 33.64
1665. **LOAD FROM STAIR CASE
1666. 101 FX 0.014 FY 26.558 FZ 9.645
1667. 102 FX -0.014 FY 27.257 FZ 9.297
1668. 103 FX 0.089 FY -27.257 FZ 9.3
1669. 104 FX -0.089 FY -26.558 FZ 9.64
1670. 201 FX 0.014 FY 26.558 FZ 9.645
1671. 202 FX -0.014 FY 27.257 FZ 9.297
1672. 203 FX 0.089 FY -27.257 FZ 9.3
1673. 204 FX -0.089 FY -26.558 FZ 9.64
1674. 301 FX 0.001 FY 4.471 FZ 4.452
1675. 302 FX -0.001 FY 5.099 FZ 4.265
1676. 303 FX 0.002 FY -5.099 FZ 4.385
1677. 304 FX -0.002 FY -4.471 FZ 4.628
1678. 401 FX 0.001 FY 4.471 FZ 4.452
1679. 402 FX -0.001 FY 5.099 FZ 4.265
1680. 403 FX 0.002 FY -5.099 FZ 4.385
1681. 404 FX -0.002 FY -4.471 FZ 4.628
1682. 501 FX 0.001 FY 4.471 FZ 4.452
1683. 502 FX -0.001 FY 5.099 FZ 4.265
1684. 503 FX 0.002 FY -5.099 FZ 4.385
1685. 504 FX -0.002 FY -4.471 FZ 4.628
1686. LOAD R12 LOADTYPE WIND  TITLE WIND LOAD -Z(WL-Z)
1687. **LOAD FROM PIPE RACK
1688. JOINT LOAD
1689. 1 FX -7.62 FY -79.89 FZ -0.05
1690. 2 FX 0.16 FY 57.85 FZ -31.5
1691. 3 FX 0.57 FY -147.66 FZ -0.48
1692. 4 FX 0.09 FY 171.56 FZ -85.24
1693. 5 FX -0.03 FY -252.29 FZ 0.16
1694. 6 FX 0.04 FY 251.16 FZ -56.46
1695. 7 FX -0.03 FY -386.04 FZ 0.34
1696. 8 FX 0.16 FY 385.52 FZ -92.03
1697. 9 FX -17.53 FY -399.11 FZ 0.31
1698. 10 FX 17.25 FY 398.72 FZ -97.6
1699. 11 FX 12.49 FY -327.93 FZ 0.31
1700. 12 FX -13.08 FY 327.95 FZ -86.93
1701. 13 FX -0.76 FY -183.38 FZ -0.14
1702. 14 FX -0.25 FY 202 FZ -79.31
1703. 15 FX 8.48 FY -136.11 FZ 0.3
1704. 16 FX 0.07 FY 117.67 FZ -34.57
1705. **LOAD FROM STAIR CASE
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1706. 101 FX -0.014 FY -26.558 FZ -9.645
1707. 102 FX 0.014 FY -27.257 FZ -9.297
1708. 103 FX -0.089 FY 27.257 FZ -9.3
1709. 104 FX 0.089 FY 26.558 FZ -9.64
1710. 201 FX -0.014 FY -26.558 FZ -9.645
1711. 202 FX 0.014 FY -27.257 FZ -9.297
1712. 203 FX -0.089 FY 27.257 FZ -9.3
1713. 204 FX 0.089 FY 26.558 FZ -9.64
1714. 301 FX -0.001 FY -4.471 FZ -4.452
1715. 302 FX 0.001 FY -5.099 FZ -4.265
1716. 303 FX -0.002 FY 5.099 FZ -4.385
1717. 304 FX 0.002 FY 4.471 FZ -4.628
1718. 401 FX -0.001 FY -4.471 FZ -4.452
1719. 402 FX 0.001 FY -5.099 FZ -4.265
1720. 403 FX -0.002 FY 5.099 FZ -4.385
1721. 404 FX 0.002 FY 4.471 FZ -4.628
1722. JOINT LOAD
1723. 501 FX -0.001 FY -4.471 FZ -4.452
1724. 502 FX 0.001 FY -5.099 FZ -4.265
1725. 503 FX -0.002 FY 5.099 FZ -4.385
1726. 504 FX 0.002 FY 4.471 FZ -4.628
1727. LOAD R13 LOADTYPE WIND  TITLE WIND LOAD +X(WL+X)
1728. **LOAD FROM PIPE RACK
1729. JOINT LOAD
1730. 1 FX 25.57 FY 60.4 FZ -0.01
1731. 2 FX 4.62 FY -0.04 FZ 0.06
1732. 3 FX -0.77 FY -56.79 FZ 0.01
1733. 4 FX 2.1 FY 2.27 FZ 1
1734. 5 FX -0.25 FY -1.61 FZ 0
1735. 6 FX -0.34 FY -1.02 FZ -0.44
1736. 7 FX -0.32 FY -2.51 FZ 0
1737. 8 FX -0.51 FY 3.41 FZ -0.57
1738. 9 FX 16.42 FY 24.98 FZ 0.03
1739. 10 FX 17.23 FY 27.02 FZ -0.04
1740. 11 FX 16.49 FY -23.34 FZ -0.04
1741. 12 FX 18.1 FY -30.28 FZ 0.62
1742. 13 FX -0.18 FY 57.1 FZ 0.02
1743. 14 FX -0.61 FY 2.32 FZ -0.44
1744. 15 FX 13.83 FY -60.35 FZ -0.03
1745. 16 FX -0.71 FY -1.57 FZ -0.17
1746. **LOAD FROM STAIR CASE
1747. 101 FX 12.187 FY 62.052 FZ 0.564
1748. 102 FX 12.052 FY -62.052 FZ -0.004
1749. 103 FX 11.308 FY -63.038 FZ 0.004
1750. 104 FX 11.576 FY 63.038 FZ -0.564
1751. 201 FX 12.187 FY 62.052 FZ 0.564
1752. 202 FX 12.052 FY -62.052 FZ -0.004
1753. 203 FX 11.308 FY -63.038 FZ 0.004
1754. 204 FX 11.576 FY 63.038 FZ -0.564
1755. 301 FX 6.915 FY 17.991 FZ 0.132
1756. 302 FX 6.508 FY -17.991 FZ -0.001
1757. 303 FX 7.014 FY -20.561 FZ 0.001
1758. 304 FX 7.263 FY 20.561 FZ -0.132
1759. 401 FX 6.915 FY 17.991 FZ 0.132
1760. 402 FX 6.508 FY -17.991 FZ -0.001
1761. 403 FX 7.014 FY -20.561 FZ 0.001
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1762. 404 FX 7.263 FY 20.561 FZ -0.132
1763. JOINT LOAD
1764. 501 FX 6.915 FY 17.991 FZ 0.132
1765. 502 FX 6.508 FY -17.991 FZ -0.001
1766. 503 FX 7.014 FY -20.561 FZ 0.001
1767. 504 FX 7.263 FY 20.561 FZ -0.132
1768. LOAD R14 LOADTYPE WIND  TITLE WIND LOAD -X(WL-X)
1769. **LOAD FROM PIPE RACK
1770. JOINT LOAD
1771. 1 FX -6.05 FY -22.26 FZ 0
1772. 2 FX 0.47 FY 0.01 FZ 0
1773. 3 FX 0.15 FY 17.48 FZ -0.02
1774. 4 FX 0.54 FY -1.5 FZ -1.51
1775. 5 FX 0.39 FY 3.79 FZ 0.01
1776. 6 FX 0.58 FY 2.25 FZ 0.59
1777. 7 FX 0.42 FY 4.05 FZ 0
1778. 8 FX 0.75 FY -5.26 FZ 0.9
1779. 9 FX -12.38 FY -17.9 FZ -0.02
1780. 10 FX -12.98 FY -19.77 FZ 0.03
1781. 11 FX -11.05 FY 15.07 FZ 0.03
1782. 12 FX -12.53 FY 23.35 FZ -0.8
1783. 13 FX 0.11 FY -51.81 FZ -0.01
1784. 14 FX 0.77 FY -2.6 FZ 0.31
1785. 15 FX -11.01 FY 54.64 FZ 0.02
1786. 16 FX 1.2 FY 0.44 FZ 0.48
1787. **LOAD FROM STAIR CASE
1788. 101 FX -12.187 FY -62.052 FZ -0.564
1789. 102 FX -12.052 FY 62.052 FZ 0.004
1790. 103 FX -11.308 FY 63.038 FZ -0.004
1791. 104 FX -11.576 FY -63.038 FZ 0.564
1792. 201 FX -12.187 FY -62.052 FZ -0.564
1793. 202 FX -12.052 FY 62.052 FZ 0.004
1794. 203 FX -11.308 FY 63.038 FZ -0.004
1795. 204 FX -11.576 FY -63.038 FZ 0.564
1796. 301 FX -6.915 FY -17.991 FZ -0.132
1797. 302 FX -6.508 FY 17.991 FZ 0.001
1798. 303 FX -7.014 FY 20.561 FZ -0.001
1799. 304 FX -7.263 FY -20.561 FZ 0.132
1800. 401 FX -6.915 FY -17.991 FZ -0.132
1801. 402 FX -6.508 FY 17.991 FZ 0.001
1802. 403 FX -7.014 FY 20.561 FZ -0.001
1803. 404 FX -7.263 FY -20.561 FZ 0.132
1804. JOINT LOAD
1805. 501 FX -6.915 FY -17.991 FZ -0.132
1806. 502 FX -6.508 FY 17.991 FZ 0.001
1807. 503 FX -7.014 FY 20.561 FZ -0.001
1808. 504 FX -7.263 FY -20.561 FZ 0.132
1809. LOAD R15 LOADTYPE TEMPERATURE  TITLE TEMPERATURE LOAD_+20(T+20)
1810. **LOAD FROM PIPE RACK
1811. JOINT LOAD
1812. 1 FX -85.3 FY -230.33 FZ -0.56
1813. 2 FX -2.55 FY 0.25 FZ 0.45
1814. 3 FX 6.44 FY 223.87 FZ 1.19
1815. 4 FX -0.98 FY 8.09 FZ -7.13
1816. 5 FX -1.79 FY 1.68 FZ -0.34
1817. 6 FX -1.3 FY 0.05 FZ 2.8
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1818. 7 FX -1.63 FY 17.26 FZ -0.38
1819. 8 FX -0.66 FY -15.88 FZ 3.57
1820. 9 FX -45.01 FY -9.07 FZ -0.42
1821. 10 FX -43.34 FY -14.54 FZ 0.68
1822. 11 FX 31.89 FY 6.95 FZ -0.35
1823. 12 FX 25.25 FY 19.1 FZ -0.58
1824. 13 FX -4.22 FY 221.08 FZ 2.67
1825. 14 FX 2.43 FY 7.08 FZ -3.96
1826. 15 FX 118.05 FY -224.72 FZ -0.82
1827. 16 FX 2.71 FY -10.87 FZ 3.18
1828. LOAD R16 LOADTYPE TEMPERATURE  TITLE TEMPERATURE LOAD_-20(T-20)
1829. **LOAD FROM PIPE RACK
1830. JOINT LOAD
1831. 1 FX 63.78 FY 162.5 FZ 0.57
1832. 2 FX 2.62 FY -0.11 FZ -0.5
1833. 3 FX -5.72 FY -150.19 FZ -1.3
1834. 4 FX 1.06 FY -11.46 FZ 9.09
1835. 5 FX 1.62 FY -19.65 FZ 0.35
1836. 6 FX 1.36 FY 15.16 FZ -6.26
1837. 7 FX 1.39 FY -10.07 FZ 0.38
1838. 8 FX 0.79 FY 10.31 FZ -2.46
1839. 9 FX 53.32 FY 21.52 FZ 0.42
1840. 10 FX 41.09 FY 10.83 FZ -0.71
1841. 11 FX -23.33 FY -23.37 FZ 0.35
1842. 12 FX -27.85 FY -12.06 FZ -0.15
1843. 13 FX 4.15 FY -189.26 FZ -2.66
1844. 14 FX -2.33 FY -6.03 FZ 4.26
1845. 15 FX -109.37 FY 195.44 FZ 0.79
1846. 16 FX -2.59 FY 6.42 FZ -2.18
1847. LOAD R17 LOADTYPE SEISMIC TITLE MAX SEISMIC FORCE
1848. **SEISMIC LOAD
1850. JOINT LOAD
1851. 1 FX 1.51 FY -60.04 FZ -2.24
1852. 2 FX 1.51 FY -49.93 FZ -2.22
1853. 3 FX 2.62 FY -25.36 FZ -2.29
1854. 4 FX 1.63 FY -38.55 FZ -13.26
1855. 5 FX 0.72 FY -35.55 FZ -1.08
1856. 6 FX 1.36 FY -31.31 FZ -10.75
1857. 7 FX 0.69 FY -42.52 FZ -1.6
1858. 8 FX 1.79 FY -46.89 FZ -15.07
1859. 9 FX -5.21 FY -44.33 FZ -1.7
1860. 10 FX -3.76 FY -53.27 FZ -17.53
1861. 11 FX 7.47 FY -38.92 FZ -1.69
1862. 12 FX 16.42 FY -17.72 FZ -17.32
1863. 13 FX -0.56 FY -47.3 FZ -0.49
1864. 14 FX 6.75 FY -41.68 FZ -11.15
1865. 15 FX 5.29 FY -23.5 FZ -0.23
1866. 16 FX 2.47 FY -25.5 FZ -4.72
1867. 101 FX 0.014 FY 26.558 FZ 9.645
1868. 102 FX -0.014 FY 27.257 FZ 9.297
1869. 103 FX 0.089 FY -27.257 FZ 9.3
1870. 104 FX -0.089 FY -26.558 FZ 9.64
1871. 201 FX 0.014 FY 26.558 FZ 9.645
1872. 202 FX -0.014 FY 27.257 FZ 9.297
1873. 203 FX 0.089 FY -27.257 FZ 9.3
1874. 204 FX -0.089 FY -26.558 FZ 9.64
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1875. 301 FX 0.001 FY 4.471 FZ 4.452
1876. 302 FX -0.001 FY 5.099 FZ 4.265
1877. 303 FX 0.002 FY -5.099 FZ 4.385
1878. 304 FX -0.002 FY -4.471 FZ 4.628
1879. 401 FX 0.001 FY 4.471 FZ 4.452
1880. 402 FX -0.001 FY 5.099 FZ 4.265
1881. 403 FX 0.002 FY -5.099 FZ 4.385
1882. 404 FX -0.002 FY -4.471 FZ 4.628
1883. 501 FX 0.001 FY 4.471 FZ 4.452
1884. 502 FX -0.001 FY 5.099 FZ 4.265
1885. 503 FX 0.002 FY -5.099 FZ 4.385
1886. 504 FX -0.002 FY -4.471 FZ 4.628
1887. LOAD R18 LOADTYPE SEISMIC TITLE MIN SEISMIC FORCE
1888. **SEISMIC LOAD
1889. JOINT LOAD
1890. 1 FX -17.13 FY -193.98 FZ -0.63
1891. 2 FX -0.53 FY -155.25 FZ 5.13
1892. 3 FX 0.64 FY -177.06 FZ -0.52
1893. 4 FX -0.56 FY -220.15 FZ 11.89
1894. 5 FX -0.01 FY -284.09 FZ -0.35
1895. 6 FX -0.19 FY -273.26 FZ 14.08
1896. 7 FX 0 FY -336.89 FZ -0.56
1897. 8 FX -0.34 FY -355.51 FZ 19.38
1898. 9 FX -49.22 FY -376.33 FZ -0.59
1899. 10 FX -52.06 FY -378.42 FZ 19.88
1900. 11 FX -8.54 FY -304.47 FZ -0.51
1901. 12 FX -21.23 FY -309.3 FZ 17.59
1902. 13 FX -1.74 FY -290.42 FZ -0.07
1903. 14 FX 1.33 FY -222.87 FZ 12.02
1904. 15 FX -4.08 FY -49.82 FZ -0.03
1905. 16 FX -0.24 FY -110.18 FZ 6.62
1906. 101 FX 0.014 FY 26.558 FZ 9.645
1907. 102 FX -0.014 FY 27.257 FZ 9.297
1908. 103 FX 0.089 FY -27.257 FZ 9.3
1909. 104 FX -0.089 FY -26.558 FZ 9.64
1910. 201 FX 0.014 FY 26.558 FZ 9.645
1911. 202 FX -0.014 FY 27.257 FZ 9.297
1912. 203 FX 0.089 FY -27.257 FZ 9.3
1913. 204 FX -0.089 FY -26.558 FZ 9.64
1914. 301 FX 0.001 FY 4.471 FZ 4.452
1915. 302 FX -0.001 FY 5.099 FZ 4.265
1916. 303 FX 0.002 FY -5.099 FZ 4.385
1917. 304 FX -0.002 FY -4.471 FZ 4.628
1918. 401 FX 0.001 FY 4.471 FZ 4.452
1919. 402 FX -0.001 FY 5.099 FZ 4.265
1920. 403 FX 0.002 FY -5.099 FZ 4.385
1921. 404 FX -0.002 FY -4.471 FZ 4.628
1922. 501 FX 0.001 FY 4.471 FZ 4.452
1923. 502 FX -0.001 FY 5.099 FZ 4.265
1924. 503 FX 0.002 FY -5.099 FZ 4.385
1925. 504 FX -0.002 FY -4.471 FZ 4.628
1926. END DEFINE REFERENCE LOADS
1927. **UNFACTORED LOAD COMBINATIONS
1928. **D
1929. LOAD 101 D+(D1+D2)
1930. REFERENCE LOAD
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1931. R1 1.0 R2 1.0 R3 1.0
1932. **D+T+L
1933. LOAD 102 (D+D1+D2)+T(+20)+L1
1934. REFERENCE LOAD
1935. R1 1.0 R2 1.0 R3 1.0 R4 1.0 R15 1.0
1936. LOAD 103 (D+D1+D2)+T(+20)+L2
1937. REFERENCE LOAD
1938. R1 1.0 R2 1.0 R3 1.0 R5 1.0 R15 1.0
1939. LOAD 104 (D+D1+D2)+T(+20)+L3
1940. REFERENCE LOAD
1941. R1 1.0 R2 1.0 R3 1.0 R8 1.0 R15 1.0
1942. **
1943. LOAD 105 (D+D1+D2)+T(-20)+L1
1944. REFERENCE LOAD
1945. R1 1.0 R2 1.0 R3 1.0 R4 1.0 R16 1.0
1946. LOAD 106 (D+D1+D2)+T(-20)+L2
1947. REFERENCE LOAD
1948. R1 1.0 R2 1.0 R3 1.0 R5 1.0 R16 1.0
1949. LOAD 107 (D+D1+D2)+T(-20)+L3
1950. REFERENCE LOAD
1951. R1 1.0 R2 1.0 R3 1.0 R8 1.0 R16 1.0
1952. **D+W
1953. LOAD 108 D+D1+D2+W(+X)
1954. REFERENCE LOAD
1955. R1 1.0 R2 1.0 R3 1.0 R13 1.0
1956. LOAD 109 D+D1+D2+W(-X)
1957. REFERENCE LOAD
1958. R1 1.0 R2 1.0 R3 1.0 R14 1.0
1959. LOAD 110 D+D1+D2+W(+Z)
1960. REFERENCE LOAD
1961. R1 1.0 R2 1.0 R3 1.0 R11 1.0
1962. LOAD 111 D+D1+D2+W(-Z)
1963. REFERENCE LOAD
1964. R1 1.0 R2 1.0 R3 1.0 R12 1.0
1965. ***D+0.75W+0.75L
1966. LOAD 112 D+D1+D2+0.75L1+0.75W(+X)
1967. REFERENCE LOAD
1968. R1 1.0 R2 1.0 R3 1.0 R4 0.75 R13 0.75
1969. LOAD 113 D+D1+D2+0.75L1+0.75W(-X)
1970. REFERENCE LOAD
1971. R1 1.0 R2 1.0 R3 1.0 R4 0.75 R14 0.75
1972. LOAD 114 D+D1+D2+0.75L1+0.75W(+Z)
1973. REFERENCE LOAD
1974. R1 1.0 R2 1.0 R3 1.0 R4 0.75 R11 0.75
1975. LOAD 115 D+D1+D2+0.75L1+0.75W(-Z)
1976. REFERENCE LOAD
1977. R1 1.0 R2 1.0 R3 1.0 R4 0.75 R12 0.75
1978. **
1979. LOAD 116 D+D1+D2+0.75L2+0.75W(+X)
1980. REFERENCE LOAD
1981. R1 1.0 R2 1.0 R3 1.0 R5 0.75 R13 0.75
1982. LOAD 117 D+D1+D2+0.75L2+0.75W(-X)
1983. REFERENCE LOAD
1984. R1 1.0 R2 1.0 R3 1.0 R5 0.75 R14 0.75
1985. LOAD 118 D+D1+D2+0.75L2+0.75W(+Z)
1986. REFERENCE LOAD
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1987. R1 1.0 R2 1.0 R3 1.0 R5 0.75 R11 0.75
1988. LOAD 119 D+D1+D2+0.75L2+0.75W(-Z)
1989. REFERENCE LOAD
1990. R1 1.0 R2 1.0 R3 1.0 R5 0.75 R12 0.75
1991. **
1992. LOAD 120 D+D1+D2+0.75L3+0.75W(+X)
1993. REFERENCE LOAD
1994. R1 1.0 R2 1.0 R3 1.0 R8 0.75 R13 0.75
1995. LOAD 121 D+D1+D2+0.75L3+0.75W(-X)
1996. REFERENCE LOAD
1997. R1 1.0 R2 1.0 R3 1.0 R8 0.75 R14 0.75
1998. LOAD 122 D+D1+D2+0.75L3+0.75W(+Z)
1999. REFERENCE LOAD
2000. R1 1.0 R2 1.0 R3 1.0 R8 0.75 R11 0.75
2001. LOAD 123 D+D1+D2+0.75L3+0.75W(-Z)
2002. REFERENCE LOAD
2003. R1 1.0 R2 1.0 R3 1.0 R8 0.75 R12 0.75
2004. **0.6D+W
2005. LOAD 124 0.6(D1+D2)+W(+X)+D
2006. REFERENCE LOAD
2007. R1 1.0 R2 0.6 R3 0.6 R13 1.0
2008. LOAD 125 0.6(D1+D2)+W(-X)+D
2009. REFERENCE LOAD
2010. R1 1.0 R2 0.6 R3 0.6 R14 1.0
2011. LOAD 126 0.6(D1+D2)+W(+Z)+D
2012. REFERENCE LOAD
2013. R1 1.0 R2 0.6 R3 0.6 R11 1.0
2014. LOAD 127 0.6(D1+D2)+W(-Z)+D
2015. REFERENCE LOAD
2016. R1 1.0 R2 0.6 R3 0.6 R12 1.0
2017. LOAD 128 D+D1+D2+0.7E_MAX
2018. REFERENCE LOAD
2019. R1 1.0 R2 1.0 R3 1.0 R17 0.7
2020. LOAD 129 D+D1+D2+0.7E_MIN
2021. REFERENCE LOAD
2022. R1 1.0 R2 1.0 R3 1.0 R18 0.7
2023. **D+0.75L+0.75*0.7E
2024. LOAD 130 D+D1+D2+0.75L1+0.75*0.7E_MAX
2025. REFERENCE LOAD
2026. R1 1.0 R2 1.0 R3 1.0 R4 0.75 R17 0.525
2027. LOAD 131 D+D1+D2+0.75L1+0.75*0.7E_MIN
2028. REFERENCE LOAD
2029. R1 1.0 R2 1.0 R3 1.0 R4 0.75 R18 0.525
2030. **
2031. LOAD 132 D+D1+D2+0.75L2+0.75*0.7E_MAX
2032. REFERENCE LOAD
2033. R1 1.0 R2 1.0 R3 1.0 R5 0.75 R17 0.525
2034. LOAD 133 D+D1+D2+0.75L2+0.75*0.7E_MIN
2035. REFERENCE LOAD
2036. R1 1.0 R2 1.0 R3 1.0 R5 0.75 R18 0.525
2037. **
2038. LOAD 134 D+D1+D2+0.75L3+0.75*0.7E_MAX
2039. REFERENCE LOAD
2040. R1 1.0 R2 1.0 R3 1.0 R8 0.75 R17 0.525
2041. LOAD 135 D+D1+D2+0.75L3+0.75*0.7E_MIN
2042. REFERENCE LOAD
```

```
2043. R1 1.0 R2 1.0 R3 1.0 R8 0.75 R18 0.525
2044. **0.6D+0.7E
2045. LOAD 136 0.6(D1+D2)+0.7E_MAX+D
2046. REFERENCE LOAD
2047. R1 1.0 R2 0.6 R3 0.6 R17 0.7
2048. LOAD 137 0.6(D1+D2)+0.7E_MIN+D
2049. REFERENCE LOAD
2050. R1 1.0 R2 0.6 R3 0.6 R18 0.7
2053. **FACTORED LOAD COMBINATIONS
2054. **1.4D
2055. LOAD 201 1.4D+1.4(D1+D2)
2056. REFERENCE LOAD
2057. R1 1.4 R2 1.4 R3 1.4
2058. **1.4(D+T)+1.7L
2059. LOAD 202 1.4(D+D1+D2)+1.4T(+20)+1.7L1
2060. REFERENCE LOAD
2061. R1 1.4 R2 1.4 R3 1.4 R4 1.7 R15 1.4
2062. LOAD 203 1.4(D+D1+D2)+1.4T(+20)+1.7L2
2063. REFERENCE LOAD
2064. R1 1.4 R2 1.4 R3 1.4 R5 1.7 R15 1.4
2065. LOAD 204 1.4(D+D1+D2)+1.4T(+20)+1.7L3
2066. REFERENCE LOAD
2067. R1 1.4 R2 1.4 R3 1.4 R8 1.7 R15 1.4
2068. **
2069. LOAD 205 1.4(D+D1+D2)+1.4T(-20)+1.7L1
2070. REFERENCE LOAD
2071. R1 1.4 R2 1.4 R3 1.4 R4 1.7 R16 1.4
2072. LOAD 206 1.4(D+D1+D2)+1.4T(-20)+1.7L2
2073. REFERENCE LOAD
2074. R1 1.4 R2 1.4 R3 1.4 R5 1.7 R16 1.4
2075. LOAD 207 1.4(D+D1+D2)+1.4T(-20)+1.7L3
2076. REFERENCE LOAD
2077. R1 1.4 R2 1.4 R3 1.4 R8 1.7 R16 1.4
2078. **1.2D+0.8W
2079. LOAD 208 1.2(D+D1+D2)+0.8W(+X)
2080. REFERENCE LOAD
2081. R1 1.2 R2 1.2 R3 1.2 R13 0.8
2082. LOAD 209 1.2(D+D1+D2)+0.8W(-X)
2083. REFERENCE LOAD
2084. R1 1.2 R2 1.2 R3 1.2 R14 0.8
2085. LOAD 210 1.2(D+D1+D2)+0.8W(+Z)
2086. REFERENCE LOAD
2087. R1 1.2 R2 1.2 R3 1.2 R11 0.8
2088. LOAD 211 1.2(D+D1+D2)+0.8W(-Z)
2089. REFERENCE LOAD
2090. R1 1.2 R2 1.2 R3 1.2 R12 0.8
2091. ***1.2D+1.6W+1.0L
2092. LOAD 212 1.2(D+D1+D2)+1.0L1+1.6W(+X)
2093. REFERENCE LOAD
2094. R1 1.2 R2 1.2 R3 1.2 R4 1.0 R13 1.6
2095. LOAD 213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
2096. REFERENCE LOAD
2097. R1 1.2 R2 1.2 R3 1.2 R4 1.0 R14 1.6
2098. LOAD 214 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
2099. REFERENCE LOAD
2100. R1 1.2 R2 1.2 R3 1.2 R4 1.0 R11 1.6
```

```
2101. LOAD 215 1.2(D+D1+D2)+1.0L1+1.6W(-Z)
2102. REFERENCE LOAD
2103. R1 1.2 R2 1.2 R3 1.2 R4 1.0 R12 1.6
2104. **
2105. LOAD 216 1.2(D+D1+D2)+1.0L2+1.6W(+X)
2106. REFERENCE LOAD
2107. R1 1.2 R2 1.2 R3 1.2 R5 1.0 R13 1.6
2108. LOAD 217 1.2(D+D1+D2)+1.0L2+1.6W(-X)
2109. REFERENCE LOAD
2110. R1 1.2 R2 1.2 R3 1.2 R5 1.0 R14 1.6
2111. LOAD 218 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
2112. REFERENCE LOAD
2113. R1 1.2 R2 1.2 R3 1.2 R5 1.0 R11 1.6
2114. LOAD 219 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
2115. REFERENCE LOAD
2116. R1 1.2 R2 1.2 R3 1.2 R5 1.0 R12 1.6
2117. **
2118. LOAD 220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
2119. REFERENCE LOAD
2120. R1 1.2 R2 1.2 R3 1.2 R8 1.0 R13 1.6
2121. LOAD 221 1.2(D+D1+D2)+1.0L3+1.6W(-X)
2122. REFERENCE LOAD
2123. R1 1.2 R2 1.2 R3 1.2 R8 1.0 R14 1.6
2124. LOAD 222 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
2125. REFERENCE LOAD
2126. R1 1.2 R2 1.2 R3 1.2 R8 1.0 R11 1.6
2127. LOAD 223 1.2(D+D1+D2)+1.0L3+1.6W(-Z)
2128. REFERENCE LOAD
2129. R1 1.2 R2 1.2 R3 1.2 R8 1.0 R12 1.6
2130. **0.9D+1.6W
2131. LOAD 224 0.9(D1+D2)+W(+X)+D
2132. REFERENCE LOAD
2133. R1 1.0 R2 0.9 R3 0.9 R13 1.6
2134. LOAD 225 0.9(D1+D2)+W(-X)+D
2135. REFERENCE LOAD
2136. R1 1.0 R2 0.9 R3 0.9 R14 1.6
2137. LOAD 226 0.9(D1+D2)+W(+Z)+D
2138. REFERENCE LOAD
2139. R1 1.0 R2 0.9 R3 0.9 R11 1.6
2140. LOAD 227 0.9(D1+D2)+W(-Z)+D
2141. REFERENCE LOAD
2142. R1 1.0 R2 0.9 R3 0.9 R12 1.6
2143. **1.2D+1.0L+1.0E
2144. LOAD 228 1.2(D+D1+D2)+1.0L1+1.0E_MAX
2145. REFERENCE LOAD
2146. R1 1.2 R2 1.2 R3 1.2 R4 1.0 R17 1.0
2147. LOAD 229 1.2(D+D1+D2)+1.0L1+1.0E_MIN
2148. REFERENCE LOAD
2149. R1 1.2 R2 1.2 R3 1.2 R4 1.0 R18 1.0
2150. **
2151. LOAD 230 1.2(D+D1+D2)+1.0L2+1.0E_MAX
2152. REFERENCE LOAD
2153. R1 1.2 R2 1.2 R3 1.2 R5 1.0 R17 1.0
2154. LOAD 231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
2155. REFERENCE LOAD
2156. R1 1.2 R2 1.2 R3 1.2 R5 1.0 R18 1.0
```

```
2157. **
2158. LOAD 232 1.2(D+D1+D2)+1.0L3+1.0E_MAX
2159. REFERENCE LOAD
2160. R1 1.2 R2 1.2 R3 1.2 R8 1.0 R17 1.0
2161. LOAD 233 1.2(D+D1+D2)+1.0L3+1.0E_MIN
2162. REFERENCE LOAD
2163. R1 1.2 R2 1.2 R3 1.2 R8 1.0 R18 1.0
2164. **0.9D+1.0E
2165. LOAD 234 0.9(D1+D2)+1.0E_MAX(+X)+D
2166. REFERENCE LOAD
2167. R1 1.0 R2 0.9 R3 0.9 R17 1.0
2168. LOAD 235 0.9(D1+D2)+1.0E_MIN(+X)+D
2169. REFERENCE LOAD
2170. R1 1.0 R2 0.9 R3 0.9 R18 1.0
2172. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**2444**	**NUMBER OF MEMBERS**	**36**
NUMBER OF PLATES	**2009**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2408**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 2258/ 30/ 93 DOF
TOTAL PRIMARY LOAD CASES = 72, TOTAL DEGREES OF FREEDOM = 7440
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 692 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 37.2/ 194102.0 MB

```
  **NOTE-Tension/Compression converged after  1 iterations, Case=  101


*** LOAD CASE   102 -- START ITERATION NO.   2
*** LOAD CASE   102 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  102


*** LOAD CASE   103 -- START ITERATION NO.   2

  **NOTE-Tension/Compression converged after  2 iterations, Case=  103


*** LOAD CASE   104 -- START ITERATION NO.   2
```

**NOTE-Tension/Compression converged after 2 iterations, Case= 104

*** LOAD CASE 105 -- START ITERATION NO. 2
*** LOAD CASE 105 -- START ITERATION NO. 3
*** LOAD CASE 105 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 105

*** LOAD CASE 106 -- START ITERATION NO. 2
*** LOAD CASE 106 -- START ITERATION NO. 3
*** LOAD CASE 106 -- START ITERATION NO. 4
*** LOAD CASE 106 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 106

*** LOAD CASE 107 -- START ITERATION NO. 2
*** LOAD CASE 107 -- START ITERATION NO. 3
*** LOAD CASE 107 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 107

*** LOAD CASE 108 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 108

*** LOAD CASE 109 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

*** LOAD CASE 111 -- START ITERATION NO. 2
*** LOAD CASE 111 -- START ITERATION NO. 3
*** LOAD CASE 111 -- START ITERATION NO. 4
*** LOAD CASE 111 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 111

*** LOAD CASE 112 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 112

*** LOAD CASE 113 -- START ITERATION NO. 2
*** LOAD CASE 113 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 113

*** LOAD CASE 114 -- START ITERATION NO. 2
*** LOAD CASE 114 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 114

*** LOAD CASE 115 -- START ITERATION NO. 2
*** LOAD CASE 115 -- START ITERATION NO. 3
*** LOAD CASE 115 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 117

**NOTE-Tension/Compression converged after 1 iterations, Case= 118

*** LOAD CASE 119 -- START ITERATION NO. 2
*** LOAD CASE 119 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 119

**NOTE-Tension/Compression converged after 1 iterations, Case= 120

**NOTE-Tension/Compression converged after 1 iterations, Case= 121

**NOTE-Tension/Compression converged after 1 iterations, Case= 122

*** LOAD CASE 123 -- START ITERATION NO. 2
*** LOAD CASE 123 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 123

*** LOAD CASE 124 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 124

*** LOAD CASE 125 -- START ITERATION NO. 2
*** LOAD CASE 125 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 125

**NOTE-Tension/Compression converged after 1 iterations, Case= 126

*** LOAD CASE 127 -- START ITERATION NO. 2
*** LOAD CASE 127 -- START ITERATION NO. 3
*** LOAD CASE 127 -- START ITERATION NO. 4
*** LOAD CASE 127 -- START ITERATION NO. 5
*** LOAD CASE 127 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 127

**NOTE-Tension/Compression converged after 1 iterations, Case= 128

**NOTE-Tension/Compression converged after 1 iterations, Case= 129

*** LOAD CASE 130 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 130

*** LOAD CASE 131 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 131

**NOTE-Tension/Compression converged after 1 iterations, Case= 132

**NOTE-Tension/Compression converged after 1 iterations, Case= 133

**NOTE-Tension/Compression converged after 1 iterations, Case= 134

**NOTE-Tension/Compression converged after 1 iterations, Case= 135

**NOTE-Tension/Compression converged after 1 iterations, Case= 136

**NOTE-Tension/Compression converged after 1 iterations, Case= 137

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

*** LOAD CASE 202 -- START ITERATION NO. 2
*** LOAD CASE 202 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 202

*** LOAD CASE 203 -- START ITERATION NO. 2
*** LOAD CASE 203 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 203

*** LOAD CASE 204 -- START ITERATION NO. 2
*** LOAD CASE 204 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 204

*** LOAD CASE 205 -- START ITERATION NO. 2
*** LOAD CASE 205 -- START ITERATION NO. 3
*** LOAD CASE 205 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 205

*** LOAD CASE 206 -- START ITERATION NO. 2
*** LOAD CASE 206 -- START ITERATION NO. 3
*** LOAD CASE 206 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 206

*** LOAD CASE 207 -- START ITERATION NO. 2
*** LOAD CASE 207 -- START ITERATION NO. 3
*** LOAD CASE 207 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

*** LOAD CASE 211 -- START ITERATION NO. 2
*** LOAD CASE 211 -- START ITERATION NO. 3
*** LOAD CASE 211 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 211

*** LOAD CASE 212 -- START ITERATION NO. 2
*** LOAD CASE 212 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 212

*** LOAD CASE 213 -- START ITERATION NO. 2
*** LOAD CASE 213 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 213

*** LOAD CASE 214 -- START ITERATION NO. 2
*** LOAD CASE 214 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 214

*** LOAD CASE 215 -- START ITERATION NO. 2
*** LOAD CASE 215 -- START ITERATION NO. 3
*** LOAD CASE 215 -- START ITERATION NO. 4
*** LOAD CASE 215 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 215

*** LOAD CASE 216 -- START ITERATION NO. 2
*** LOAD CASE 216 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 216

*** LOAD CASE 217 -- START ITERATION NO. 2
*** LOAD CASE 217 -- START ITERATION NO. 3
*** LOAD CASE 217 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 217

*** LOAD CASE 218 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 218

*** LOAD CASE 219 -- START ITERATION NO. 2
*** LOAD CASE 219 -- START ITERATION NO. 3
*** LOAD CASE 219 -- START ITERATION NO. 4
*** LOAD CASE 219 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 219

*** LOAD CASE 220 -- START ITERATION NO. 2
*** LOAD CASE 220 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 220

*** LOAD CASE 221 -- START ITERATION NO. 2
*** LOAD CASE 221 -- START ITERATION NO. 3
*** LOAD CASE 221 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 221

*** LOAD CASE 222 -- START ITERATION NO. 2
*** LOAD CASE 222 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 222

*** LOAD CASE 223 -- START ITERATION NO. 2
*** LOAD CASE 223 -- START ITERATION NO. 3
*** LOAD CASE 223 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 223

*** LOAD CASE 224 -- START ITERATION NO. 2
*** LOAD CASE 224 -- START ITERATION NO. 3
*** LOAD CASE 224 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 224

*** LOAD CASE 225 -- START ITERATION NO. 2
*** LOAD CASE 225 -- START ITERATION NO. 3
*** LOAD CASE 225 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 225

*** LOAD CASE 226 -- START ITERATION NO. 2
*** LOAD CASE 226 -- START ITERATION NO. 3
*** LOAD CASE 226 -- START ITERATION NO. 4
*** LOAD CASE 226 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 226

*** LOAD CASE 227 -- START ITERATION NO. 2
*** LOAD CASE 227 -- START ITERATION NO. 3
*** LOAD CASE 227 -- START ITERATION NO. 4
*** LOAD CASE 227 -- START ITERATION NO. 5
*** LOAD CASE 227 -- START ITERATION NO. 6
*** LOAD CASE 227 -- START ITERATION NO. 7
*** LOAD CASE 227 -- START ITERATION NO. 8

**NOTE-Tension/Compression converged after 8 iterations, Case= 227

*** LOAD CASE 228 -- START ITERATION NO. 2
*** LOAD CASE 228 -- START ITERATION NO. 3
*** LOAD CASE 228 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 228

*** LOAD CASE 229 -- START ITERATION NO. 2
*** LOAD CASE 229 -- START ITERATION NO. 3
*** LOAD CASE 229 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 229

**NOTE-Tension/Compression converged after 1 iterations, Case= 230

**NOTE-Tension/Compression converged after 1 iterations, Case= 231

**NOTE-Tension/Compression converged after 1 iterations, Case= 232

**NOTE-Tension/Compression converged after 1 iterations, Case= 233

**NOTE-Tension/Compression converged after 1 iterations, Case= 234

**NOTE-Tension/Compression converged after 1 iterations, Case= 235

2173. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 18,2020 TIME= 16:37: 4 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

ATTACHMENT 3 STAAD OUTPUT : MD-216-3520-EG-CV-CAL-1285_PIT IN NEUTRALIZATION AREA

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAY 18, 2020                *
*           Time=    16:23:44                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\samrat dey 14.10.15\7. Maaden\1.CCD _Neutralization area\Pipe rack\Neutralization area.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 13-JAN-17
    4. END JOB INFORMATION
    5. INPUT WIDTH 79
    6. UNIT METER KN
    7. JOINT COORDINATES
    8. 1 -0.05 -1.84 -0.05; 2 1.55 -1.84 -0.05; 3 -0.05 -1.84 1.55; 4 1.55 -1.84 1.55
    9. 5 0.05 -1.84 1.55; 6 0.05 -1.84 1.45; 7 -0.05 -1.84 1.45; 8 0.4 -1.84 1.55
   10. 9 0.4 -1.84 1.45; 10 0.75 -1.84 1.55; 11 0.75 -1.84 1.45; 12 1.1 -1.84 1.55
   11. 13 1.1 -1.84 1.45; 14 1.45 -1.84 1.55; 15 1.45 -1.84 1.45; 16 1.55 -1.84 1.45
   12. 17 0.05 -1.84 1.1; 18 -0.05 -1.84 1.1; 19 0.4 -1.84 1.1; 20 0.75 -1.84 1.1
   13. 21 1.1 -1.84 1.1; 22 1.45 -1.84 1.1; 23 1.55 -1.84 1.1; 24 0.05 -1.84 0.75
   14. 25 -0.05 -1.84 0.75; 26 0.4 -1.84 0.75; 27 0.75 -1.84 0.75; 28 1.1 -1.84 0.75
   15. 29 1.45 -1.84 0.75; 30 1.55 -1.84 0.75; 31 0.05 -1.84 0.4; 32 -0.05 -1.84 0.4
   16. 33 0.4 -1.84 0.4; 34 0.75 -1.84 0.4; 35 1.1 -1.84 0.4; 36 1.45 -1.84 0.4
   17. 37 1.55 -1.84 0.4; 38 0.05 -1.84 0.05; 39 -0.05 -1.84 0.05; 40 0.4 -1.84 0.05
   18. 41 0.75 -1.84 0.05; 42 1.1 -1.84 0.05; 43 1.45 -1.84 0.05; 44 1.55 -1.84 0.05
   19. 45 0.05 -1.84 -0.05; 46 0.4 -1.84 -0.05; 47 0.75 -1.84 -0.05
   20. 48 1.1 -1.84 -0.05; 49 1.45 -1.84 -0.05; 50 -0.05 0 -0.05; 51 1.55 0 -0.05
   21. 52 -0.05 0 1.55; 53 1.55 0 1.55; 54 0.05 -1.59 -0.05; 55 -0.05 -1.59 -0.05
   22. 56 0.4 -1.59 -0.05; 57 0.75 -1.59 -0.05; 58 1.1 -1.59 -0.05
   23. 59 1.45 -1.59 -0.05; 60 1.55 -1.59 -0.05; 61 0.05 -1.19 -0.05
   24. 62 -0.05 -1.19 -0.05; 63 0.4 -1.19 -0.05; 64 0.75 -1.19 -0.05
   25. 65 1.1 -1.19 -0.05; 66 1.45 -1.19 -0.05; 67 1.55 -1.19 -0.05
   26. 68 0.05 -0.75 -0.05; 69 -0.05 -0.75 -0.05; 70 0.4 -0.75 -0.05
   27. 71 0.75 -0.75 -0.05; 72 1.1 -0.75 -0.05; 73 1.45 -0.75 -0.05
   28. 74 1.55 -0.75 -0.05; 75 0.05 -0.5 -0.05; 76 -0.05 -0.5 -0.05
   29. 77 0.4 -0.5 -0.05; 78 0.75 -0.5 -0.05; 79 1.1 -0.5 -0.05; 80 1.45 -0.5 -0.05
   30. 81 1.55 -0.5 -0.05; 82 0.05 -0.25 -0.05; 83 -0.05 -0.25 -0.05
   31. 84 0.4 -0.25 -0.05; 85 0.75 -0.25 -0.05; 86 1.1 -0.25 -0.05
   32. 87 1.45 -0.25 -0.05; 88 1.55 -0.25 -0.05; 89 0.05 0 -0.05; 90 0.4 0 -0.05
   33. 91 0.75 0 -0.05; 92 1.1 0 -0.05; 93 1.45 0 -0.05; 94 1.55 -1.59 0.05
   34. 95 1.55 -1.59 0.4; 96 1.55 -1.59 0.75; 97 1.55 -1.59 1.1; 98 1.55 -1.59 1.45
   35. 99 1.55 -1.59 1.55; 100 1.55 -1.19 0.05; 101 1.55 -1.19 0.4
   36. 102 1.55 -1.19 0.75; 103 1.55 -1.19 1.1; 104 1.55 -1.19 1.45
   37. 105 1.55 -1.19 1.55; 106 1.55 -0.75 0.05; 107 1.55 -0.75 0.4
   38. 108 1.55 -0.75 0.75; 109 1.55 -0.75 1.1; 110 1.55 -0.75 1.45
```

39. 111 1.55 -0.75 1.55; 112 1.55 -0.5 0.05; 113 1.55 -0.5 0.4; 114 1.55 -0.5 0.75
40. 115 1.55 -0.5 1.1; 116 1.55 -0.5 1.45; 117 1.55 -0.5 1.55; 118 1.55 -0.25 0.05
41. 119 1.55 -0.25 0.4; 120 1.55 -0.25 0.75; 121 1.55 -0.25 1.1
42. 122 1.55 -0.25 1.45; 123 1.55 -0.25 1.55; 124 1.55 0 0.05; 125 1.55 0 0.4
43. 126 1.55 0 0.75; 127 1.55 0 1.1; 128 1.55 0 1.45; 129 1.45 -1.59 1.55
44. 130 1.1 -1.59 1.55; 131 0.75 -1.59 1.55; 132 0.4 -1.59 1.55
45. 133 0.05 -1.59 1.55; 134 -0.05 -1.59 1.55; 135 1.45 -1.19 1.55
46. 136 1.1 -1.19 1.55; 137 0.75 -1.19 1.55; 138 0.4 -1.19 1.55
47. 139 0.05 -1.19 1.55; 140 -0.05 -1.19 1.55; 141 1.45 -0.75 1.55
48. 142 1.1 -0.75 1.55; 143 0.75 -0.75 1.55; 144 0.4 -0.75 1.55
49. 145 0.05 -0.75 1.55; 146 -0.05 -0.75 1.55; 147 1.45 -0.5 1.55
50. 148 1.1 -0.5 1.55; 149 0.75 -0.5 1.55; 150 0.4 -0.5 1.55; 151 0.05 -0.5 1.55
51. 152 -0.05 -0.5 1.55; 153 1.45 -0.25 1.55; 154 1.1 -0.25 1.55
52. 155 0.75 -0.25 1.55; 156 0.4 -0.25 1.55; 157 0.05 -0.25 1.55
53. 158 -0.05 -0.25 1.55; 159 1.45 0 1.55; 160 1.1 0 1.55; 161 0.75 0 1.55
54. 162 0.4 0 1.55; 163 0.05 0 1.55; 164 -0.05 -1.59 1.45; 165 -0.05 -1.59 1.1
55. 166 -0.05 -1.59 0.75; 167 -0.05 -1.59 0.4; 168 -0.05 -1.59 0.05
56. 169 -0.05 -1.19 1.45; 170 -0.05 -1.19 1.1; 171 -0.05 -1.19 0.75
57. 172 -0.05 -1.19 0.4; 173 -0.05 -1.19 0.05; 174 -0.05 -0.75 1.45
58. 175 -0.05 -0.75 1.1; 176 -0.05 -0.75 0.75; 177 -0.05 -0.75 0.4
59. 178 -0.05 -0.75 0.05; 179 -0.05 -0.5 1.45; 180 -0.05 -0.5 1.1
60. 181 -0.05 -0.5 0.75; 182 -0.05 -0.5 0.4; 183 -0.05 -0.5 0.05
61. 184 -0.05 -0.25 1.45; 185 -0.05 -0.25 1.1; 186 -0.05 -0.25 0.75
62. 187 -0.05 -0.25 0.4; 188 -0.05 -0.25 0.05; 189 -0.05 0 1.45; 190 -0.05 0 1.1
63. 191 -0.05 0 0.75; 192 -0.05 0 0.4; 193 -0.05 0 0.05; 194 -0.05 0.35 -0.05
64. 195 1.55 0.35 -0.05; 196 -0.05 0.35 1.55; 197 1.55 0.35 1.55
65. 198 0.05 0.35 -0.05; 199 0.4 0.35 -0.05; 200 0.75 0.35 -0.05
66. 201 1.1 0.35 -0.05; 202 1.45 0.35 -0.05; 203 1.55 0.35 0.05; 204 1.55 0.35 0.4
67. 205 1.55 0.35 0.75; 206 1.55 0.35 1.1; 207 1.55 0.35 1.45; 208 1.45 0.35 1.55
68. 209 1.1 0.35 1.55; 210 0.75 0.35 1.55; 211 0.4 0.35 1.55; 212 0.05 0.35 1.55
69. 213 -0.05 0.35 1.45; 214 -0.05 0.35 1.1; 215 -0.05 0.35 0.75
70. 216 -0.05 0.35 0.4; 217 -0.05 0.35 0.05; 218 -0.05 0.6 -0.05
71. 219 1.55 0.6 -0.05; 220 -0.05 0.6 1.55; 221 1.55 0.6 1.55; 222 0.05 0.6 -0.05
72. 223 0.4 0.6 -0.05; 224 0.75 0.6 -0.05; 225 1.1 0.6 -0.05; 226 1.45 0.6 -0.05
73. 227 1.55 0.6 0.05; 228 1.55 0.6 0.4; 229 1.55 0.6 0.75; 230 1.55 0.6 1.1
74. 231 1.55 0.6 1.45; 232 1.45 0.6 1.55; 233 1.1 0.6 1.55; 234 0.75 0.6 1.55
75. 235 0.4 0.6 1.55; 236 0.05 0.6 1.55; 237 -0.05 0.6 1.45; 238 -0.05 0.6 1.1
76. 239 -0.05 0.6 0.75; 240 -0.05 0.6 0.4; 241 -0.05 0.6 0.05
77. 242 -0.05 0.85 -0.05; 243 1.55 0.85 -0.05; 244 -0.05 0.85 1.55
78. 245 1.55 0.85 1.55; 246 0.05 0.85 -0.05; 247 0.4 0.85 -0.05
79. 248 0.75 0.85 -0.05; 249 1.1 0.85 -0.05; 250 1.45 0.85 -0.05
80. 251 1.55 0.85 0.05; 252 1.55 0.85 0.4; 253 1.55 0.85 0.75; 254 1.55 0.85 1.1
81. 255 1.55 0.85 1.45; 256 1.45 0.85 1.55; 257 1.1 0.85 1.55; 258 0.75 0.85 1.55
82. 259 0.4 0.85 1.55; 260 0.05 0.85 1.55; 261 -0.05 0.85 1.45; 262 -0.05 0.85 1.1
83. 263 -0.05 0.85 0.75; 264 -0.05 0.85 0.4; 265 -0.05 0.85 0.05
84. ELEMENT INCIDENCES SHELL
85. 1 3 5 6 7; 2 5 8 9 6; 3 8 10 11 9; 4 10 12 13 11; 5 12 14 15 13; 6 14 4 16 15
86. 7 7 6 17 18; 8 6 9 19 17; 9 9 11 20 19; 10 11 13 21 20; 11 13 15 22 21
87. 12 15 16 23 22; 13 18 17 24 25; 14 17 19 26 24; 15 19 20 27 26; 16 20 21 28 27
88. 17 21 22 29 28; 18 22 23 30 29; 19 25 24 31 32; 20 24 26 33 31; 21 26 27 34 33
89. 22 27 28 35 34; 23 28 29 36 35; 24 29 30 37 36; 25 32 31 38 39; 26 31 33 40 38
90. 27 33 34 41 40; 28 34 35 42 41; 29 35 36 43 42; 30 36 37 44 43; 31 39 38 45 1
91. 32 38 40 46 45; 33 40 41 47 46; 34 41 42 48 47; 35 42 43 49 48; 36 43 44 2 49
92. 37 1 45 54 55; 38 45 46 56 54; 39 46 47 57 56; 40 47 48 58 57; 41 48 49 59 58
93. 42 49 2 60 59; 43 55 54 61 62; 44 54 56 63 61; 45 56 57 64 63; 46 57 58 65 64
94. 47 58 59 66 65; 48 59 60 67 66; 49 62 61 68 69; 50 61 63 70 68; 51 63 64 71 70

95. 52 64 65 72 71; 53 65 66 73 72; 54 66 67 74 73; 55 69 68 75 76; 56 68 70 77 75
96. 57 70 71 78 77; 58 71 72 79 78; 59 72 73 80 79; 60 73 74 81 80; 61 76 75 82 83
97. 62 75 77 84 82; 63 77 78 85 84; 64 78 79 86 85; 65 79 80 87 86; 66 80 81 88 87
98. 67 83 82 89 50; 68 82 84 90 89; 69 84 85 91 90; 70 85 86 92 91; 71 86 87 93 92
99. 72 87 88 51 93; 73 2 44 94 60; 74 44 37 95 94; 75 37 30 96 95; 76 30 23 97 96
100. 77 23 16 98 97; 78 16 4 99 98; 79 60 94 100 67; 80 94 95 101 100
101. 81 95 96 102 101; 82 96 97 103 102; 83 97 98 104 103; 84 98 99 105 104
102. 85 67 100 106 74; 86 100 101 107 106; 87 101 102 108 107; 88 102 103 109 108
103. 89 103 104 110 109; 90 104 105 111 110; 91 74 106 112 81; 92 106 107 113 112
104. 93 107 108 114 113; 94 108 109 115 114; 95 109 110 116 115; 96 110 111 117 116
105. 97 81 112 118 88; 98 112 113 119 118; 99 113 114 120 119; 100 114 115 121 120
106. 101 115 116 122 121; 102 116 117 123 122; 103 88 118 124 51
107. 104 118 119 125 124; 105 119 120 126 125; 106 120 121 127 126
108. 107 121 122 128 127; 108 122 123 53 128; 109 4 14 129 99; 110 14 12 130 129
109. 111 12 10 131 130; 112 10 8 132 131; 113 8 5 133 132; 114 5 3 134 133
110. 115 99 129 135 105; 116 129 130 136 135; 117 130 131 137 136
111. 118 131 132 138 137; 119 132 133 139 138; 120 133 134 140 139
112. 121 105 135 141 111; 122 135 136 142 141; 123 136 137 143 142
113. 124 137 138 144 143; 125 138 139 145 144; 126 139 140 146 145
114. 127 111 141 147 117; 128 141 142 148 147; 129 142 143 149 148
115. 130 143 144 150 149; 131 144 145 151 150; 132 145 146 152 151
116. 133 117 147 153 123; 134 147 148 154 153; 135 148 149 155 154
117. 136 149 150 156 155; 137 150 151 157 156; 138 151 152 158 157
118. 139 123 153 159 53; 140 153 154 160 159; 141 154 155 161 160
119. 142 155 156 162 161; 143 156 157 163 162; 144 157 158 52 163; 145 3 7 164 134
120. 146 7 18 165 164; 147 18 25 166 165; 148 25 32 167 166; 149 32 39 168 167
121. 150 39 1 55 168; 151 134 164 169 140; 152 164 165 170 169; 153 165 166 171 170
122. 154 166 167 172 171; 155 167 168 173 172; 156 168 55 62 173
123. 157 140 169 174 146; 158 169 170 175 174; 159 170 171 176 175
124. 160 171 172 177 176; 161 172 173 178 177; 162 173 62 69 178
125. 163 146 174 179 152; 164 174 175 180 179; 165 175 176 181 180
126. 166 176 177 182 181; 167 177 178 183 182; 168 178 69 76 183
127. 169 152 179 184 158; 170 179 180 185 184; 171 180 181 186 185
128. 172 181 182 187 186; 173 182 183 188 187; 174 183 76 83 188
129. 175 158 184 189 52; 176 184 185 190 189; 177 185 186 191 190
130. 178 186 187 192 191; 179 187 188 193 192; 180 188 83 50 193; 181 50 89 198 194
131. 182 89 90 199 198; 183 90 91 200 199; 184 91 92 201 200; 185 92 93 202 201
132. 186 93 51 195 202; 187 51 124 203 195; 188 124 125 204 203
133. 189 125 126 205 204; 190 126 127 206 205; 191 127 128 207 206
134. 192 128 53 197 207; 193 53 159 208 197; 194 159 160 209 208
135. 195 160 161 210 209; 196 161 162 211 210; 197 162 163 212 211
136. 198 163 52 196 212; 199 52 189 213 196; 200 189 190 214 213
137. 201 190 191 215 214; 202 191 192 216 215; 203 192 193 217 216
138. 204 193 50 194 217; 205 194 198 222 218; 206 198 199 223 222
139. 207 199 200 224 223; 208 200 201 225 224; 209 201 202 226 225
140. 210 202 195 219 226; 211 195 203 227 219; 212 203 204 228 227
141. 213 204 205 229 228; 214 205 206 230 229; 215 206 207 231 230
142. 216 207 197 221 231; 217 197 208 232 221; 218 208 209 233 232
143. 219 209 210 234 233; 220 210 211 235 234; 221 211 212 236 235
144. 222 212 196 220 236; 223 196 213 237 220; 224 213 214 238 237
145. 225 214 215 239 238; 226 215 216 240 239; 227 216 217 241 240
146. 228 217 194 218 241; 229 218 222 246 242; 230 222 223 247 246
147. 231 223 224 248 247; 232 224 225 249 248; 233 225 226 250 249
148. 234 226 219 243 250; 235 219 227 251 243; 236 227 228 252 251
149. 237 228 229 253 252; 238 229 230 254 253; 239 230 231 255 254
150. 240 231 221 245 255; 241 221 232 256 245; 242 232 233 257 256

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151. 243 233 234 258 257; 244 234 235 259 258; 245 235 236 260 259
152. 246 236 220 244 260; 247 220 237 261 244; 248 237 238 262 261
153. 249 238 239 263 262; 250 239 240 264 263; 251 240 241 265 264
154. 252 241 218 242 265
155. START GROUP DEFINITION
156. MEMBER
157. FLOOR
158. ELEMENT
159. _FDN 8 TO 11 14 TO 17 20 TO 23 26 TO 29
160. _WALL 38 TO 41 44 TO 47 50 TO 53 56 TO 59 62 TO 65 68 TO 71 74 TO 77 -
161. 80 TO 83 86 TO 89 92 TO 95 98 TO 101 104 TO 107 110 TO 113 116 TO 119 122 -
162. 123 TO 125 128 TO 131 134 TO 137 140 TO 143 146 TO 149 152 TO 155 158 TO 161 -
163. 164 TO 167 170 TO 173 176 TO 179 182 TO 185 188 TO 191 194 TO 197 -
164. 200 TO 203 206 TO 209 212 TO 215 218 TO 221 224 TO 227 230 TO 233 -
165. 236 TO 239 242 TO 245 248 TO 251
166. END GROUP DEFINITION
167. ELEMENT PROPERTY
168. 1 TO 36 THICKNESS 0.2
169. 37 TO 252 THICKNESS 0.2
170. DEFINE MATERIAL START
171. ISOTROPIC CONCRETE
172. E 2.95804E+007
173. POISSON 0.17
174. DENSITY 25
175. ALPHA 5.5E-006
176. DAMP 0.05
177. TYPE CONCRETE
178. STRENGTH FCU 35000
179. END DEFINE MATERIAL
180. CONSTANTS
181. MATERIAL CONCRETE ALL
182. SUPPORTS
183. 1 TO 36 PLATE MAT DIRECT Y SUBGRADE 117600 PRINT
```

JOINT	INFLUENCE AREA
1	0.002500
2	0.002500
3	0.002500
4	0.002500
5	0.011250
6	0.050625
7	0.011250
8	0.017500
9	0.078750
10	0.017500
11	0.078750
12	0.017500
13	0.078750
14	0.011250
15	0.050625
16	0.011250
17	0.078750
18	0.017500
19	0.122500

20	0.122500
21	0.122500
22	0.078750
23	0.017500
24	0.078750
25	0.017500
26	0.122500
27	0.122500
28	0.122500
29	0.078750
30	0.017500
31	0.078750
32	0.017500
33	0.122500
34	0.122500
35	0.122500
36	0.078750
37	0.017500
38	0.050625
39	0.011250
40	0.078750
41	0.078750
42	0.078750
43	0.050625
44	0.011250
45	0.011250
46	0.017500
47	0.017500
48	0.017500
49	0.011250
TOTAL	2.559999

```
184. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD(DL)
185. SELFWEIGHT Y -1
186. LOAD 2 LOADTYPE NONE  TITLE SOIL PR (DRY COND)(SDC)
187. ELEMENT LOAD
188. 37 TO 42 73 TO 78 109 TO 114 145 TO 150 TRAP JT 18.43 18.43 15.9259 15.9259
189. 43 TO 48 79 TO 84 115 TO 120 151 TO 156 TRAP JT 15.9259 15.9259 11.9194 11.9194
190. 49 TO 54 85 TO 90 121 TO 126 157 TO 162 TRAP JT 11.9194 11.9194 7.51223 7.51223
191. 55 TO 60 91 TO 96 127 TO 132 163 TO 168 TRAP JT 7.51223 7.51223 5.00815 5.00815
192. 61 TO 66 97 TO 102 133 TO 138 169 TO 173 -
193. 174 TRAP JT 5.00815 5.00815 2.50408 2.50408
194. 67 TO 72 103 TO 108 139 TO 144 175 TO 180 TRAP JT 2.50408 2.50408 0 0
195. LOAD 3 LOADTYPE NONE  TITLE SURCHARGE PR(SP)
196. ELEMENT LOAD
197. 37 TO 180 PR 5
198. LOAD 4 LOADTYPE NONE  TITLE WATER PRESSURE(WP)
199. ELEMENT LOAD
200. 37 TO 42 73 TO 78 109 TO 114 145 TO 150 TRAP JT -27.9 -27.9 -25.3071 -25.3071
201. 43 TO 48 79 TO 84 115 TO 120 151 TO 155 -
202. 156 TRAP JT -25.3071 -25.3071 -21.1584 -21.1584
203. 49 TO 54 85 TO 90 121 TO 126 157 TO 161 -
204. 162 TRAP JT -21.1584 -21.1584 -16.5948 -16.5948
205. 55 TO 60 91 TO 96 127 TO 132 163 TO 167 -
206. 168 TRAP JT -16.5948 -16.5948 -14.0019 -14.0019
207. 61 TO 66 97 TO 102 133 TO 138 169 TO 173 -
208. 174 TRAP JT -14.0019 -14.0019 -11.4089 -11.4089
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209. 67 TO 72 103 TO 108 139 TO 144 175 TO 179 -
210. 180 TRAP JT -11.4089 -11.4089 -8.81599 -8.81599
211. 181 TO 204 TRAP JT -8.81599 -8.81599 -5.18587 -5.18587
212. 205 TO 228 TRAP JT -5.18587 -5.18587 -2.59294 -2.59294
213. 229 TO 252 TRAP JT -2.59294 -2.59294 -1.23641E-006 -1.23641E-006
214. 1 TO 36 PR GY -27.9
215. **
216. LOAD COMB 101 DL
217. 1 1.0
218. LOAD COMB 102 DL+SDC+SP
219. 1 1.0 2 1.0 3 1.0
220. LOAD COMB 103 DL+WP+SDC+SP
221. 1 1.0 2 1.0 3 1.0 4 1.0
222. LOAD COMB 201 1.4DL
223. 1 1.4
224. LOAD COMB 202 1.4DL + 1.7(SDC+SP)
225. 1 1.4 2 1.7 3 1.7
226. LOAD COMB 203 1.4(DL+WP) + 1.7(SDC+SP)
227. 1 1.4 2 1.7 3 1.7 4 1.4
228. PERFORM ANALYSIS PRINT STATICS CHECK
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**265**	**NUMBER OF MEMBERS**	**0**
NUMBER OF PLATES	**252**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**49**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 167/ 28/ 174 DOF
TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 1443
TOTAL LOAD COMBINATION CASES = 6 SO FAR.
SIZE OF STIFFNESS MATRIX = 252 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 16.1/ 194103.3 MB

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE NONE TITLE DEAD LOAD(DL)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.749999952E+00
Y = -0.669110018E+00
Z = 0.749999952E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -98.88
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 74.16 MY= 0.00 MZ= -74.16

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 98.88
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -74.16 MY= -0.00 MZ= 74.16

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X	-3.86378E-04	166
Y	-3.41052E-02	263
Z	-3.86378E-04	57
RX	-3.97591E-05	41
RY	-4.72565E-06	65
RZ	3.97591E-05	24

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE NONE TITLE SOIL PR (DRY COND)(SDC)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 1.2732117E-16
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = 1.0590806E-15

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4.0140275E-16 MY= -3.0019329E-08 MZ= -2.8850822E-15

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -2.9939766E-15
SUMMATION FORCE-Y = -2.4996136E-13
SUMMATION FORCE-Z = 2.2840646E-14

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2.5348583E-13 MY= 3.0019316E-08 MZ= -1.7528207E-13

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	7.52355E-04	171
Y =	2.43423E-04	1
Z =	7.52355E-04	64
RX=	1.35058E-05	57
RY=	-9.94817E-06	170
RZ=	-1.35058E-05	166

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE NONE TITLE SURCHARGE PR(SP)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 6.5589694E-17
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -3.6653064E-16

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4.2649042E-16 MY= -2.0258295E-15 MZ= 2.7910035E-16

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 7.4077772E-15
SUMMATION FORCE-Y = -1.0825241E-13
SUMMATION FORCE-Z = 1.5417436E-14

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 9.2874560E-14 MY= -1.5454006E-14 MZ= -6.5530566E-14

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	4.47722E-04	176
Y =	1.02249E-04	239
Z =	4.47722E-04	71
RX=	6.19058E-06	57
RY=	6.21988E-06	72
RZ=	-6.19058E-06	166

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE NONE TITLE WATER PRESSURE(WP)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.749999940E+00
Y = -0.184000007E+01
Z = 0.749999940E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -71.42
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 53.57 MY= 0.00 MZ= -53.57

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 71.42
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -53.57 MY= -0.00 MZ= 53.57

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X	-1.55956E-03	176
Y	-2.41548E-02	1
Z	-1.55956E-03	71
RX	-2.49429E-05	57
RY	2.18371E-05	175
RZ	2.49429E-05	166

********** END OF DATA FROM INTERNAL STORAGE **********

229. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 18,2020 TIME= 16:23:47 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
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